

MAGNETOM

Troubleshooting Guide System

Gradient

Harmony syngo MR
Symphony syngo MR
Sonata syngo MR
Trio
Upgrades to syngo MR
from Impact/Expert
from Vision

Document Version

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1.1 General

In this section you will find the strategies and test procedures for trouble shooting the Gradient System.

The enclosed information applies to all **MAGNETOM** MR systems equipped with the **gradient cabinets K2209 (GPA) or K2217 (KAS)**.

It contains information about possible errors as well as troubleshooting charts to assist you during troubleshooting.

1.1.1 Overview about the Gradient Systems:

System	Harmony/ Symphony							Sonata
Option	Turbo	Ultra	Turbo	Ultra	Ultra with Quantum HW	Sprint	Quantum	n.a.
Gradient Cabinet	K2209 3 amplifiers	K2209 6 amplifiers	K2217 Cas. light	K2217 Cas. light	K2217 Cas. Quantum	K2217 Cas. Quantum	K2217 Cas. Quantum	K2217 Cas. Quantum
Software	N3.5/N4	N3.5/N4	N4	N4	>=N4 VA12B	>=N4 VA15A	N3.5/N4	N3.5/N4
Field Strength [mT/m]	20mT/m	20mT/m	20mT/m	20mT/m	20mT/m	30mT/m	30mT/m	40mT/m
Software configuration	400V/300A	800V/300A	500V/300A	1000V/300A	2000V/380A/ SR50 for VA15A; 1000V/380A for VA12B	2000V/ 380A/ SR75	2000V/380A	2000V/500A
Rise time [μsec]	800μsec	400μsec	800μsec	400μsec	400μsec	400μsec	300μsec	200μsec
Slew rate [mT/m/msec]	25	50	25	50	50	75	125 for Z, 100 for X and Y)	200
Field of View	500	500	500	500	500	500	500	400
Board Ensemble	D1, 3 x D2, D3, D4, D6, D7 (fi- ber optic), D8 (Backpl.)	D1, 3 x D2, D3, D4, D5, D6, D7 (fiber optic), D8 (Backpl.)	D111, D124, D142, D16 (CAN) D17, D18 (Backplane)	D111, D124, D142, D16 (CAN) D17, D18 (Backplane)	D11, D12, D14, D16 (CAN) D17, D18 (Backplane)	D11, D12, D14, D16 (CAN), D17, D18 (Back- plane)	D11, D12, D14, D16 (CAN), D17, D18 (Back- plane)	D11, D12, D14, D16 (CAN), D17, D18 (Back- plane)
Regulator Adjustment	N4: software based (tuneup step regulator adjustment) N3.5: soft- ware based (tuneup step regula- tor adjustment)		N4: software based (tuneup step regulator adjustment)				N4: software based (tuneup step regulator adjustment) N3.5: ac- cording to the jumper settings mentioned in the diagrams	
Stimulation Monitor	no	no	no	no	SW-Stim.	SW-Stim.	N4: SW-Stim. in the MMC; N3.5: HW-Stim. D13	N4: SW-Stim. in the MMC; N3.5: HW-Stim. D131
Gradient Coil	AS39S, AS39RS	AS39S, AS39RS	AS39RS	AS39RS	AS39T	AS39T	AS39T	AS Sonata

System	Harmony/ Symphony							Sonata
Option	Turbo	Ultra	Turbo	Ultra	Ultra with Quantum HW	Sprint	Quantum	n.a.
Gradient Cabinet	K2209 3 amplifiers	K2209 6 amplifiers	K2217 Cas. light	K2217 Cas. light	K2217 Cas. Quantum	K2217 Cas. Quantum	K2217 Cas. Quantum	K2217 Cas. Quantum
Amplifier	3x output stage (4769258) or 3x Papillon 400 (5939447)	6x output stage (4769258) or 6x Papillon 400 (5939447)	3x power stage light (5938720)	3x power stage light (5938720)	3x power stage (5938142)	3x power stage (5938142)	3x power stage (5938142)	3x power stage (5938142)
Chokes	6x (3058278)		6x (5725887)					
Body Resonator	BC024 BC047	BC024 BC047	BC024 BC047	BC024 BC047	BC024 BC047Q BC047-2	BC024 BC047Q BC047-2	BC024 BC047Q BC047-2	BC Sonata BC2 Sonata
Body Tune Box	BTB024 BTB047	BTB024 BTB047	BTB024 BTB047syngo	BTB024 BTB047syngo BTB047-2 for BC047-2	BTB024 BTB047syngo BTB047-2 for BC047-2	BTB024 BTB047syngo BTB2 for BC047-2	BTB024 BTB047 BTB047syngo BTB047-2 for BC047-2	BTB Sonata BTB047-2 for BC2 Sonata
Cooler (RCA)	1	1	1	1	1	1	1	2
Cooling: Heat dissipation	30 kW	30 kW	30 kW	30 kW	30 kW	30 kW	30 kW (45 kW for research)	50kW
Flow reducer	yes	yes	yes	yes	yes	yes	yes	no

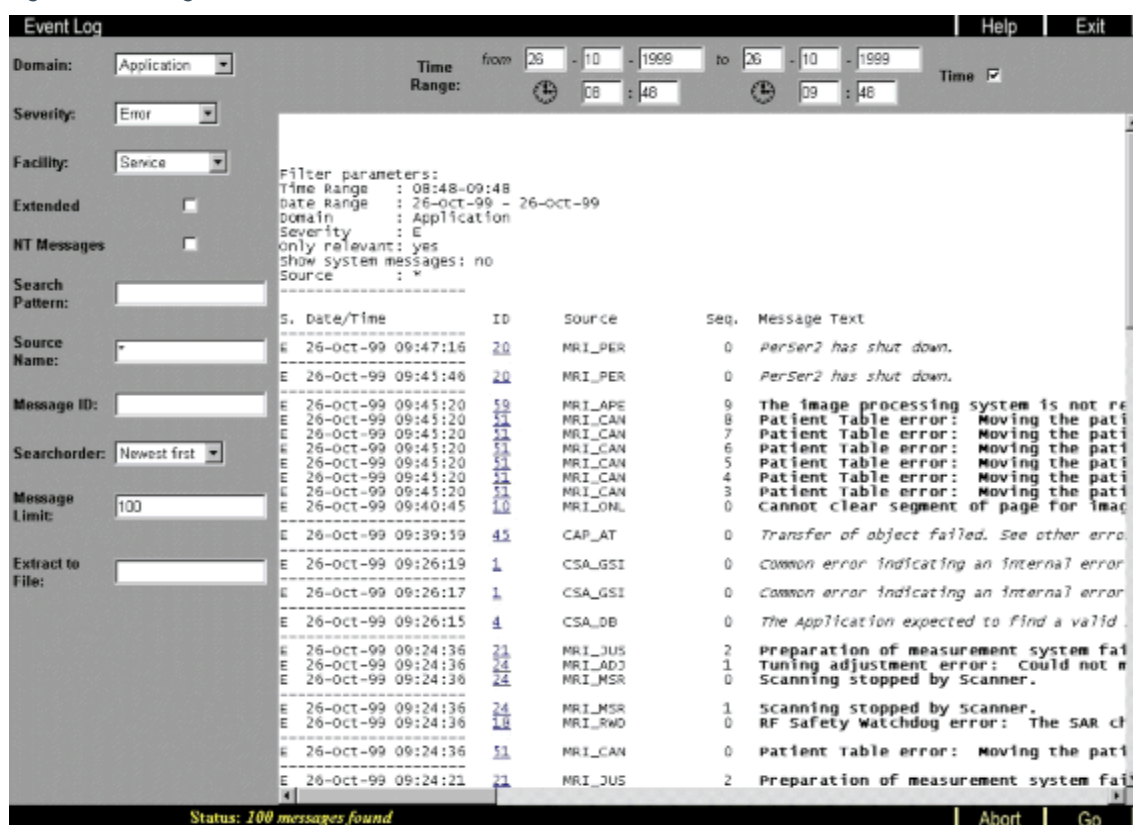
1.2 Error Messages - Error Codes

The error messages are listed in the (→ SKB, <https://skb.healthcare.siemens.com/>).

1.2.1 Error Messages in the Event Log

In the **Event Log window**, available from the Service Software Homepage, you can select detailed information from the error logger. (→ Fig. 1 Page 8) shows the window with the possible selections.

Fig. 1: Event Log window



Each Message ID has a hypertext-link to an **Extended Error Message** explanation (→ Fig. 2 Page 8). Here you will find some more information about recommended action and an explanation of the error message.

Fig. 2: Example of an Extended Error Message

Message:	E	_23	_KAS_RECTIFY_TEMP
Text	Gradient Power Amplifier error: Temperature of rectifier too high. Check sequence parameter, reboot Scanner.		
Explanation:	The temperature of the rectifier assembly on D32 is too high.		
Action:	1.a) Check the RCA cooling system, check water hoses to D32. 1.b) Check flat cable between D31 and D32. Replace defective rectifier board D32; for further checks refer to TSG GPA, TSG Chart 03. 2.) Check for monitoring error (refer to TSG GPA, TSG Chart 03).		

The **Explanation** line gives you more detailed description of the error

The **Action** line gives you a recommended action to solve the error.

Additional information may be found in the information line.

1.2.2 Sources for errors in the Gradient System

As compared to most of the other system components, the gradient system contains a number of assemblies that generate and process analog signals. These assemblies, all field replaceable units, (FRU's), form a closed current control circuit, which makes it at times difficult to limit the source of error to a single FRU.

Errors are usually looped across several assemblies. For example the backplane D8/D18 of the gradient small signal unit (GSSU) as well as all plug-in connections of the boards (PCB's) may be the source for an error.

Usually, if error messages are sent by the application software (ASW) to the MMC via the CAN-bus, at the same time the modulators will be blocked. The gradient current control is disabled and the **Regulation error message** is generated as a consequence.

Except for image quality errors, the following major types of errors are stored in the system error log (system/report). Several of them also are indicated via LED's on the assemblies of the GSSU:

- **Dynamic errors:**

These errors occur during measurements: They are either reproducible under the same conditions or they appear sporadically. In either case, the measurement is terminated. In most cases, the error message generated corresponds to the actual situation.

- **Static errors:**

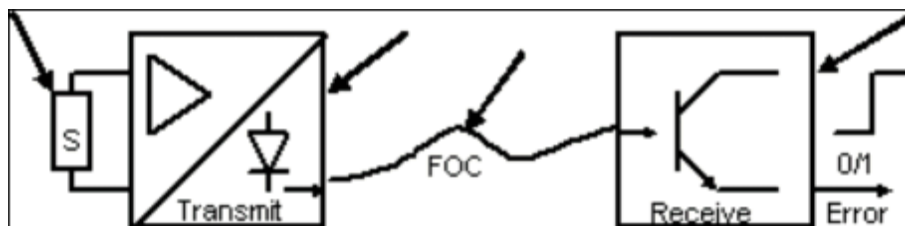
These errors messages are generated during GPA/KAS switch-on. Reset will not remove these errors. The source for error is an actual defect in the component or an error in the monitoring/transfer path (monitoring error).

- **Image quality errors:**

Image quality depends on the error-free interactions between all system components, the most important of these are the RF and gradient systems. No specific error message will be generated when image quality degraded. Refer to (→ Image Quality - Test Strategy / Page 88).

When errors continue to occur, monitoring errors have to be ruled out (refer to (→ Fig. 3 Page 9), the arrows show the range of error sources). There is a relatively simple test that traces the signal path, e.g. it tests the light emitter and the fiber optic cables (FOC). "Light off" means that "there is an error" (fail safe-principle).

Fig. 3: Error signal sources in a transmission line



Example:

The components involved in the error **COIL_NOT_RDY** (refer to (→ Troubleshooting Charts GPA K2209 / Page 13), (→ Troubleshooting Charts KAS K2217 / Page 48), (→ Chart 01 / Page 15) are:

- NTC in gradient coil
- Coil plug X2
- Line to D9 in patient table electronics
- Plug in location X20 on D9
- D9 monitoring assembly with fiber optic cable transmitter, including supply voltage (from patient table electronics)
- FOC D9-->D7/D17 (in gradient small signal unit, GSSU)
- D7/D17 FOC assembly in GSSU including Power Supply E4A
- Backplane D8/D18 in the GSSU
- D6/D16 Service and CAN assembly in the GSSU
- CAN-Bus transmission path to MPCU



All error signals could be influenced by a defective backplane D8/D18 of the GSSU and/or by bad connections between D8/D18 and the relevant board.

Error messages are forwarded directly or via software-controlled multiplexer (on D4 K2209/D14/D142 K2217, modulator PCB) to the CAN processor.

All occurring changes of the GPA/KAS-status are forwarded from the CAN processor on D6 K209/D16 K2217 to the MPCU via a CAN telegram such as "Alarm", "General Error" or "Warning".

The type of telegram to be used is established in the application software of the GPA/KAS CAN unit. The definitions for the three types of telegrams are:

Alarm:	Fatal error or error status which will terminate a measurement in progress or prevent the start of a measurement. Whenever possible, the error will be indicated independent of the MPCU or MPCU Software (e.g. red LED's in the GSSU).
General Error:	Serious error or error status which will terminate a measurement in progress or prevent the start of a measurement.
Warning:	Error status which neither affects the measurement in progress nor prevents the start of a measurement. However the image quality may be affected !

1.2.2.1 Different STOP-modes for the KAS:

MOS:

All modulators will be inhibited via loadware by "SOFTSTOP", (i.e. freewheeling of power stage/gradient coil circuit, avoiding stimulation of the patient).

MOD OFF:

All modulators will be inhibited by hardware.

POW OFF:

The KAS will be switched into stand-by mode by loadware, i.e. transformer T1 off, power stage voltages are off.

TS OFF:

The KAS will be directly switched into stand-by mode, i.e. transformer T1 off, power stage voltages are off.

1.2.3 Test of CAN communication and GPA/KAS status

If you select "GPA/KAS Status" within the platform "Test Tools" of the Service Software the following two quick tests will be performed:

- Test of the CAN communication between GSSU/D6/D16 with MMC/ISA-CAN.
- A listing of the GPA/KAS status will be displayed. Error conditions are marked with "!!!".

In case of errors the displayed diagnosis will give you hints for further trouble shooting.

1.2.4 Error Reset and warm restart of GPA/KAS

The latched error signals (displayed by error LED's on the GSSU) can be reset via the "Reset" button S1 on D3 K2209/D17 K2217, except for permanent errors.

Pressing the "CAN reset" button S404 on D6 K2209/D16 K2217 provokes a warm restart, i.e. "Power on" sequence of the GPA/KAS. Static errors then will be logged and displayed again.

1.3 Troubleshooting Charts GPA

1.3.1 General Explanation

- **Message Id/error characterization:**
Number of the error which occurred (in System/Report) and error characterization
 - **SymbolicName:**
Name of error signal in the application software
 - **Error Description frame:**
Description of the original meaning of the error and routing of the error signal
 - **Symptom:**
How you can recognize the error
 - **Area of Trouble (Components/function):**
Localization of the error in the system, a component or a system function.
 - **Probable Cause:**
Description of probable reasons of the error.
 - **Service Action:**
Recommended action for remedy
 - **Reference:**
Further support for troubleshooting
 - **Remarks:**
Additional information
- Comments for using the following troubleshooting charts:**
- Check all error messages and the time sequence of their appearance.
 - Refer to the (→ Troubleshooting Charts GPA K2209 / Page 13), (→ Troubleshooting Charts KAS K2217 / Page 48)
 - If you do not have a clear idea of the cause of an error, proceed step by step through the recommended service actions of the (→ Troubleshooting Charts GPA K2209 / Page 13), (→ Troubleshooting Charts KAS K2217 / Page 48).

1.3.2 Troubleshooting Charts GPA K2209

1.3.2.1 Survey of GPA K2209 Error Messages

GPA Error Message (Id)	TSG Chart
Gradient coil not ready (Id=18)	(→ Chart 01 / Page 15)
Resonant Converter not ready (Id=19,20)	(→ Chart 02 / Page 16)
Resonant Converter Over temperature (Id=21)	(→ Chart 03 / Page 18)
Resonant Converter Output in Current Trip (Id=22)	(→ Chart 04 / Page 20)
Resonant Converter defective (Id=23)	(→ Chart 05 / Page 22)
Driver current supply defective (Id=24)	(→ Chart 06 / Page 24)
No Reaction of Modulator (Id=25,26)	(→ Chart 07 / Page 25)
No Reaction of Resonant Converter (Id=27)	(→ Chart 08 / Page 26)
Synchronization fault (Id=28,29)	(→ Chart 09 / Page 27)
Triangle voltage error (Id=30)	(→ Chart 10 / Page 29)
Plus 15V/ Minus 15V/ Plus 5V defective (Id=31,32,33)	(→ Chart 11 / Page 30)
Upgrade component error (Id=34)	(→ Chart 12 / Page 31)

GPA Error Message (Id)	TSG Chart
Power stage defective (Id=35)	(→ Chart 13 / Page 32)
Regulation error on X,Y,Z axis (Id=36,37,58,59,80,81)	(→ Chart 14 / Page 33)
Current actual value error on X,Y,Z axis (Id=38,60,82)	(→ Chart 15 / Page 36)
High current on X,Y,Z axis (Id=39,61,83)	(→ Chart 16 / Page 38)
Duty cycle for power stage X1(2),Y1(2),Z1(2) (Id=40,46,62,68,84,90)	(→ Chart 17 / Page 39)
Controlling error for power stage X1(2),Y1(2),Z1(2) (Id=41,47,63,69,85,91)	(→ Chart 18 / Page 41)
Temperature error power stage X1(2),Y1(2),Z1(2) (Id=42,48,64,70,86,91)	(→ Chart 19 / Page 42)
High voltage by power stage X1(2),Y1(2),Z1(2) (Id=43,49,65,71,87,93)	(→ Chart 20 / Page 45)

1.3.2.2 Chart 01

Message Id=18

Gradient coil not ready

SymbolicName: GPA_COIL_NOT_RDY

The gradient coil temperature exceeds 76 degrees C. The temperature is measured with six NTC resistors located in the gradient coil. The monitoring is done on D9 K2209 (patient table) and the error signal is routed via FOC to D7 K2209, D6 K2209 and via CAN Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error during normal operation of the system: FOC from D9 to U41/D7 is glowing weakly; right red LED, V413/D6 is on.	a. Gradient coil temperature too high	Thermal overload caused by specific measurement conditions	Wait 10 minutes until the coil temperature is in safe range. Check sequence parameters.	
		Error of cooling system	Check cooling system: Pipes, water flow, function of RCA.	(→ Cooling / MR-000.840.18)
		NTC/cable defective	Measure the NTC resistance	(→ Supervision D9/Grad.coil Test Procedure / Page 111)
		Sensor cable not connected	Check cable and connectors	
		D9 defective or supply voltage is missing if green LED V55 on D9 is off.	Check D9: Red LED V4,12,20 and position of DIP-switches. Replace defective D9	
		FOC from D9 to D7 defective or not connected	Check FOC from U1/D9 to U41/D7 (light off = error)	

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
02. Error exists permanently: FOC from D9 to U41/D7 is glowing weakly; right red LED V413/D6 is on.	a. Monitoring error	D7 defective (D8 defective)	Replace defective D7 (Replace D8)	
		D6 defective	Check CAN-Bus communication Replace defective D6	(→ Service/CAN D6 Test Procedures / Page 105)
Remarks: At a coil temperature of about 20°, the NTC resistance should be approx. 25 - 42 kOhm and at the temperature limit lower than 4 kOhm. A defective NTC can be bypassed by DIP switch at the D9. There must be at least four working NTC's in the coil to ensure safety.				

1.3.2.3 Chart 02

Message Id=19,20

Resonant Converter not ready

SymbolicName: GPA_RC_NOTRDY_ERR/_WARN

The voltage of the RC does not exceed 60% of the nominal voltage (e.g. 250 VDC_{act} at 415 VDC_{nom}). The voltage is monitored internally in the RC and the error signal is routed via FOC to D7 K2209, forwarded to D6 K2209 and via CAN-Bus to the MPCU..

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. FOC "RDY" from RC to U53/D7 is dark; green LED "RDY" at RC is off. Eventually red LED's "PHASE" and/or "VDIP" and/or "VPRI" and/or "WATER TEMP" and/or "AIR TEMP" and/or "ILIM" at RC are on.	a. Communication error/ no enable signal	Defective FOC signal lines between GSSU/D7/U51; RC or D6 defective	Reboot scanner, if no success: Press button S402 at GSSU/D6 and check FOC D7/U51; if there is no light, check D6, D7 or D8.	(→ Service/CAN D6 Test Procedures / Page 105)
		Defective FOC signal line between RC and GSSU/D7/U53	Check FOC signal line between RC and GSSU/D7/U53.	
	b. 230 VAC aux. supply of RC	Missing "VDIP" AC power supply or VDIP < 185 VAC or > 252VAC or defective RC or control unit	Check AC power supply line; Check green voltage LED's (P15V, N15V, HF15V). Check plug (3-pin) on front panel; replace defective RC	(→ Gradient / MR-000.841.16), RC (→ RC Test Procedure / Page 113)
	c. Three-phase supply and supervision	Three-phase AC supply voltage error (missing phase) or "VPRI" supply voltage higher than 528 VAC	Check circuit breaker F9/50A; Provide stable AC supply.	
	d. RC overload	Overtemperature RC or Current limit/trip RC	Check FOC signal lines (light off = error); Check LED's on RC.	(→ Chart 03 / Page 18), (→ Chart 04 / Page 20), (→ Chart 20 / Page 45)
	e. Monitoring error	Defective control board in RC	Replace RC.	

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
02. Service mode is activated: RC is expected to be on; no error message but warning; LED "RDY" on RC lights up	a. Communication error	Defective "RDY" FOC connection	Check FOC at U53/ D7 and RDY/RC: Off: FOC or RC display board defective. Replace defective components.	
Remarks: In service mode this error is treated as a warning and will not be evaluated, i.e. modulators and regulators will be not disabled. In case of overtemperature, overvoltage or current-limit (primary stage of RC), the ready-signal will become low, when the output-voltage becomes less than 60% of the nominal value. In case of further error messages related to the errors mentioned above, refer to the corresponding Troubleshooting Charts (→ Chart 03 / Page 18), (→ Chart 04 / Page 20), (→ Chart 20 / Page 45). Note: After a supply voltage error for the RC or the GSSU you must switch the GPA into service mode (right switch S1 on D3 in upward position), to reactivate the RC via the button S402 on D6. Afterwards switch back to normal mode and perform "Reload/All").				

1.3.2.4 Chart 03

Message Id=21

Resonant Converter Overtemperature

SymbolicName: GPA_RC_TEMP

The temperature of the chokes or of the RC power stages is too high. The choke temperature is measured by 6 temperature sensors installed in the chokes (Update Instruction MR050/01/S), the RC temperature is measured near the IGBT's/diodes in the RC and monitored internally. The error signal is routed via FOC to D7 K2209, forwarded to D6 and via CAN Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error occurs during normal operation FOC "TEMP" from RC to U52/D7 is dark; red LED's "WATER TEMP" and/or "AIR TEMP" at RC is on.	a. Overtemperature of the power stage of the RC	Error of the cooling system: Chilled ´water and/or fans beside the RC inside the cabinet	Check cooling system: Pipes, water flow/temp., function of RCA, Internal cabinet coolers (fans).	(→ Cooling / MR-000.840.18)
		IGBT's/diodes in the RC are not driven in a correct manner (control PCB).	Try to switch on after a few minutes; if no success, check all other reasons 01.a,02.a,03a; if negative, replace the RC.	(→ Gradient / MR-000.841.16), RC

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
Error occurs during normal operation; RC was switched off	a. Chokes temperature switch was activated.	A high leakage current of the final stage leads to an overtemperature at one of the 6 chokes (valid for Papillon final stages 5939477)	Measure the resistance of the final stage against the housing.	(→ Final Stage (Papillon type 5939447) / Page 117)
		One of the chokes was overheated due to excessive measurements	Ask the customer whether any exotic, or high performance sequence was used	
		broken/ disconnected cables at the choke temperature switches	Remove all final stages and check the connections	

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
02. Error exists permanently; red LED "TEMP" at RC is off	a. Monitoring error	FOC from RC to D7 defective or not connected	Check FOC from RC to U52/D7 (light off = error).	
03. Error exists permanently; red LED "TEMP" at RC is on		D7 defective	Replace defective D7.	
		D6 defective	Check CAN-Bus communication; Replace defective D6.	(→ Service/CAN D6 Test Procedures / Page 105)
		Monitoring(sensors) of RC defective	Replace display board at RC if available; Replace RC.	(→ Gradient / MR-000.841.16), RC
Remarks: If an overtemperature occurs, the RC will be switched off. After the output voltage becomes less than 60% of the nominal voltage, the ready-signal of the RC will become low. The GPA (modulators/regulators) will be inhibited.				

1.3.2.5 Chart 04

Message Id=22

Resonant Converter Output in Current Trip

SymbolicName: GPA_RC_CURTRIP

One of the six output stages of RC is in current_trip condition. The current is measured internally in RC. The RC will be disabled internally. The error signal is routed via FOC to D7 K2209, forwarded to D6 K2209 and via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error during normal operation: FOC "ITRIP" from RC to U54/D7 becomes dark; a red LED at RC (e.g. IX1) is on, showing the overloaded stage.	a. At least one of the output stages of RC was overloaded	Overload of RC caused by specific measurement conditions, i.e. overload of one of the power stages of the GPA.	Check sequence parameters; Restart the measurement	
		Wrong gradient data from GCTX/MPCU	Check gradient data cable; Run Test DAC/Functional, data transmission between GCTX and DAC	SeSo: Test Tools; Gradient
		D1 (DAC) or D2 (Regulator) defective	Check D1 or D2/ Replace defective D1 or D2	(→ Regulator D2 Test Procedure / Page 98)
02. (only in Service mode) Error occurs again after an attempt to switch on GPA a second time. (Disable modulators of GPA by switches on D4)	a. Power stage of GPA	Defective power stage (indicated by the corresponding error LED at RC (IX1...IZ2).	Replace the power stage	(→ Gradient / MR-000.841.16), PS
	b. Secondary rectifier of RC	Defective secondary rectifier of RC	Pull out the corresponding PS and switch on RC; if the error occurs again, replace the RC	(→ Gradient / MR-000.841.16), RC

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
03. Permanent error: FOC "ITRIP" from RC to U54/D7 is dark; other red LED's at the RC may be on or of	a. Monitoring error	Monitoring(sensors) of RC defective	Replace display board at RC; Replace RC.	(→ Gradient / MR-000.841.16), RC
		FOC "ITRIP" from RC to U54/D7 defective	Check FOC from RC to U54/D7 (light off = error).	
		D7/D8/D6 defective	Check CAN-Bus communication; Replace D6, Replace D7.	(→ Service/CAN D6 Test Procedures / Page 105)
Remarks: If the yellow LED "ILIM" on RC lights up during measurement, the primary power stage of the RC is in current limit. This may be caused if the rating of a measurement sequence is too high. If the current limit remains active for some time, the output voltage of the RC may become less than 60% of the nominal value. Then the ready signal becomes low and switches off the modulators of the GPA; refer to (→ Chart 02 / Page 16). As the current limit signal is not latched within the RC it will normally come back to the OK-status if the overload is removed. The next measurement may successfully restart the GPA. Refer also to (→ Chart 13 / Page 32), "power stage defective".				

1.3.2.6 Chart 05

Message Id=23

Resonant Converter defective

SymbolicName: GPA_RESCON_DEF

The three/six signals "High voltage by power stage X1...Z2" (depending on installation of three or six power stages in the gradient system) are present at the same time ($U_{nom} > 425V$). The signals are generated on the RC, routed via FOC's to D7, further via D4 -supervision and error multiplexer to the gradient processor on D6. The gradient processor creates the final error signal which is routed via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. All FOC's (VX1...VZ2) from RC to U153/253/353//163/263/363/D7 are dark. All red error LED's "Overvoltage" at D4/5 are on.	a. RC control	U _{nom} of RC misadjusted or RC control electronic defective	Check the U _{nom} adjustment on the RC (with pot R1 on the left side of the display board); Replace RC	(→ Gradient / MR-000.841.16), RC (→ RC Test Procedure / Page 113)
		AC supply for RC control UDIP exceeds 252 VAC. (Tolerance range: 207...252 VAC)	Check the correct adaptation of line supply voltage at transformer T1 to the local line voltage.	(→ Start-up / M1-010.815.01); "Checking the line voltage"

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
02. RC in standby Error red LED's "Overvoltage" at D4/5 may be on or off	a. Monitoring error	FOC's from RC to U153/253/353/163/263/363/D7 are defective	Check FOC's (light off = error)	
		D7 defective	Replace defective D7	
		D4/5 (Error multiplexer) defective	Check red error LED's on D4/5; Replace defective D4/5	(→ Modulator D4 Test Procedures / Page 102)
		D6 defective	Check CAN-Bus communication; Replace defective D6	(→ Service/CAN D6 Test Procedures / Page 105)

Remarks: The RC is switched into standby mode by the gradient processor, i.e. no output voltage is present. The resonant circuit of the RC will not be inhibited internally.

1.3.2.7 Chart 06

Message Id=24

Driver power supply defective

SymbolicName: GPA_DRV_PS_DEF

The driver voltage is not ok. The monitoring is done via the three/six temperature error signals (FOC/T), (depending on installation of three or six power stages in the gradient system). If all temperature error signals are present at the same time (routed from PS via FOC to D7, further to D4 supervision and error multiplexer to gradient processor on D6), the gradient processor creates the final error signal which is routed via CAN Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. All three/six FOC's (T) from PS to D7 are dark; green LED at D10 is off.	a. Synchronization of D10	No synchronization clock, D3/D7 defective, FOC between D7 and D10 is missing or defective. Jumper X4 on D10 is missing	Check red error LED V37/D3 (refer to (→ Chart 09 / Page 27) Replace D3/D7 Check FOC between D7 and D10 Check correct jumper position 1- 2	
	b. GSSU Power Supply E4A, Slot 2, 15 V supply	E4A defective	Remove the 4-pin connector X1 on D10. Measure supply voltage 15V at X1, pin 1- 2 or on terminal of E4A/slot 2; replace defective E4A.	
	c. Power supply connection between E4A and D10	Power cable disconnected, bad connector X1	Check power cable and connector X1.	
	d. D10 Driver supply board	D10 defective	Remove the 24-pin connectors (X2,X3) on D10; if LED (at D10) does not go on, replace D10.	

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
02. Green LED lights up after removing the 24-pin connector from D10	a. Power stage driver board	Driver board of power stage defective	Reconnect the 24-pin connector at D10. Remove the 9-pin X1 connectors at the power stages, one after another, until LED on D10 is on. Replace the corresponding power stage.	
03. Green LED at D10 is on.	a. Driver supply connections	Defective driver supply connections to PS	Check at X1 at PS:15V rectangular signal between 1-4, 2-5, 3-6, 7-8. Check X2, X3 at D10.	
	b. Monitoring error	D4/5, D6 defective FOC's (T) between PS and D7 defective	Check CAN-Bus communication; Replace D6/D4/5 Check FOC's (all lights off = error)	(→ Service/CAN D6 Test Procedures / Page 105)
Remarks: Check jumper on D10 as well: X4 Pin 1/2 external clock control (normal operation), Pin 2/3 internal clock control.				

1.3.2.8 Chart 07

Message Id=25,26

No Reaction of Modulator

SymbolicName: GPA_MOD_SET_ERR/_WARN

The modulators of the three axes x,y,z can be controlled by the Service Software or by switches on D4, i.e. they can be switched on and off. If after a control command from the MPCU the modulator is not in the desired condition, this error is generated by the gradient processor, using a combination of several control signals (switch signals from D4/5 and control signals from D6). The final error signal will be routed via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Normal operation of MR measurement system or Service Software tests running (e.g. gradient tests): Modulator disabled, yellow left LED V1/A2 at D4 is on. (only if modulators disabled by application software of GPA or switches S1/2 on D4). GPA_Status message: X,Y,Z modulator inhibit = yes	a. Modulator board D4/5	Switch position: S1 left, S2 left and right: Upper position means modulator x,y,z-axis disabled	Set switches in "on" position (downward position) Loading and starting a sequence enables the modulators if no other error condition exists. Eventually reboot scanner..	SeSo: Test Tools Gradient GPA_Status (Online Help)
		D4/D5 defective	Replace defective D4/5.	
	b. D6	D6 defective	Check CAN-Bus communication Replace defective D6.	(→ Service/CAN D6 Test Procedures / Page 105)
Remarks: In the service mode (right switch S1 on D4 in upper position or service mode enabled by the Service Software) this error is treated as a warning and will not be evaluated, i.e. modulators and regulators will not be disabled (with the exception of manual disabling via switches on D4/5).				

1.3.2.9 Chart 08

Message Id=27

No Reaction of Resonant Converter

SymbolicName: GPA_RC_SET_ERR

Within five seconds after an "Enable" request (either by NUMARIS or by manually pressing the push button "RC on" S402/D6), the RC does not respond with the "RDY" message. The gradient processor generates the final error message which is routed via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error after GPA "Power On" command	a. Communication between gradient processor and RC	FOC's "EN" and/or "RDY" between RC and D7 missing or defective	Check FOC connections by manually switching the RC on/off (pressing S402 on D6). The green LED's between S402 and S 403 must be on then.	
		D7 defective	Check FOC connections on D7: U51 (RC_ENABLE_FOC) U53 (RC_READY_FOC) (no light = error).	
	b. Three-phase supply and supervision	Three-phase AC supply voltage error or missing phase; red LED "PHASE" may be on.	Check circuit breaker F9/50A; Provide stable AC supply.	(→ Chart 02 / Page 16)
	c. RC control	RC control defective	Replace the RC.	
	d. Monitoring error	D6 defective	Check CAN-Bus communication Replace defective D6.	(→ Service/CAN D6 Test Procedures / Page 105)
Remarks: If the RC cannot be switched on by pressing S402 (caused by application software of GPA), proceed as follows: Switch on the "service mode": right switch S1 at D4 in upward position. Press reset-button S 404 at D6. Now press S402, RC on. If the RC is switched on (green LED "RDY" at RC is on) switch the "service mode" off again (right switch S1 at D4 in downward position). Afterwards reboot scanner.				

1.3.2.10 Chart 09

Message Id=28,29

Synchronization fault

SymbolicName: GPA_SYNC_ERR/_WARN

The synchronized clock signal from MPCU(GCTX) is faulty or missing. This 4MHz clock signal is transmitted via the gradient waveform interface cable from the GCTX/MC4C40 to the D1 K2209. Here the signal is divided into a 50 kHz signal. If the phase detector (PLL) on D3 K2209 detects an error, the error signal will be routed to D6 K2209 and via CAN-Bus to the MPCU; a red LED, V37 "SYNC Error" at D3 will be on.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error during normal operation; red error LED V37 "SYNC Error" at D3 is on. Image quality may be degraded.	a. Communication between MMC and D1	Error on MMC (MC4C40/GCTX)	Run MMC Tests (MC4C40/GCTX)	SeSo: Test Tools MMC
		Gradient data cable and/or connectors defective	Check gradient data cable; Run Test DAC/Functional, data transmission between GCTX and DAC	SeSo: Test Tools Gradient
	b. D1	D1, wrong switch positions (S3/4); D1 defective	Check switch positions on D1: S3-pos.7, S4-pos.2 Replace defective D1.	

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
02. Red error LED at D3 is on or off.	a. Adjustment error and/or monitoring error	D3 misadjusted or defective	Measure the signal "TRIA"/"FREQ1"(52 kHz) at D3 X10/12; Adjust D3/R66 (during adjustment remove gradient data cable from MMC to D1); Replace D3.	(→ Synchronization D3 Test Procedures / Page 100)
		D6 defective	Check CAN-Bus communication Replace defective D6.	(→ Service/CAN D6 Test Procedures / Page 105)

Remarks: In the service mode this error is treated as a warning and will not be evaluated, i.e. modulators and regulators will be not disabled. Image quality may be degraded.

1.3.2.11 Chart 10

Message Id=30

Triangle voltage error

SymbolicName: GPA_TRIA_ERR

The triangle voltage amplitude is out of tolerance or the offset is too high. The error signal is generated on D3 K2209 in combination with the supervision of the signal F_SU; this signal is generated on D4 (Jumper S3) from combination of three clock signals, generated on D3. The final error signal "TRIA ERR" is routed to D6 and via CAN-Bus to the MPCU; a red LED, V38 "TRIA Error" at D3 will be on.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error during normal operation; red error LED V38 "TRIA Error" at D3 is on.	a. D3, generation of triangle voltage (amplitude/offset)	D3 defective	Measure the triangle voltage: "TRIA" (50kHz) at D3/X10 Replace defective D3.	(→ Synchronization D3 Test Procedures / Page 100)
	b. D4, generation of the signal F_SU (25kHz)	D4 defective or incorrect position of DIP switches S3	Check position of DIP switches on D4: S3: 5=off, 7=on. Replace defective D4.	(→ Modulator D4 Test Procedures / Page 102), (→ Turbo Modulator D5 Test Procedures / Page 104)
	c. D2	D2 defective	Replace defective D2.	(→ Gradient / MR-000.841.16), GSSU
02. Red error LED at D3 is on or off.	a. Monitoring error	D3 defective	Measure the triangle voltage "TRIA" (50 kHz) at D3/X10; Replace defective D3.	(→ Synchronization D3 Test Procedures / Page 100)
		D6 defective	Check CAN-Bus communication; Replace defective D6.	(→ Service/CAN D6 Test Procedures / Page 105)

1.3.2.12 Chart 11

Message Id=31,32,33 **Plus 15V/ Minus 15V/ Plus 5V defective**

SymbolicName: GPA_P15/ N15/ VCC_ERR

DC supply voltages from E4A (+5V VCC, Slot1 and +/-15V, Slot3) are out of tolerance (+/- 9%). The voltage is monitored on D3 K2209; the error signals are routed to D6 and via CAN-Bus to the MPCU. In case of a +/-15V error red LED's V32/31 "+/-15V Error" are on; in case of a +5V error, green LED V30 "+5V OK" is off.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error after GPA "Power On" Green LED V30 at D3 off Red LED's V32 (+15V) and/or V31 (-15V) are on.	a. Power supply unit E4A	Power supply E4A defective; Fan defective	Measure/adjust the output voltages directly at the outputs of the power supply unit slots. If not adjustable replace the power supply unit.	(→ Small Signal Power Supply E4A / Page 108)
	b. Power supply lines	D7,8 defective or not plugged correctly; Cabling between D7 and E4A defective.	Measure the +/-15V and +5V supply voltages at D7/X5; if ok, the backplane D8 is probably defective.	
	c. Boards of GSSU and D10	Defective PCB	Replace the defective board.	Remarks
	d. Monitoring error	Supervision on D3 defective	Measure the supply voltages at D3 X16: +15 V, X18: -15V; X20: +5V (+/- 5% tolerance range) if ok, replace defective D3.	(→ Synchronization D3 Test Procedures / Page 100)
		D6 defective	Check CAN-Bus communication Replace defective D6.	(→ Service/CAN D6 Test Procedures / Page 105)

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
Remarks: In principle each defective board in the GSSU including the D10 and the backplane D8 could be the reason for the supply voltage breakdown and should be checked. Also check fuse F14/4A in the CCA cabinet. The hum of the supply voltages should be lower than 10 mV _{pp} .				

1.3.2.13 Chart 12

Message Id=34

Upgrade component error

SymbolicName: GPA_EPI_FAIL

The upgrade component is defective or not available. The error signal is generated on the upgrade component and routed via FOC or wire (D7/X5) to D7 K 2209 forwarded to D6 and via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01 FOC D7/U14 is dark. Left red LED V413/D6 is on.	a. Optional upgrade component	Error of upgrade component	Troubleshooting on upgrade component; Replacement of defective components	
	a. Monitoring error	Defective upgrade interface between D7 (connector X5) and upgrade component	Check connection line.	
		D7 defective	Check FOC U14 at D7; (light off = error). Replace defective D7.	
02. FOC D7/U14 lights up		D6 defective	Check CAN-Bus communication Replace defective D6.	(→ Service/CAN D6 Test Procedures / Page 105)
Remarks: If the Upgrade component is not installed, a short link must be installed on D7, connector X5, between PIN 8 and 4. If the error occurs, the RC will be switched to standby mode.				

1.3.2.14 Chart 13

Message Id=35

Power stage defective

SymbolicName: GPA_PST_DEF

One or more of the six output stages of the RC is in current_trip condition. The current is measured internally in the RC. The RC will be disabled internally. The error signal is routed via FOC to D7 K2209, forwarded to D6 K2209 and via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. The error occurs after the second attempt to switch on the GPA, especially if the modulators are disabled (e.g. in service mode). FOC "ITRIP" from RC to U54/D7 becomes dark; one or more red LED "Current trip" at RC (IX1...IZ2) is on, indicating the overloaded stage.	a. Power stage	Power stage(s) defective	Remove the corresponding power stage (the defective power stage is indicated by a glowing red LED's "Current trip" (IX1 ... IZ2) on the RC) and try to switch on the RC again. If "Current trip" doesn't occur again, replace the gradient power stage.	(→ Gradient / MR-000.841.16), PS
	b. RC	Secondary Rectifier in RC defective	Remove corresponding gradient power stage and try to switch on the RC again; if "Current trip" occurs again, replace RC.	(→ Gradient / MR-000.841.16), RC

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
02. No error LED at RC lights up. The RC is disabled, i.e. green "Ready" LED is off.	a. Monitoring error	FOC "ITRIP" defective Supervision of RC defective	Check FOC connection "ITRIP" at RC; if light is off, replace RC.	
		FOC "ITRIP" between RC and D7 defective; D7 defective	Check FOC U54 "N_RC_CUR-TRIP_FOC" at D7 (light off = error); replace D7 if FOC "ITRIP" is lighted up at RC.	
		D6 defective	Check CAN-Bus communication Replace defective D6.	(→ Service/CAN D6 Test Procedures / Page 105)
Remarks: If the current trip occurs again, even the modulators are disabled (e.g. via service mode switch at the D4). This is a strong indication of a defective RC or defective power stage .				

1.3.2.15 Chart 14

Message Id=
36,37,58,59,80,81

Regulation error on X,Y,Z-axis

SymbolicName: GPA_LOOP_ERR/_WARN_X/_Y/_Z

The error signal and the regulator output signal are monitored in the closed loop error logic on D2 K2209. If one of these signals reaches the upper limit (10V), the yellow LED V27 on D2 lights up and an error signal is generated on D2, routed to D6 and via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Dynamic error while running a measurement; the yellow LED V27 "Regulation Error" on D2 lights up.	a. Measurement sequence	Gradient rise time is too fast	Switch on the Service mode to find out the affected axis. Run the specific sequence again; check the sequence parameters, if other than Siemens-released sequences are used.	
	b. Gradient data transmission	MC4C40/GCTX defective, Data cable and/or connectors defective, D1 K2209 defective	Run MMC Test; run Gradient Test for DAC; check gradient data cable and connectors. Replace defective D1.	SeSo: Test Tools MMC (→ Gradient / MR-000.841.16)
	c. Load circuit gradient coil interrupted	Gradient coil, cable connections, RF filters defective	Check the continuity of the load circuit; check for loose cable connections. Replace defective components.	
	d. Closed control loop	D2, D4/5 defective; D7 defective; FOC's to power stages defective	Perform Regulator Adjust/Check; cyclically replace the D2's; check for other error messages.	SeSo: Tune-up/QA and (→ Regulator D2 Test Procedure / Page 98)

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. continued Dynamic error....	e. Measurement of actual current	SENS, D2 defective	If no CAV_ERR_X,Y,Z (refer to TSG Chart 15), cyclically replace D2 (x,y,z); replace defective D2. Otherwise replace defective SENS	(→ Regulator D2 Test Procedure / Page 98), (→ Gradient / MR-000.841.16) GSSU, (→ Chart 15 / Page 36)
	f. Power stage	Power stage defective	Check for other error messages; cyclically replace power stages.	
	g. Resonant converter	Too low output voltage	Check for error messages from RC. Measure the output voltage of the RC and adjust it if necessary	(→ RC Test Procedure / Page 113)

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
02. Error after GPA "Power On"; (in the idle mode), the yellow LED V27 "Regulation Error" on D2 lights up.	a. DAC D1	D1 defective/ high offset	Check voltage at D2, X3 in idle mode: Offset < 1 mV	
	b. Regulator D2	D2 defective	Cyclically Replace the D2's; replace defective D2.	(→ Gradient / MR-000.841.16) GSSU
	c. Modulator D4/5	D4/5 defective	Replace defective D4/5.	
	d. Current sensors (SENS)	Current sensors (SENS) defective	Replace current sensors (SENS).	(→ Gradient / MR-000.841.16) SENS
	e. Monitoring error	D6 defective	Check CAN Bus communication Replace defective D6.	(→ Service/CAN D6 Test Procedures / Page 105)
Remarks: The regulation error occurs as a consequence of every other error but will be suppressed in this case; you will see only the causal errors in the error report. The regulation error will only be recorded in the system report if it occurs as a primary error. Refer to (→ Chart 15 / Page 36), (Current actual value error) or (→ Chart 16 / Page 38), High current value error). In Service mode, the regulation error is treated as a warning and will not be evaluated, i.e. modulators and regulators will be not disabled. In case that the Service Software program "Regulator Adjust" does not work - overshoot too large and termination of the program with the error message - "Calculated Amp (p) pot is out of allowed range. Setting to default value." - check the Unom adjustment on the RC (with pot R1 on the left side of the display board). Refer to repair proc./ (→ RC Test Procedure / Page 113).				

1.3.2.16 Chart 15

Message Id=38,60,82 **Current actual value error on X,Y,Z-axis**

SymbolicName: GPA_CAV_ERR_X/_Y/_Z

The current sensor signal (actual current value) is not correct. The signal is monitored on D2 K2209 (red LED V24 "SENS Error" lights up) and is forwarded to D6 and routed via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error after GPA "Power On"; the red LED V24 "SENS Error" on D2 lights up.	a. Current sensor supply	Break in signal line or power supply disconnected or supply line shorted	Check supply/signal line from SENS/X1 to D7/X3 for continuity and shorts.	
	b. Power line from GPA to the gradient coil	Ground short of a gradient power circuit	Check the gradient coil circuit for ground shorts (gradient coil, gradient cables, gradient filters).	
	c. D2 Regulator	D2 defective/ defective SENS soft start control (the error is detected, if the sensors are switched off)	Replace defective D2.	(→ Gradient / MR-000.841.16) GSSU
	d. D7 FOC board	D7/8 defective	Check connectors; replace defective D7.	
	e. (Type IT600S/LA205-S) Synchronization via FOC	FOC's between D7,U22(X), U24(Y), or U25(Z) defective	Check the FOC's between D7 and SENS.	(→ Gradient / MR-000.841.16) SENS
	f. Current sensors (SENS) Type LB200-S/SP4 paired or Type IT600S combined with LA 205-S	SENS defective: Difference of compensation currents of two paired sensors is higher than 24 mA or in case of the IT600S/LA 205-S combination to high	Measure the sensor signals at D2, X707 and X709 Replace current sensors (SENS).	(→ Gradient / MR-000.841.16) SENS

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
No error LED lights up:	g. Monitoring error	D6 defective	Check CAN-Bus communication Replace defective D6.	(→ Service/CAN D6 Test Procedures / Page 105)

1.3.2.17 Chart 16

Message Id=39,61,83 High current on X,Y,Z-axis

SymbolicName: GPA_HI_CURR_X/_Y/_Z

The actual current exceeds the limit of 300A +/- 9A; (red LED's "Overcurrent" for x,y or z axis on D4/5 K2209 light up). The monitoring is done at D4 and an error signal is forwarded to D6 K2209. The final error signal is generated on D6 and routed via CAN Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Dynamic error while running a measurement; the left red LED V4,5 or 6 "Overcurrent X,Y,Z" on D4 lights up	a. Measurement sequence	Sporadic overload	First run the duty_cycle sequence: If ok, check the measurement sequence parameters.	
	b. D2	Regulator D2 oscillates due to wrong parameters; D2 defective	Run the Regulator Adjust/Check; cyclically replace the D2's. Replace defective D2.	SeSo: QA/Tune-up: Regulator Adjust/Check
	c. Gradient load circuit	Short in the gradient load circuit (gradient coil, cables, filters)	Measure the resistance of the load circuit: Short, if only a few mOhm.	
	d. Communication between MMC and D1	Error on MMC (MC4C40/GCTX)	Run MMC Tests.	SeSo: Test Tools MMC
		Gradient data cable and/or connectors defective	Check gradient data cable; Run Gradient tests for DAC.	SeSo: Test Tools Gradient
		D1 defective	Replace defective D1.	(→ Gradient / MR-000.841.16) GSSU
	e. D4/5	Defective Modulators	Replace defective D4/5.	(→ Modulator D4 Test Procedures / Page 102)

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. continued Dynamic error ...	f. D7/FOC's to PS's	Defective Lght-board and/or FOC's	Replace D7 and or FOC's.	
	g. current sensors (SENS)	Measured values incorrect by defective current sensors (SENS)	Replace current sensors (SENS).	(→ Chart 15 / Page 36) (→ Gradient / MR-000.841.16) SENS
02. Error after GPA "Power On"; the left red LED V4,5 or 6 "Overcurrent X,Y,Z" on D4 lights up	a. Monitoring error	D4 defective	Replace defective D4.	(→ Modulator D4 Test Procedures / Page 102)
No error LED lights up.		D6 defective	Check CAN-Bus communication Replace defective D6.	(→ Service/CAN D6 Test Procedures / Page 105)
Remarks: Check also for accompanying error messages, e.g. (→ Chart 15 / Page 36) "Current actual value error" or (→ Chart 14 / Page 33) "Regulation error on X,Y,Z-axis" and refer to the corresponding TSG-Chart (→ Troubleshooting Charts GPA K2209 / Page 13).				

1.3.2.18 Chart 17

Message Id=
40,46,62,68,84,90

Duty cycle for power stage X1(2),Y1(2),Z1(2)

SymbolicName: GPA_DUTYCYCLE_X1(2)/_Y1(2)/_Z1(2)

The duty cycle of the actual current is too high; (red LED's "Duty Cycle Error" for x,y or z axis on D4 K2209 light up). The monitoring is done on D4 and an error signal is forwarded to D6 K2209. The final error signal is generated on D6 and routed via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Dynamic error while running a measurement; one or more of the the right red LED V5,6,or 7 "Duty Cycle Error X1,Y1,Z1" on D4 lights up.	a. Measurement sequence	Sporadic overload	Check the measurement sequence parameters Check the system by running a high duty cycle sequence.	
	b. Communication between MMC and D1	Error on MMC (MC4C40/GCTX)	Run MMC Tests.	SeSo: Test Tools
		Gradient data cable and/or connectors defective	Check gradient data cable; Run Gradient Tests for DAC.	SeSo Test Tools: (→ Gradient / MR-000.841.16) DAC/Functional
		D1 defective	Replace defective D1.	
	c. D2	Regulator D2 oscillates due to incorrect parameters; D2 defective	Run Regulator Adjust/Check; Cyclically replace the D2's but remember their original positions. Replace defective D2.	SeSo: QA Tune-up: Regulator Adjust/Check
	d. D4/5	Defective Modulator boards	Replace defective D4/5.	
	e. D7/FOC's to power stages	Defective Light-board D7 and/or FOC's to power stages	Replace D7 and or FOC's to power stages.	
Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. continued Dynamic error	f. Current sensors (SENS)	Incorrect values measured by defective current sensors (SENS)	Replace current sensors (SENS).	(→ Gradient / MR-000.841.16), SENS

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
02. Error after GPA "Power On"; one or more of the right red LED V5,6,or 7 "Duty cycle Error Xi,Yi,Zi" on D4/5 lights up	a. Monitoring error	D4/5 defective, especially if a X,Y,Z_2/_3 error occurs.	Replace defective D4/5.	(→ Modulator D4 Test Procedures / Page 102), (→ Turbo Modulator D5 Test Procedures / Page 104)
No error LED lights up.		D6 defective	Check CAN-Bus communication Replace defective D6.	(→ Service/CAN D6 Test Procedures / Page 105)
Remarks: Check also for accompanying error messages, e.g. (→ Chart 15 / Page 36) "Current actual value error" -> , and refer to the corresponding TSG-Chart (→ Troubleshooting Charts GPA K2209 / Page 13).				

1.3.2.19 Chart 18

Message Id=
41,47,63,69,85,91

Controlling error for power stage X1(2),Y1(2),Z1(2)

SymbolicName: GPA_CTRL_ERR_X1(2)/_Y1(2)/_Z1(2)

The control signals for the power transistors of a specific power stage are not ok. The monitoring is done on D4/5 K2209 (one red LED "Drive Error" lights up) and an error signal is forwarded to D6 K2209. The final error signal is generated on D6 and routed via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Dynamic error while running a measurement; the right red LED V2/A2 "Drive Error" on D4/D5 lights up.	a. D4/D5 (attempt to switch the power stage transistors without dead time; refer to FUN/ Gradient system)	D4/D5 defective	Replace defective D4/D5.	(→ Modulator D4 Test Procedures / Page 102), (→ Turbo Modulator D5 Test Procedures / Page 104)

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
02. Error after GPA "Power On"; the right red LED V2/A2 "Drive Error" on D4/5 lights up.	a. Monitoring error	D2/D4/D5 defective	Replace the D2 of affected axis with another D2. Replace defective D2/D4/D5.	(→ Modulator D4 Test Procedures / Page 102), (→ Turbo Modulator D5 Test Procedures / Page 104)
03. Error after GPA "Power On"; the right red LED V2/A2 "Drive Error" on D4/D5 are off		D6 defective	Check CAN-Bus communication Replace defective D6.	(→ Service/CAN D6 Test Procedures / Page 105)
Remarks: Because of the possibility of D4/5 errors (i.e. the modulators cannot be disabled), the RC will be switched to stand-by. Check also for accompanying error messages, e.g. (→ Chart 16 / Page 38) "High current error on...-axis" and refer to the corresponding TSG-Chart (→ Troubleshooting Charts GPA K2209 / Page 13).				

1.3.2.20 Chart 19

Message Id= **Temperature error for power stage X1(2),Y1(2),Z1(2)** SymbolicName: GPA_TEMP_ERR_X1(2)/_Y1(2)/_Z1(2)
42,48,64,70,86,92

The power stage temperature exceeds the limit of 56C. The temperature is measured inside the power stages (at power stage transistors) and an error signal is sent via FOC to D7 K2209. The error signal is monitored on D4/5 (one red LED "Overtemp" lights up) and forwarded to D6 K2209. The final error signal is generated on D6 and routed via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Dynamic error while running a measurement; the left red LED V2/A1 "Over-temperature" on D4/5 lights up.	a. Measurement sequence	Load of GPA by measurement parameters or measurement conditions are too high	Check the sequence parameters; Run an approved high duty cycle sequence	
	b. Cooling system	No water flow or water flow in the cooling system is too low or cabinet cooling system (RCA) defective	Check for other error messages from RCA or MMC (cabinet temperature control) Check RCA, water pipes, flow rate; Check the cabinet cooling system (CCS).	(→ Cooling / MR-000.840.18)
	c. GSSU of GPA	Any component of the GSSU defective	Check for other error messages	Corresponding (→ Troubleshooting Charts GPA K2209 / Page 13)
	d. D7	D7 defective	Replace defective D7.	
	e. Power stage	"Drive" FOC's from D7 to power stage defective	Check and replace FOC's.	

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
02. Continuous error, trying GPA "Power On" after waiting period of approx. 30 sec; the left red LED's V2/A1 "Overtemperature" on D4/5 lights up. One or more of the FOC's "T" from the power stages to D7 are dark.	a. Monitoring error	FOC's "T" from power stage to D7 defective	Check the FOC at power stage "T" and at D7/U152, 252, 352 and D7/U161, 262, 362 (Light off = error)	(→ FOC D7/Power Supply Test Procedures / Page 107)
		Temperature sensors of power stage defective	Replace power stage if "T" is continuously off in idle state.	(→ Gradient / MR-000.841.16), PS
		D7 defective	Check FOC's at D7/U152, 252, 352 and D7/U162, 262, 362 (Light off = error) Replace D7 if light is on.	(→ FOC D7/Power Supply Test Procedures / Page 107)
		D4/D5 defective, especially if a X,Y,Z_3 error occurs	Replace defective D4/5.	(→ Modulator D4 Test Procedures / Page 102), (→ Turbo Modulator D5 Test Procedures / Page 104)
No error LED lights up.		D6 defective	Check CAN-Bus communication Replace defective D6.	(→ Service/CAN D6 Test Procedures / Page 105)
Remarks: If the FOC's "T" of all power stages are dark, then probably the driver supply board D10 K2209 supplies no output power; refer to (→ Chart 06 / Page 24). If the duty cycle error for the same axis (refer to (→ Chart 17 / Page 39)) occurs at the same time, check at first the used measurement parameters, otherwise check at first the cooling system.				

1.3.2.21 Chart 20

Message Id=
43,49,65,71,87,93

High voltage by power stage X1(2),Y1(2),Z1(2)

SymbolicName: GPA_HIVOLT_X1(2)/_Y1(2)/_Z1(2)

The output voltage of one channel of the RC exceeds the upper limit 425V. The voltages are measured on the RC. The error signal is sent via FOC to D7 K2209 and is monitored at D4/5 K2209 (dependent on the axis a red LED, e.g. "Overvoltage X" lights up) and is forwarded to D6 K2209. The final error signal is generated on D6 and routed via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Dynamic error while running a measurement; one or more of the the right red LED's V3,4 "Overvoltage Xi,Yi,Zi" on D4/5 lights up.	a. Power stage	A single defective power stage produces overvoltage spikes.	Swap the power stage with another one. Replace defective power stage.	(→ Gradient / MR-000.841.16), PS
	b. Drive control (only when D5 installed)	D4/5 defective (possible only if an upgrade component is installed)	Replace defective D4/5.	(→ Modulator D4 Test Procedures / Page 102), (→ Turbo Modulator D5 Test Procedures / Page 104)
	c. RC	RC defective or misadjusted	Measure the threshold voltage for the overvoltage supervision (6 test points behind the front panel of the RC). Replace defective RC.	(→ RC Test Procedure / Page 113) (→ Gradient / MR-000.841.16), RC

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
02. Simultaneous error messages "High voltage" from all power stages	a. RC	RC defective or misadjusted	Adjust the actual output voltage of the RC.	(→ RC Test Procedure / Page 113), (→ Chart 05 / Page 22) (→ Gradient / MR-000.841.16), RC
			Replace RC.	
	b. Line voltage out of tolerance or wrong tapping of primary transformer	Line voltage out of tolerance range	Provide correct line input voltage.	

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
03. Error after GPA "Power On"; one or more of the right red LED's V3,4 "Overvoltage Xi,Yi,Zi" on D4/5 lights up. One or more of the FOC's VXi,Yi,Zi from RC to D7 are dark. (i = 1,2)	a. Monitoring error	RC supervision defective	Measure the RC output voltages. Check FOC outputs at RC: VX1...VZ2 (Light off = error, i.e. voltage too high) Replace RC.	(→ RC Test Procedure / Page 113)
		FOC's between RC and D7 defective	Check FOC's at D7/U153, 253, 353 and D7/U163, 263, 363 (Light on = ok)	(→ FOC D7/Power Supply Test Procedures / Page 107)
		D7 defective	Check FOC's at D7/U153, 253, 353 and D7/U163, 263, 363 (Light off = error) Replace D7 if light on.	(→ FOC D7/Power Supply Test Procedures / Page 107)
		D4/5 defective	Replace defective D4/5.	(→ Modulator D4 Test Procedures / Page 102), (→ Turbo Modulator D5 Test Procedures / Page 104)
04. No error LED lights up.		D6 defective	Check CAN-Bus communication Replace defective D6.	(→ Service/CAN D6 Test Procedures / Page 105)
Remarks: In this error status, the RC will be switched to stand-by within 0.1 ms by the gradient processor which disables the RC ("Enable"-signal -> off). If all RC stages signal overvoltage at the same time, the RC is probably defective -> refer also to (→ Chart 05 / Page 22).				

1.3.3 Troubleshooting Charts KAS K2217

1.3.3.1 Survey of KAS K2217 Error Messages

KAS Error Message (Id)	TSG Chart
Gradient coil not ready (Id=18)	(→ Chart 01 / Page 50)
Main power supply not ready (Id=19)	(→ Chart 02 / Page 51)
Rectifier overtemperature (Id=23)	(→ Chart 03 / Page 53)
Transformer overtemperature (Id=21)	(→ Chart 04 / Page 55)
No reaction of main power supply (Id=27)	(→ Chart 05 / Page 56)
Synchronization error (Id=28,29)	(→ Chart 06 / Page 58)
Modulator clock error (Id=30)	(→ Chart 07 / Page 59)
Plus 15V/Minus 15V/Plus 5V defective (Id=31,32,33)	(→ Chart 08 / Page 59)
GSSU-Power supply defective (Id=86)	(→ Chart 09 / Page 61)
Driver supply voltage error (X, Y, Z) (Id=43,61,79)	(→ Chart 10 / Page 62)
FAST OFF (Id=34)	(→ Chart 11 / Page 65)
Regulation error on X,Y,Z-axis (Id=35,36,53,54,88,72)	(→ Chart 12 / Page 66)

KAS Error Message (Id)	TSG Chart
Current actual value error on X,Y,Z-axis (Id=37,55,73)	(→ Chart 13 / Page 70)
High current on X,Y,Z-axis (Id=38,56,74)	(→ Chart 14 / Page 71)
Duty cycle error for power stage X,Y,Z (Id=39,57,75)	(→ Chart 15 / Page 73)
Controlling frequency error for power stage X,Y,Z-axis (Id=40,58,76)	(→ Chart 16 / Page 75)
No reaction of modulator (Id=25,26)	(→ Chart 17 / Page 77)
Modulator defective (Id=24)	(→ Chart 18 / Page 78)
Temperature error for power stage X,Y,Z-axis (Id=41,59,77)	(→ Chart 19 / Page 78)
High voltage error by power stage X,Y,Z-axis (Id=42,60,78)	(→ Chart 20 / Page 80)
Pull protection error (Id=22)	(→ Chart 21 / Page 82)
Danger of stimulation (Id=45)	(→ Chart 22 / Page 83)
Stimulation monitor error (Id=46)	(→ Chart 23 / Page 84)
Filterfan error (Id=47)	(→ Chart 24 / Page 85)
Ohm power losses in X,Y,Z-axis (Id=62)	(→ Chart 25 / Page 86)

1.3.3.2 Chart 01

Message ID=18

Gradient coil not ready

SymbolicName: KAS_COIL_NOT_RDY

The gradient coil temperature exceeds 76 C. The temperature is measured with six NTC-resistors located in the gradient coil. The monitoring is done on supervision board D9 K2209 (patient table). The error signal is routed via FOC from U1/D9 to U7/D17 K2217, through D18 to D16 and via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error during normal operation of the system: FOC from U1/D9 to U7/D17 is glowing weakly; red LED, V413/D16 lights up. The red LED V38/A1 on D17 lights up.	a. Gradient coil temperature too high	Thermal overload caused by specific measurement conditions	Wait 10 minutes until the coil temperature is in safe range. Check sequence parameters.	
		Error of cooling system	Check cooling system: Pipes, water flow, function of RCA.	(→ Cooling / MR-000.840.18)
02. Error exists permanently: FOC from U1/D9 to U7/D17 is glowing weakly; right red LED, V413/D16 "COILERR" is on or off. lights up. The red LED V38/A1 on D17 is off.	a. Monitoring error	NTC/cable defective	Measure the NTC resistance.	(→ Supervision D9/Grad.coil Test Procedure / Page 111)
		Sensor cable not connected	Check cable and connectors.	
		D9 defective or supply voltage is missing if green LED V55 on D9 is off.	Check D9: Red LED V4,12,20 and position of DIP-switches. Replace D9.	(→ Supervision D9/Grad.coil Test Procedure / Page 111)
		FOC from D9 to D17 defective or not connected or connectors soiled, (not enough light passed through the FOC).	Check FOC from U1/D9 to U7/D17 (light off = error). Clean FOC connectors or replace FOC.	

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
02. continued Error exists....	a. Monitoring error	D17 defective (D18 defective)	Replace defective D17,(D18).	
		D16 defective	Check CAN-Bus communication; replace defective D16.	(→ CAN-Service D16 Test Procedures / Page 128)
03. The right red LED V413/D16 "COILERR" lights up, the red LED V38/A1 on D17 is off.		D18 and/or D16 defective	Check D18 and D16.	
Remarks: All modulators will be stopped by "SOFTSTOP" (MOS). The regulators will be inhibited. At a coil temperature of about 20C, the NTC resistance should be approx. 25 - 42 kOhm and at the temperature limit lower than 4 kOhm. A defective NTC can be by-passed by a DIP switch at the D9. There must be at least four working NTC's in the coil to ensure safety.				

1.3.3.3 Chart 02

Message Id=19

Main power supply not ready

SymbolicName: KAS_LT_NOTRDY_ERR/_WARN

The 400 VDC supply voltages for the power stages are switched off. The error signal is generated by opening of relay contact K3 on D31 K2217(open circuit) as a consequence of a not closed main contactor K2 (main supply voltage for KAS transformer). The error signal is routed via cable to D17 K2217, through D18 to D16 and via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. The two green LED's V407 on D16 are off.	a. Communication error: No enable signal from D16 for D31, or no acknowledge signal from Soft Start Unit D31	Defective cable connection between GSSU X3/D17 and X1/D31	Reboot the scanner, if no success: Check cable between X3/D17 and X1/D31. Check D16, D17/D18 and D31. Replace defective components. Check K1. Check K2 coil voltage (A1/A2): 24 VDC. Check cable between X2/D31 and K2 contacts 13/14. Replace D31.	(→ CAN-Service D16 Test Procedures / Page 128) (→ I/O D17/Power supply Test Procedures / Page 130)
		D31 or D16 or D17/18 defective		
	b. Soft Start Unit D31	Cable connection between K2 and D31 defective Defective contactor K1 or K2 on D31		(→ D31 Soft start Unit Test Procedure / Page 135) (→ Replacing the D31 K2217 / MR-000.841.16)
	c. Power Supply of D31 from GSSU Power Supply	Line voltage for GSSU Power Supply is missing or fuse F14 in the CCA is blown. Cable between GSSU/D17 and D31 defective	Check voltage LED's on D14/D142. Check AC power supply line to GSSU Power Supply via fuse F14 in the CCA. Check cable between GSSU/D17 and D31/X1.	(→ Chart 08 / Page 59), (→ Chart 09 / Page 61)

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01 continued The two green LED's ...	c. Voltage supply of D31 from GSSU Power Supply	Line voltage for GSSU Power Supply is missing or fuse F14 in the CCA is blown. Cable between GSSU/D17 and D31 defective	Check voltage LED's on D14/D142. Check AC power supply line to GSSU Power Supply via fuse F14 in the CCA. Check cable between GSSU/D17 and D31/X1.	(→ Chart 08 / Page 59), (→ Chart 09 / Page 61)
	d. Monitoring error	Defective relay K3 on D31 or cable between K2 and D31	Check cable between X2/D31 and K2 contacts 13/14. Replace D31.	(→ D31 Soft start Unit Test Procedure / Page 135)
		Defective D16	Check CAN-Bus communication; replace defective D16.	(→ CAN-Service D16 Test Procedures / Page 128)
Remarks: In case of this error during normal operation (contactor K1, K2 or K3 opens), the main transformer will be switched off by contactor K2, i.e. the KAS is switched to stand-by (POW OFF -> 400 VDC supply for power stages is off). Refer also to (→ Chart 05 / Page 56).				

1.3.3.4 Chart 03

Message Id=23

Rectifier overtemperature

SymbolicName: KAS_RECTIFY_TEMP

The temperature of the rectifier assembly on D32 K2217 is too high. The temperature is measured at the cooling plate of the rectifiers on D32 by a NTC-resistor. The monitoring is done on D31 K2217. The error signal is routed via cable from X1/D32 to X8/D31, from X1/D31 to X3/D17 K2217, forwarded through D18 to D16 and via CAN-Bus to the MPCU

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error occurs during normal operation. Red LED, V27/A1 on D17 "RC_TEMP" lights up	a. Cooling system	Error of the cooling system: Chilled water and/or fans beside the D32 inside the KAS cabinet	Check cooling system: Hoses to D32, water flow/temp., function of RCA/CCS, internal cabinet fans.	(→ Cooling / MR-000.840.18)
	b. Rectifier board D32	Defective rectifier on D32	Try to switch on after a few minutes; if no success, check all other reasons 01.a,02.a; if negative, replace the D32.	(→ Replacing the D32 K2217 / MR-000.841.16)
02. Error exists permanently. Red LED, V27/A1 on D17 "RC_TEMP" lights up	a. Monitoring error	Cable connection between X1/D32 and X8/D31 and/or between X1/ D31 and X3/D17 defective.	Check the cable connections. Check flat cable between D31 and D32.	(→ I/O D17/Power supply Test Procedures / Page 130)
		NTC-resistor on D32 defective	Measure the NTC-resistance. Replace defective D32.	(→ Replacing the D32 K2217 / MR-000.841.16)

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
02. continued Error exists ...		D17 defective (D18 defective)	Replace defective D17,(18)	(→ CAN-Service D16 Test Procedures / Page 128)
		D16 defective	Check CAN-Bus communication; replace defective D16.	

Remarks: All modulators will be stopped by "SOFTSTOP" (MOS). The regulators will be inhibited.

1.3.3.5 Chart 04

Message ID=21

Transformer overtemperature

SymbolicName: KAS_LT_TEMP

The transformer temperature is not in the supervision at now. If the error occurs, check the jumper setting on D17!

The main supply transformer T1 is in overtemperature condition. The monitoring is done on D31. The error signal is routed via cable to X5/D31, from X1/D31 to X3/D17 K2217, forwarded through D18 to D16 and via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error occurs during normal operation Red LED, V27/A2 on D17 "TR_TEMP" lights up	a. Main transformer T1 in the KAS cabinet	Transformer T1 thermal overload caused by specific measurement conditions	Wait for cool down.	
		Error of the CCS air cooling system or fans inside the KAS cabinet	Check function of the CCS and internal cabinet fans.	(→ Cabinet Cooling System / MR-000.840.18)
		Main transformer T1 in the KAS cabinet is defective	Try to go on after a few minutes; if no success, check all other reasons 01.a,02.a; if negative, replace the transformerT1.	

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
02. Error exists permanently Red LED, V27/A2 on D17 "TR_TEMP" lights up	a. Monitoring error	Temperature sensor in the transformer defective	Check the temperature sensor.	(→ I/O D17/Power supply Test Procedures / Page 130)
		Cable connection between temperature sensor and X1/D31 and/or X1/D31 and X3/D17 defective	Check cable connections.	
		D17 defective (D18 defective)	Replace defective D17, (D18).	
		D16 defective	Check CAN-Bus communication; replace defective D16.	(→ CAN-Service D16 Test Procedures / Page 128)
Remarks: All modulators will be stopped by "SOFTSTOP" (MOS). The regulators will be inhibited.				

1.3.3.6 Chart 05

Message Id=27

No reaction of main power supply

SymbolicName: KAS_LT_SET_ERR

Within 3 seconds after an "Enable" request (either by NUMARIS or by manually pressing the push-button "Power on" S402/D16), the main power supply does not respond with the "RDY" message (refer to (→ Chart 02 / Page 51)). The gradient processor on D16 K2217 generates the final error message which is routed via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error after KAS "Power on" command. The two green LED's V407 on D16 stay off.	a. Communication error: No enable signal from D16 for D31, or no acknowledge signal from Soft Start Unit D31.	Defective cable connection between GSSU X3/D17 and X1/D31 D31 or D16 defective	Reboot the scanner, if no success: Check cable between X3/D17 and X1/D31. Check D16, D18/D17 and D31. Replace defective components.	(→ CAN-Service D16 Test Procedures / Page 128) (→ I/O D17/Power supply Test Procedures / Page 130)
	b. Soft Start Unit D31	Cable connection between K2 and D31 defective Defective contactor K1 or K2 on D31	Check K1. Check K2 coil voltage (A1/A2): 24 VDC. Check cable between X2/D31 and K2 contacts 13/14. Replace D31.	(→ D31 Soft start Unit Test Procedure / Page 135) (→ Replacing the D31 K2217 / MR-000.841.16)
	c. Voltage supply of D31 from GSSU Power Supply	Cable between GSSU/D17 and D31 defective	Check cable between GSSU/D17 and D31/X1.	

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. continued Error after KAS ...	d. Monitoring error	Defective contactor K3 on D31 or cable between K2 and D31	Check cable between X2/D31 and K2 contacts 13/14. Replace D31.	(→ D31 Soft start Unit Test Procedure / Page 135)
		Defective D16	Check CAN-Bus communication; replace D16.	(→ CAN-Service D16 Test Procedures / Page 128)

Remarks: As a consequence of this error the KAS will be directly switched to stand-by mode by hardware. (TS OFF -> K2 off, i.e. primary voltage of transformer T1 off).

Refer also to (→ Chart 02 / Page 51).

1.3.3.7 Chart 06

Message Id=28

Synchronization fault

SymbolicName: KAS_SYNC_ERR/_WARN

The synchronizing clock signal from MPCU(GCTX) is faulty or missing. This 4MHz clock signal is transmitted via the gradient waveform interface cable from the GCTX/MC4C40 to the D11//D111 K2217. Here the signal is divided into a 20kHz signal. If the supervision on D14/D142 K2217 detects an error, the error signal will be routed through D18 to D16 and via CAN-Bus to the MPCU; a yellow LED, V59/A1 "SYNC Error" at D14/D142 will be on.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error during normal operation; yellow error LED V59/A1 "SYNC Error" at D14/D142 is on. Image quality may be degraded.	a. Communication between MMC and D11/D111	Error on MMC (MC4C40/GCTX)	Run MMC Tests (MC4C40/GCTX).	SeSo: Test Tools MMC
		Gradient data cable and/or connectors defective	Check gradient data cable; Run Test DAC/Functional, data transmission between GCTX and DAC.	SeSo: Test Tools Gradient
	b. DAC board D11/D111	D11/D111, wrong switch positions (S3/4) D11/D111 defective	Check switch positions on D11/D111: S3-pos.3, S4-pos.6. Replace defective D11/D111.	(→ DAC D112/111/11 Test Procedures / Page 122)
02. Yellow error LED V59/A1 "SYNC Error" at D14/D142 is on or off.	a. Monitoring error	D14/D142 defective	Replace D14/D142.	
		D16 defective	Check CAN-Bus communication; replace defective D16.	(→ CAN-Service D16 Test Procedures / Page 128)
Remarks: In case of this error all modulators will be stopped by "SOFTSTOP" (MOS). The regulators will be inhibited. In the service mode this error is treated as a warning and will not be evaluated, i.e. modulators and regulators will be not disabled. The image quality may be degraded.				

1.3.3.8 Chart 07

Message Id=30

Modulator clock error

SymbolicName: KAS_MOD_SYNC_ERR

The internal modulator clock is faulty. The error signal is generated on D14/D142 K2217 in combination with the supervision of Power-up condition and the signals CONFIG_DONE and READY of the programmable D14/D142 chips. The final error signal "CLOCK_ERR" is routed through D18 to D16 and via CAN-Bus to the MPCU; a red LED, V57/A1 "Clock error" at D14/D142 will be on.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error during normal operation; red error LED V57/A1 "Clock Error" at D14/D142 lights up.	a. Modulator board D14/D142	D14/D142 defective	Replace D14/D142.	
02. Red error LED V57/A1 "Clock Error" at D14/D142 is on or off.	a. Monitoring error	D14/D142 defective	Replace D14/D142.	
		D16 defective	Check CAN-Bus communication; replace defective D16.	(→ CAN-Service D16 Test Procedures / Page 128)
Remarks: In case of this error the KAS will be switched to stand-by (POW OFF -> 400 VDC supply for power stages is off).				

1.3.3.9 Chart 08

Message Id=31,32,33

Plus 15V/ Minus 15V/ Plus 5V defective

SymbolicName: KAS_P15/ N15/ VCC_ERR

DC supply voltages from E4A (+5V VCC, Slot1 and +/-15V, Slot3) are out of tolerance (+/- 9%). The voltage is monitored on D14/D142 K2217; the error signals are routed through D18 to D16 and via CAN-Bus to the MPCU. In case of a +/-15V error red LED's on D14/D142, V58/A1/A2 "+/-15V Error" are on; in case of a +5V error, green LED on D14/D142, V59/A2 "+5V OK" is off.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error after KAS "Power On" Green LED V59/A2 at D14/D142 off and/or Red LED's at D14/D142, V58/A1 (+15V) and/or V58/A2 (-15V) is on.	a. Power supply unit E4A	Power Supply E4A defective; fan defective	Measure/adjust the output voltages directly at the outputs of the power supply unit slots. If not adjustable replace the power supply unit.	(→ Small Signal Power Supply E4A Procedures / Page 132)
	b. Power supply lines	D17, (D18) defective or not plugged correctly; cabling between D17 and E4A defective	Measure the +/-15V and +5V supply voltages at D17/X4 and Test pins at front of D17; if ok, the backplane D18 is probably defective.	(→ I/O D17/Power supply Test Procedures / Page 130)
	c. Boards of GSSU,D31,32	Defective PCB	Replace the defective board.	Remarks

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. continued Error after KAS ...	d. Monitoring error	Supervision on D14/D142 defective	Measure the supply voltages at D17 (+/- 5% tolerance range) if ok, replace D14/D142.	(→ I/O D17/Power supply Test Procedures / Page 130)
		D16 defective	Check CAN-Bus communication; replace D16.	(→ CAN-Service D16 Test Procedures / Page 128)

Remarks: In principle each defective board in the GSSU including the D31/32 and the backplane D18 could be the reason for the supply voltage breakdown and should be checked. Check circuit breaker F14/4A in the CCA cabinet. The hum of the supply voltages should be lower than 10 mV_{pp}.

In case of this error the KAS will be switched to stand-by (POW OFF -> 400 VDC supply for power stages is off). Refer also to (→ Chart 09 / Page 61)

1.3.3.10 Chart 09

Message Id=86

GSSU power supply error

SymbolicName: KAS_SMALL_SIG_DEF

The supply voltage from GSSU power supply, slot 2 is missing. All driver supply errors (x, y, z) are present at the same time (refer to (→ Chart 10 / Page 62)). The driver voltages are measured and evaluated in the power stages (D21). The error signal is routed via cables to D14/ D142 and through D18 to D16. Here the final error signal is created by the gradient processor and routed via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. The red LED V64/A2 on D14/D142 is on during state1,2 and 3. The green LED V86 on power stage/D21 are off. If the green LED V5 is off, check cable connections between E4A/D31 and power stage.	a. Power supply unit E4A	Power Supply E4A defective; fan defective	Measure/adjust the output voltages directly at the outputs of the power supply unit slots. If not adjustable replace the power supply unit.	(→ Small Signal Power Supply E4A Procedures / Page 132)
	b. Power supply lines	Cable connections between E4A and X10/D31 and X12,13,14/D31 and power stages	Check cable connections. Measure the voltage on connector X10/D31 (+16,1V to +16,9VDC)	

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
02. Error LED's on D14/D142 in arbitrary conditions	a. Monitoring error	Supervision on power stage (D21) defective	Replace power stage.	(→ Gradient / MR-000.841.16), PS
		D14/D142 defective	Check D14/D142. Replace D14/ D142, if green LED's V86 on power stages are on.	(→ Modulator D14/142 Test Procedures / Page 126)
		D16 defective	Check CAN-Bus communication; replace defective D16.	(→ CAN-Service D16 Test Procedures / Page 128)
Remarks: The KAS will be switched to stand-by (POW OFF -> 400 VDC supply for power stages is off). Refer also to (→ Chart 08 / Page 59).				

1.3.3.11 Chart 10

Message Id=43,61,79

Driver supply voltage error (X, Y, Z)

SymbolicName: KAS_DRV_VOLT_X/_Y/_Z

The driver voltage of some axis is not ok. The monitoring is done on power stage/D21; (green LED V86 on -> ok). The error signals are routed via cable to D14/D142 K2217 (supervision and error multiplexer), through D18 to D16 and via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. The red LED V64/A2 on D14/D142 is on during conditions 1,2 and/or 3. The green LED V86 on power stage/D21 is off If the green LED V5 is off, check cable connections between E4A/D31 and power stage.	a. Driver supply/D21 inside power stage	D21/driver board (inside PS) defective	V6/7 and V86 on D21 off: Replace the corresponding power stage.	(→ Gradient / MR-000.841.16), PS
	b. Power supply lines	Cable connections between E4A and X10/D31 and X12,13,14/ D31 and power stages (V5=off)	Check cable connections. Measure the voltage on connector X10,X12,X13, X14 on D31 and X2 on PS (16.5 VDC).	
	c. Synchronization of driver supply voltage	D14/D142 defective	Replace defective D14/D142.	
	d. Cable connection D14/D142 -> power stage	Cable connection between power stage and D14/D142 defective	Check/replace the cable.	

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
02. Error LED's on D14/D142 in arbitrary conditions	a. Monitoring error	Supervision on power stage (D21) defective	Replace the power stage.	(→ Gradient / MR-000.841.16), Power stages
		Cable connection between power stage and D14/D142 defective	Check/replace the cable.	
		D14/D142 defective	Check D14/D142. Replace D14/D142, if green LED's V86 on power stages are on.	(→ Modulator D14/142 Test Procedures / Page 126)
		D16 defective	Check CAN-Bus communication; replace defective D16.	(→ CAN-Service D16 Test Procedures / Page 128)
Remarks: The KAS will be switched to stand-by (POW OFF -> 400 VDC supply for power stages is off). Refer also to (→ Chart 09 / Page 61).				
Note: The driver supply error always is coupled with a temperature error of the corresponding power stage. The reason for this behavior is that the driver supply is also the power supply unit for the temperature supervision of the power stage transistors Meaning of green LED's on D21 (LED's are on during normal operation): V5: DC supply voltage (from GSSU) ok; (LED above connector X2) V6/7: primary AC supply voltage ok V86: secondary AC supply voltage ok; (LED on right side of connector X1)				

1.3.3.12 Chart 11

Message Id=34

FAST OFF

SymbolicName: KAS_FAST_OFF

The voltage across the damping resistors during the "Switch on" sequence of KAS does not decrease in a certain time interval (after closing of contactor K2) and the contactor K2 will drop out again. The voltage is monitored on D31 and the error signal is routed via cable to D17 K2217, through D18 to D16 and via CAN-Bus to the MPCU

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error after KAS "Power On"	a. Power stages	Power stages short circuit	Replace power stages. Replace defective power stage. Check fuses between T1 and D32.	(→ Gradient / MR-000.841.16), Power stages
	b. D32 Rectifier board	D32 defective (as a consequence of PS short circuit)	Check D32 Replace defective D32. Check fuses between T1 and D32.	(→ D32 Rectifier Test Procedure / Page 138)
	c. Soft start unit D31	Defective/broken damping resistors	Check D31; replace defective D31; check power stages.	(→ D31 Soft start Unit Test Procedure / Page 135)

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. continued Error after KAS	d. Monitoring error	Monitor circuit on D31 defective	Replace defective D31.	
		Defective error signal path, cable connection between D31 and D17.	Check error signal path: D31->D17->D18->D16.	
		D16 defective	Check CAN-Bus communication; replace defective D16.	(→ CAN-Service D16 Test Procedures / Page 128)
Remarks: If a power stage has a short circuit, a high input current could destroy the damping resistors. As a consequence of this error the KAS will be directly switched to stand-by by hardware. (-> K2 off, i.e. primary voltage of transformer T1 off: POW OFF).				

1.3.3.13 Chart 12

Message Id=
35,36,53,54,88,72

Regulation error on X, Y, Z-axis

SymbolicName: KAS_LOOP_ERR/_WARN_X/_Y/_Z

The error signal and the regulator output signal are monitored in the closed loop error logic on D12/D123/D124 K2217. If one of these signals reaches the upper limit (10V), an error signal is generated on D12/D123/D124, routed through D18 to D16 and via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Dynamic error while running a measurement; the yellow LED's V135/A1, V135/A2, V136/A3 for X, Y, Z-axis "Loop Error" on D12/D123/D124 light up.	a. Measurement sequence	Gradient rise time (required from the sequence) is too fast (Refer also to 01/f.: partly defective power stage)	Switch on the Service mode to find out the affected axis. Run the specific sequence again; check the sequence parameters, if other than Siemens-released sequences are used.	
	b. Gradient data transmission	MC4C40/GCTX defective, data cable and/or connectors defective, D11/D111 K2217 defective (Gradient rise time is too fast)	Run MMC Test; run Gradient Test for DAC; check gradient data cable and connectors. Replace D11/D111.	SeSo: Test Tools MMC Gradient
	c. Load circuit gradient coil interrupted	Gradient coil, cable connections, RF filters defective	Check the continuity of the load circuit; check for loose cable connections. Replace defective components.	
	d. Closed control loop	D12/D123/D124, D14/D142 defective	Perform Regulator Adjust/Check; check for other error messages.	SeSo: Tune-up/QA

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. continued. Dynamic ...	e. Measurement of actual current	SENS and/or Regulator board D12/D123/D124 defective	If CAV_ERR_X,Y,Z (refer to TSG Chart 13), replace defective SENS, else replace D12/D123/D124.	(→ Regulator D12/123/124 Test Procedure / Page 124) (→ Gradient / MR-000.841.16), SENS; GSSU (→ Chart 13 / Page 70)
	f. Power stage	Power stage or part of power stage ("cylinder") defective (this may deteriorate image quality, because the risetime of the gradient pulses will be enlarged)	Check for other error messages; check the six green LED's through the holes in the front of the power stage. If LED's are off, the corresponding part of the power stage is defective; cyclically replace power stages. Replace defective power stage.	(→ Gradient / MR-000.841.16), Power stages
	g. Main power supply, Transformer T1. Main voltage	Too low output voltage	Measure the output voltage of the main power supply unit (D32). Check T1, voltage selection taps.	(→ D32 Rectifier Test Procedure / Page 138)

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
02. Error after KAS "Power On"; (in the idle mode)	a. DAC D11/D111	D11/D111 defective/ gradient rise time too fast.	Check voltage at D12/D123/D124, X11,12,13 in idle mode: Offset < +/- 0.05mV.	(→ Regulator D12/123/124 Test Procedure / Page 124)
	b. Regulator D12/D123/D124	D12/D123/D124 defective	Replace defective D12/D123/D124.	(→ Gradient / MR-000.841.16) (→ GPA K2217 / MR-000.841.16), and (→ Replacing the GSSU Components (E5) K2217 / MR-000.841.16)
	c. Modulator D14/D142	D14/D142 defective	Replace defective D 14.	
	d. Current sensors (SENS)	Current sensors (SENS) defective	Replace current sensors (SENS).	(→ Replacing the Current Sensor assembly (SENS/ E7) / MR-000.841.16)
	e. Monitoring error	D16 defective	Check CAN bus communication; replace defective D16.	(→ CAN-Service D16 Test Procedures / Page 128)
Remarks: The regulation error occurs as a consequence of every other error but will be suppressed in this case; you will see only the causal errors in the error report. The regulation error will only be recorded in the system report if it occurs as a primary error. All modulators will be stopped by Hardware (MOD OFF). In Service mode, the regulation error is treated as a warning and will not be evaluated, i.e. modulators and regulators will be not disabled.				

1.3.3.14 Chart 13

Message Id=37,55,73 **Current actual value error on X, Y, Z-axis**

SymbolicName: KAS_CAV_ERR_XI_YI_Z

Some current sensor signals (actual current value) are not correct. The current signals from the current sensors (in both directions) are monitored and compared on D12/D123/D124 K2217, a possible error signal is forwarded through D18 to D16 and routed via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error after KAS "Power On"; the LED's V131/A2, V132/A1, V132/A2 for X, Y, Z-axis "CAV_ERR" on D12/D123/D124 light up.	a. Current sensors (SENS) Type LB200-S/SP4 paired or Type IT600S combined with LA 205-S	SENS defective: Difference of measured currents in case of the IT600S/LA 205-S combination too high. (Difference > 23A)	Measure the sensor signals at D12/D123/D124, AV_X, AV_Y and AV_Z. Replace defective current sensors (SENS).	(→ Gradient / MR-000.841.16), SENS
	b. Current sensor supply	Break in signal line or power supply unit disconnected or supply line shorted	Check supply/signal line from SENS to X10/D12/D123/D124 for continuity and shorts.	
	c. Regulator D12/D123/D124	D12/D123/D124 defective/ defective SENS soft-start control (the error is detected, if the sensors are switched off)	Replace defective D12/D123/D124.	(→ GPA K2217 / MR-000.841.16) and (→ Replacing the GSSU Components (E5) K2217 / MR-000.841.16)
	d. Gradient cable from KAS to the Gradient Coil	Ground short of a gradient power circuit	Check the gradient coil circuit for ground shorts (gradient coil, gradient cables, gradient filters).	

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. continued Error after KAS ..	e. (Type IT600S/LA205-S) Synchronization via FOC	FOC's between D17,U22(X), U24(Y), or U25(Z) defective	Check the FOC's between D17 and SENS.	(→ Gradient / MR-000.841.16), SENS
02. No error LED lights up.	a. Monitoring error	D16 defective	Check CAN-Bus communication; replace defective D16.	(→ CAN-Service D16 Test Procedures / Page 128)
Remarks: In case of this error all modulators will be stopped by Hardware (MOD OFF).				

1.3.3.15 Chart 14

Message ID=38,56,74 **High current on X, Y, Z-axis**

SymbolicName: KAS_HI_CURR_X/_Y/_Z

The actual current exceeds the limit. The monitoring is done at D12/D123/D124 and an error signal is forwarded through to D16 K2217 and routed via CAN bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Dynamic error while running a measurement; the LED's V130/A1, V130/A2, V131/A1 for X, Y, Z-axis "IMAX" on D12/D123/D124 light up.	a. Measurement sequence	Sporadic overload	First run the duty_cycle sequence: If ok, check the measurement sequence parameters.	
	b. Regulator board D12/D123/D124	Regulator D12/D123/D124 oscillates due to wrong parameters; D12/D123/D124 defective	Run the Regulator Adjust/Check. Replace defective D12/D123/D124.	SeSo: Tune-up/QA: Regulator Adjust/Check
	c. Communication between MMC and D11/D111	Error on MMC (MC4C40/GCTX)	Run MMC Tests.	SeSo: Test Tools MMC
		Gradient data cable and/or connectors defective	Check gradient data cable; Run Gradient tests for DAC.	SeSo: Test Tools Gradient
		D11/D111 defective	Replace defective D11/D111.	(→ GPA K2217 / MR-000.841.16) and (→ Replacing the GSSU Components (E5) K2217 / MR-000.841.16)
	d. Modulator board D14/D142	D14/D142 defective	Replace defective D14/D142.	
	e. Current sensors (SENS)	Measured values incorrect by defective current sensors (SENS)	Replace defective current sensors (SENS).	(→ Chart 13 / Page 70), (→ Gradient / MR-000.841.16), SENS

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. continued Dynamic error while ...	f. Power stage (only if nominal value and actual value not equal)	Defective Power Stage transistors	Replace defective Power Stage.	(→ Gradient / MR-000.841.16), Power Stages
02. Error after KAS "Power On"	a. Monitoring error	D12/D123/D124 defective	Replace defective D12/D123/D124.	
		D16 defective	Check CAN-Bus communication; replace D16.	(→ CAN-Service D16 Test Procedures / Page 128)
Remarks: In case of this error all modulators will be stopped by hardware (MOD OFF). Check also for accompanying error messages, e.g. (→ Chart 13 / Page 70) "Current actual value error" -> and refer to the corresponding (→ Troubleshooting Charts KAS K2217 / Page 48).				

1.3.3.16 Chart 15

Message ID=39,57,75 **Duty cycle for power stage X, Y, Z**

SymbolicName: KAS_DUTYCYCLE_X/_Y/_Z

The duty cycle of the actual current is too high. The monitoring is done on D12/D123/D124 and an error signal is forwarded through D18 to D16 K2217 and routed via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Dynamic error while running a measurement; the LED's V133/A1, V133/A2, V134/A1 for X, Y, Z-axis "DuCy" on D12/D123/D124 light up.	a. Measurement sequence	Sporadic overload	Check the measurement sequence parameters Check the system by running a high duty cycle sequence.	
	b. Communication between MMC and D11/D111	Error on MMC (MC4C40/GCTX)	Run MMC Tests.	SeSo: Test Tools
		Gradient data cable and/or connectors defective	Check gradient data cable; Run Gradient Tests for DAC.	SeSo Test Tools: Gradient DAC/Functional
		D11/D111 defective	Replace defective D11/D111.	
	c. Regulator board D12/D123/D124	Regulator D12/D123/D124 oscillates due to incorrect parameters; D12/D123/D124 defective	Run Regulator Adjust/Check. Replace defective D2.	SeSo: QA Tune-up: Regulator Adjust/Check
	d. Modulator board D14/D142	Defective Modulator boards	Replace defective D14/D142.	
	e. Current sensors (SENS)	Incorrect values measured by defective current sensors (SENS)	Replace defective current sensors (SENS).	(→ Gradient / MR-000.841.16), SENS

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
02. Error after KAS "Power On"	a. Monitoring error	D14/D142 defective	Replace defective D14/D142.	
		D16 defective	Check CAN-Bus communication; replace defective D16.	(→ CAN-Service D16 Test Procedures / Page 128)
Remarks: In case of this error all modulators will be stopped by Hardware (MOD OFF). Check also for accompanying error messages, e.g. (→ Chart 13 / Page 70) "Current actual value error" -and refer to the corresponding (→ Troubleshooting Charts KAS K2217 / Page 48). The duty cycle supervision considers the rating of the power stages dependent on the direction of current flow.				

1.3.3.17 Chart 16

Message Id=40,58,76 **Controlling frequency error for power stage X, Y, Z** SymbolicName: KAS_CTRL_ERR_X/_Y/_Z

The frequency of the control signals for the power transistors of a specific power stage are not ok. The monitoring is done on D14/D142 K2217 (red LED V63 "Drive control error" lights up) and an error signal is forwarded through D18 to D16 K2217 and routed via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Dynamic error while running a measurement; Red LED V63/A1 on D14/D142 in state 1,2 and/or 3 according to the affected axis X,Y,Z lights up.	a. DAC board D11/D111 (wrong nominal value)	D11/D111 defective	Perform the DAC loop test. Replace defective D11/D111.	
	b. Regulator board D12/D123/D124	Regulator D12/D123/D124 oscillates or D12/D123/D124 defective	Check cable between D14/142 and power stage. Replace defective cable. Run the Regulator Adjust/Check. Replace defective D12/D123/D124.	SeSo: Tune-up/QA: Regulator Adjust/Check
	c. Modulator board D14/D142	Modulator board D14/D142 defective	Replace defective D14/D142.	

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
02. Error after KAS "Power On"	a. Monitoring error	D14 defective	Replace defective D14.	
03. Error after KAS "Power On"; The red LED V63/A1 on D14 is of during state 1,2 and 3.		D16 defective	Check CAN-Bus communication; replace defective D16.	(→ CAN-Service D16 Test Procedures / Page 128)

Remarks: In case of this error all modulators will be stopped by Hardware (MOD OFF).

1.3.3.18 Chart 17

Message ID=25,26

No Reaction of Modulator

SymbolicName: KAS_MOD_SET_ERR/_WARNING

The modulators of the three axes x,y,z can be controlled by the Service Software or by switches on D17, i.e. they can be switched on and off. If after a control command from the MPCU the modulator is not in the desired condition, this error is generated by the gradient processor, using a combination of several control signals (switch signals from D17 and control signals from D16). The final error signal will be routed via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Normal operation of MR measurement system or Service Software tests running (e.g. gradient tests): Red LED V61/A2 on D14/D142 is on in state 1,2 and/or 3 according to the affected axis X,Y,Z. KAS_Status message: X,Y,Z modulator inhibit = yes	a. I/O board D17	Modulator switches in "off" position (upward position) (means modulator x,y,z-axis disabled)	Set switches in "on" position (downward position). Loading and starting a sequence enables the modulators if no other error condition exists.	SeSo: Test Tools Gradient KAS_Status (On-line Help)
	b. Modulator D14/D142	D14/D142 defective	Replace defective D14/D142.	
	c. Service board D16	D16 defective	Check CAN-Bus communication; replace defective D16.	(→ CAN-Service D16 Test Procedures / Page 128)
Remarks: In case of this error the KAS will be directly switched to stand-by (TS OFF -> K2 off, i.e. primary voltage of transformer T1 off). In the service mode (switch S2/2 on D17 in upward position or service mode enabled by the Service Software) this error is treated as a warning and will not be evaluated, i.e. modulators and regulators will not be disabled (with the exception of manual disabling via switches on D17).				

1.3.3.19 Chart 18

Message ID=24

Modulator defective

SymbolicName: KAS_MOD_DEF

This error message is generated, if driver signals for S1/2 and S3/4 want to switch on their switches at the same time. The error is detected on D14/D142, the error signal is routed through D18 to D16 and via CAN-Bus to the MPCU. The red error LED V62/A2 "MOD_DEF" on D14/D142 is on during state 2.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error after KAS "Power On" or Dynamic error while running a measurement The red LED V62/A2 "MOD_DEF" on D14/D142 lights up in state 2.	a. Modulator D14/D142	D14/D142 defective	Replace defective D14/D142.	
	c. Monitoring error	D14/D142 defective	Replace defective D14/D142.	
		D16 defective	Check CAN-Bus communication; replace defective D16.	(→ CAN-Service D16 Test Procedures / Page 128)
Remarks: In case of this error the KAS will be switched to stand-by (POW OFF -> 400 VDC supply for power stages is off)				

1.3.3.20 Chart 19

Message Id=41,59,77

Temperature error for power stage X, Y, Z

SymbolicName: KAS_TEMP_ERR_X/_Y/_Z

The power stage temperature exceeds the limit of 56C. The temperature is measured inside the power stages (in the power stage transistor module).The signal is evaluated on Driver board D21 (green LED V74 on -> Temp ok) and an error signal is sent via cable to D14/D142 K2217. The error signal is monitored on D14/D142 (red LED V63/A2 "Temp" lights up), forwarded through D18 to D16 K2217 and routed via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Dynamic error while running a measurement; Red LED V63/A2 "Over-temperature" on D14/ D142 lights up in state 1,2 or 3 according to the affected axis X,Y,Z.	a. Cooling system	No water flow or water flow in the cooling system is too low or cabinet cooling system (RCA/CCS) defective.	Check for other error messages from RCA or MMC (cabinet temperature control). Check RCA, water pipes, flow rate; Check the cabinet cooling system (RCA/CCS).	(→ Cooling / MR-000.840.18) and (→ Cabinet Cooling System / MR-000.840.18)
	c. Power stage/D21 inside PS	Power stage defective.	Replace defective power stage.	(→ Gradient / MR-000.841.16), PS
	b. Measurement sequence	Load of KAS by measurement parameters or measurement conditions are too high.	Check the sequence parameters; run an approved high duty cycle sequence.	

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
02. Continuous error, trying KAS "Power On" after waiting period of approx. 30 sec; Red LED V63/A2 "Overtemperature" on D14/D142 lights up in state 1,2 or 3 according to the affected axis X,Y or Z.	a. Monitoring error	Temperature sensors (NTC resistors) of power stage defective	Replace defective power stage.	(→ Gradient / MR-000.841.16), PS
		Modulator board D14/D142 defective	Replace defective D14/D142.	
No error LED lights up.		D16 defective	Check CAN-Bus communication; replace defective D16.	(→ CAN-Service D16 Test Procedures / Page 128)
Remarks: In case of this error all modulators will be stopped by Hardware (MOD OFF). If the duty cycle error for the same axis (refer to (→ Chart 15 / Page 73)) occurs at the same time, check at first the used measurement parameters, otherwise check at first the cooling system. The temperature error is also a consequence of a driver supply error.				

1.3.3.21 Chart 20

Message Id=42,60,78 **High voltage by power stage X, Y, Z**

SymbolicName: KAS_PST_VOLT_X/_Y/_Z

The output voltage of D32 (rectifier board) exceeds the upper limit. The voltages are measured on D32. The error signal is sent via cable from X1/D32 to X8/D31 and X1/D31 to X3/D17K2217 and is monitored at D14/D142 (dependent on the axis a red LED, V64/A1 "Overvoltage" lights up in state 1,2 or 3). The error signal is forwarded through D18 to D16 and routed via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Dynamic error while running a measurement; Red LED V64/A1 "Overvoltage" on D14/D142 lights up in state 1,2 or 3 according to the affected axis X,Y or Z.	a. Main voltage	Instable main supply voltage, line overvoltage	Measure the output voltages of D32; Provide for stable main supply voltage.	(→ D32 Rectifier Test Procedure / Page 138)
	b. Main power supply unit Transformer T1	Transformer taps wrong connected.	Check the cable connections (taps) at T1.	
	c. Power stage	Power stage defective or a partial stage of a power stage doesn't work	Check if all six green LED's behind the six holes in the front of the power stage are on during normal operation; if not, check fuses between T1 and D32, check D32; swap and/or replace defective power stage.	(→ Gradient / MR-000.841.16), PS
	d. SCSI-cable between Modulator and PS	Defective SCSI-cable or connector	Swap and/or replace defective SCSI-cable.	

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. continued Dynamic error ...	e. Modulator board D14/D142	Modulator D14/D142 defective	Replace defective D14/D142.	(→ Small Signal Power Supply E4A Procedures / Page 132)

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
02. Error after KAS "Power On"; Red LED V64/A1 "Overvoltage" on D14/D142 lights up in state 1,2 or 3 according to the affected axis X,Y or Z.	a. Monitoring error	D32 voltage supervision defective	Measure the output voltages on D32; Replace defective D32.	(→ D32 Rectifier Test Procedure / Page 138)
		Cable connections defective between X1/D32 to X8/D31 and X1/D31 to X3/D17.	Check cable connections.	
		D17(D18) defective	Replace defective D17(D18)	
		Modulator D14/D142 defective	Replace defective D14/D142.	(→ Small Signal Power Supply E4A Procedures / Page 132)
03. No error LED lights up.		D16 defective	Check CAN-Bus communication; replace defective D16.	(→ CAN-Service D16 Test Procedures / Page 128)
Remarks: In case of this error the KAS will be switched to stand-by (POW OFF -> 400 VDC supply for power stages is off).				

1.3.3.22 Chart 21

Message Id=22

Pull protection error

SymbolicName: KAS_PULL_PROT

The cable connections between D14/D142 K2217 and the power stages (X, Y, Z) are not plugged correctly or are defective. The error is monitored on D14/D142. The error signal is routed through D18 to D16 and via CAN_Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Red LED V62/A2 "Cable connected" on D14/D142 lights up in state 1.	a. Cable connection between Modulator D14/ D142 and power stage	Cable connectors not correctly plugged in, bent pins on connector or cable broken.	Check cable connection, connector for bent pins and cable.	
	b. Monitoring error	D14/D142 defective	Replace defective D14/D142.	(→ Small Signal Power Supply E4A Procedures / Page 132)
		D16 defective	Check CAN-Bus communication; replace defective D16.	(→ CAN-Service D16 Test Procedures / Page 128)
Remarks: In case of this error all modulators will be stopped by Hardware (MOD OFF).				

1.3.3.23 Chart 22

Message Id=45

Danger of stimulation

SymbolicName: KAS_STIM_DANGER

If the SW stimulation monitor measures/calculates a risk of patient stimulation from the actual gradient pulses (measured by SENS/D12/D123/D124), he sends a signal to D17 K2217 (The red LED "EMMI" on D17 lights up). The signal is forwarded through D18 to D14/D142 (error multiplexer) further to D16 and via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error during imaging: The red LED "EMMI" on D17 lights up; the modulators will be stopped. The red LED V62/A1 "EMMI" on D14/D142 lights up in state 1.	a. Measurement sequence	The measurement parameters demand gradient pulses which could cause patient stimulations.	Note: This event can occur during routine imaging. Change measurement parameters.	
	b. Regulator board D12/ D123/D124/ SENS	D12/D123/D124 and/or SENS defective	Run Regulator Adjustment. Replace defective components.	Service Software / Tune-up
	c. Monitoring error	Defective cable connection between MPCU/MC1C40 and D17	Check/ replace defective cable.	
		Defective MC1C40/GPA_INT or defective D17	Check/replace MC1C40/GPA_INT	MMC tests + GPA/ Stim. Mon.
		Defective D16	Check CAN-Bus communication; replace defective D16.	(→ CAN-Service D16 Test Procedures / Page 128)
Remarks: In case of this error all modulators will be stopped by Hardware (MOD OFF).				

1.3.3.24 Chart 23

Message Id=46

Stimulation monitor error

SymbolicName: KAS_STIM_MON_ERR

The stimulation monitor detects an internal error. The error signal is routed from the MPCU/MC1C40/GPA_INT to D17 K2217, through D18 to D16 and via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error during normal operation of the system The red LED, V40/A2, "DB_WDG_ERR" on D17 lights up; the modulators will be stopped.	a. Stimulation monitor	Stimulation monitor not plugged in correctly or defective. MC1C40/GPA_INT	Replace defective component.	MMC tests + GPA/ Stim. Mon.
	b. Monitoring error	Defective cable connection between MPCU/MC1C40/GPA_INT and D17	Check/ replace defective cable.	
		Defective MC1C40 or defective D17	Check/replace defective MC1C40/ GPA_INT.	MMC tests + GPA/ Stim. Mon.
		Defective D16	Check CAN-Bus communication; replace defective D16.	(→ CAN-Service D16 Test Procedures / Page 128)
Remarks: In case of this error all modulators will be stopped by "SOFTSTOP" (MOS). The regulators will be inhibited.				

1.3.3.25 Chart 24

Message Id=47

Filterfan error

SymbolicName: KAS_FAN_ERR

The temperature of the gradient filter is not in the supervision at now. If the error occurs, check the jumper setting at D17!

The temperature of the gradient filter is out of tolerance. The monitoring is done on the RF-filter plate. The error signal is routed via FOC to D17 K2217, through D18 to D16 and via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Error during normal operation of the system The red LED, V40/A2, "FILTERFAN" on D17 lights up; the modulators will be stopped. (FOC from filter temperature supervision: Light off => Error).	a. Gradient filter temperature too high	Thermal overload caused by specific measurement conditions or too high ambient temperature.	Wait for cool down; check the air condition system.	
		Error of filter fan	Replace filter fan.	
	b. Monitoring error	Defective temperature sensor or cable from sensor to D17.	Replace defective components.	
		Defective temperature supervision	Replace defective temperature supervision.	
		FOC from RF-filter plate to D17 defective	Check/replace FOC.	
		Defective I/O board D17	Replace defective D17.	
	Defective D16	Check CAN-Bus communication; replace defective D16.	(→ CAN-Service D16 Test Procedures / Page 128)	
Remarks: In case of this error all modulators will be stopped by Hardware (MOD OFF).				

1.3.3.26 Chart 25

Message Id=80

OHM power losses in X, Y, Z-axis

SymbolicName: KAS_OHMLOSS_XI_YI_Z

The error signal is generated on D12/D123/D124 from the supervision of the actual current. The final error signal is a combination of the "OHMLOSS" error and the "DuCy" error. The error signal is routed through D18 to D16 and via CAN-Bus to the MPCU.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Dynamic error while running a measurement.	a. Gradient power circuit	Ohmic power losses in the gradient cables (KAS/filters/chokes, cables, etc. too high)	Decrease the rating of the gradient system by gradient pulsing.	
The red LED, V134/A2 "I2" and one or more of the red LED's "DuCy" on D12/D123/D124 light up (V133/A1, V133/A2 or V134/A1 according to the affected axis).	b. Monitoring error	D12/D123/D124 and/or SENS defective	Replace defective components.	
02. Error after KAS "Power On";		Defective D16	Check CAN-Bus communication; replace defective D16.	(→ CAN-Service D16 Test Procedures / Page 128)
Remarks: In case of this error all modulators will be stopped by Hardware (MOD OFF). The OHM power loss supervision considers the rating of the power stages independent on the direction of current flow. The duty cycle error will be present at the same time.				

1.4 Image Quality - Test Strategy

1.4.1 General

The stability of the MR system is critical to the quality of MR imaging. Stability Check and CalcAr are routines used in the Tune-up and Quality Assurance to monitor the stability of the MR imaging system. If the results of either or both of these routines are greater than a specific tolerance, the source of the instability can be found by investigating the various components making up the MR imaging system. The service engineer would use the Gradient Tests (DAC, X,-Y-,Z-Axis, Functional) in the Test Tools Service Software platform when there is an indication that the source of the problem, e.g. instability, is in the gradient system:

For instance, when the instability measured in the QA functions of the Service Software "Stability Check" or "CalcAr" is orientation-dependent, it is indicative of instability due to gradient instability and not RF or a magnetic field instability.

Other QA-functions check whether system parameters are adjusted correctly, e.g. Regulator check/ ECC Check or Graddelay. If the results are not within tolerance, you have to perform the corresponding measurements/adjustments of the Tune-up.

In case of **orientation-dependent RF-disturbances in the image**, which are supposed to be caused by the gradient system (switching frequency of the amplifiers), you should interchange the gradient cables from the KAS of the affected axis in order to locate the origin of the problem. If the disturbances disappear with swapped cables, the problem is probably located in the GPA/KAS. Otherwise the problem must be connected with the gradient/bodycoil system.

1.4.2 Summary of Gradient-dependent IQ - Parameters

IQ Parameter	Influence Factor	Available Tests
Spatial Resolution Contrast (Symptoms are edge rings and edge blurring)	Non-linear amplitude of gradient pulses Sequence timing related to eddy current fields	Check linearity of gradient pulses (Gradient Loops); eddy current compensation (ECC)
Slice Thickness Slice Profile Slice Position Slice Warp	Eddy currents Linearity of gradient field	ECC Depending on gradient coil design; no improvement possible
Artifacts such as smearing artifacts Ghosts: discrete/blurred	Phase encoding errors: Any kind of gradient instability; periodic unequal phase encoding steps SENS instability/nonlinearity	DAC: Data loop test Check gradient stability/linearity (Gradient loops) Regulator adjustment

IQ Parameter	Influence Factor	Available Tests
Image Uniformity Spatial Linearity	Non-linear gradient magnetic fields	No improvement possible
Signal to Noise Ratio	Gradient system: Eddy currents; Spikes generated by gradient switching	ECC, Gradient Delay; Spike Check

1.4.3 Gradient System - Image Quality TSG Chart

Normally there are no specific error messages corresponding to an image quality problem. You have to specify the individual problem according to the column "Symptom" in the table below and to perform the proposed test to localize the problem in the MR system.

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
01. Smearing artifacts in phase encoding direction: - discrete ghosts - blurred ghosts	a. Low phase encoding performance of the gradient system caused by instability	SENS instability and offset at low gradient amplitudes D11/D111 DAC defective/wrong gradient data from MPCU/GCTX Regulator oscillation	Perform the Gradient Tests; check the stability of the DAC/actual value Replace D11/D111 Check regulator adjustments	SeSo: Test Tools Tune-up/QA functions Regulator Check
02. Edge blurring/rings artifacts in phase encoding direction	a. Nonlinearity of gradient pulse steps	Slew rate limitation of the gradient pulses at high amplitudes (may be caused e.g. by partially defective power stages)	Perform the Gradient Tests; check the linearity of the DAC/actual value. Check function of power stages: The six green LED's behind the holes in the front of the power stage must be on.	Seso: Test Tools (→ Gradient / MR-000.841.16), Power Stages
03. Streaks and/or dots in images	a. RF noise caused by rapid gradient switching	Magnet room not RF-tight	Check RF-room shielding, RF-filter plate, RF-filters	

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
04. Low signal to noise ratio and/or spikes	a. Gradient system: Gradient/Bodycoil system	Eddy currents; Spikes caused by mechanical vibrations during gradient switching	Perform - Graddelay - ECC Check - Spike Check *	SeSo: Tune-up/QA functions
05. Non-uniformity of images (phantom image)	a. Gradient system	Eddy currents	Perform ECC Check	
06. Image spatially distorted	a. Adjustment of gradient system	Gradient sensitivity misadjusted	Perform Gradsens	
07. Low contrast/resolution of the image: Edge rings and blurring	a. Gradient system Non-linear amplitude switching of gradient pulses	Wrong gradient data D11/D111 DAC	Check gradient data transmission: Perform MMC Tests and Gradient Tests (check results of DAC/functional Test, GCTX-DAC) Perform the gradient linearity tests DAC/actual value	SeSo: Test Tools MMC Gradient
	b. Gradient system	Eddy currents	ECC/Check	SeSo: QA/Tune-up functions
08. Wrong slice position and/or warped slice	a. Shim condition of magnet	Distorted magnetic field	Check the shim condition of magnet/ Reshim the magnet	SeSo: Tune-up Shim/Phantom Shim
	b. Gradient coil design	Non-linear gradient field	No improvement	
09. Broadening of slice thickness/profile	a. Gradient system	Eddy currents	ECC/Check	SeSo: QA/Tune-up functions

Symptom	Area of Trouble (Components/function)	Probable Cause	Service Action	Reference
10. Insufficient Fat Saturation	a. Base magnetic field	Magnetic field inhomogeneities due to bad shim (possible reasons may be: magnet quench, magnetizable materials in the magnet bore)	Perform Shim Check. Perform Phantom Shim. Reshim the magnet.	SeSo: QA: Shim Check Tune-up: Phantom Shim
	b. Gradient offsets	Gradient offset currents have changed (linear gradients)	Perform Shim Check. Perform Phantom Shim. Reshim the magnet.	
	c. Eddy currents	Uncompensated eddy currents/ cross term eddy currents	Perform ECC Check/ ECC adjustment Perform CTC Check/ CTC adjustment	SeSo: QA: ECC check CTC check Tune-up: ECC CTC
*) Note: TRUFI and HASTE- sequences and special sequences used e.g. for EPI or BOLD- Imaging generate spikes, which cannot be verified by the normal QA "Spike Check". In these cases, the "Advanced Spike" procedure in QA Expert mode can be helpful for spike troubleshooting. Please notice: The "Advanced Spike" procedure contains a powerful gradient pulse scheme which induces high mechanical stress over a wide frequency range. It shall be therefore used very careful and only when all other means are exhausted. Risk of hardware damage! » DO NOT USE the "Advanced Spike" procedure without guidance from RSC/ HSC!				

2.1 General

This section describes the various tests and checks for field replaceable units (FRU's) of the Gradient System K2209 (GPA) and K2217 (KAS) of the MR systems Harmony/Symphony/Quantum/Sonata. The tests and checks support error localization (in response to error messages). After replacement of a FRU, the tests and checks may be required to ensure that the switch/jumper configuration or other special adjustments are correct. In a subsequent section the possible gradient test loops of the Service Software, platform "Test Tools" are described shortly, i.e.

- Gradient data transmission between MMC/GCTX and GSSU/D1/D11/D11(1)
- Test loop DAC-value: D1/D11(1)(DAC) - D6/D16 - RFSU (ADC)
- Test loop Actual value: D2/D12/D123/D124 (Regulator) - D6/16 - RFSU (ADC).

The gradient test loops verify the stability and linearity of the gradient power amplifier.

2.1.1 Required Tools

Prerequisites for successfully troubleshooting the gradient system using this document are:

- Functional Description (FUN)
- Diagrams (Chapter Gradient System)
- Set of standard tools,
- Digital multimeter, Fluke 8060A or equivalent

2.1.2 Summary of FRU's and Tests for K2209

FRU's K2209	Test procedures
DAC board D1	1. Check jumpers and switch position 2. Offset measurement/adjustment 3. Gradient data test loop GCTX/D1
Regulator board D2	1. Default regulator setting by DIP switches 2. SENS offset measurement/adjustment 3. Regulator Adjust/Check
Synchronization board D3	1. Check jumper settings 2. Measurements of clock signals and supply voltages
Modulator board D4	1. Check of DIP switch configurations
Supermodulator board D5	1. Check of DIP switch configurations

FRU's K2209	Test procedures
Service/CAN board D6	1. Check communication to RC 2. Manual loop setting 3. CAN-Bus communication (Test Tools/Gradient/GPA status).
FOC board D7	1. Board layout 2. FOC - Tests 3. Supply voltage measurement
Small signal Power Supply E4A/Driver supply D10	1. Supply voltage measurement/adjustment
Supervision board D9	1. DIP switch configurations 2. Sensor tests
Resonant converter (RC)	1. Measurement/adjust. of output volt.
Final Stages (Papillon type 5939447)	1. Measurement of the resistance to the housing (leakage current)

2.1.3 Summary of FRU's and Tests for K2217

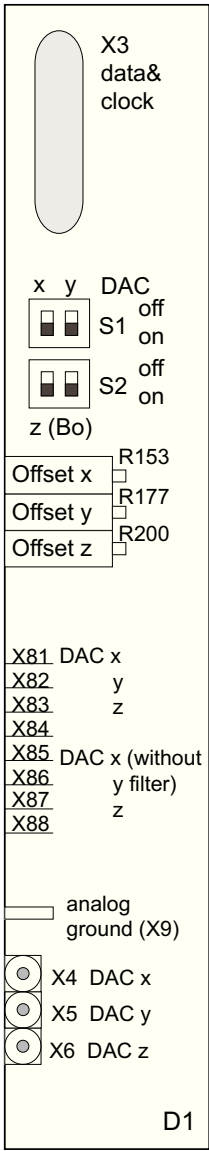
FRU's K2217	Test procedures
GSSU	
DAC board D112/111/11	1. Check jumpers and switch position 2. Offset measurement/adjustment 3. Gradient data test loop GCTX/D11
Regulator board D12/123/124	1. Default regulator setting by DIP switches 2. Jumper settings for maximum current, current regulation mode and current direction-dependent switching 3. SENS offset measurement/adjustment d. Regulator Adjust/Check
Modulator board D14/142	1. Check of slew rate jumper settings
Service/CAN board D16 (57 25 515 K2217)	1. Check of communication to the main power supply unit. 2. Manual loop setting 3. CAN-Bus communication (Test Tools/Gradient/GPA status).

FRU's K2217	Test procedures
I/O board D17 (57 26 430 K2217)	1. Switch setting of S1 and S2 2. Jumper settings X9 3. Supply voltage measurement
GPA cabinet	
Small signal Power Supply E4A (30 59 482)	1. Supply voltage measurement/adjustment
Current sensor assembly (SENS) (57 25 655 K2217)	1. Offset adjustment of D12 2. Regulator Adjustment 3. Phantom Shim 4. Grad Delay 5. Grad Sens 6. ECC Check 7. Image Orientation 8. SN DIP & CalcAr
GPA cabinet Power Stage Supply boards:	
D31 switch on assembly (57 26 414 K2217)	1. Check of connection to GSSU 2. Supply voltage measurements
D32 rectifier board (57 27 016 K2217)	1. Check of PTC temperature sensor 2. PS supply voltage measurement
GPA cabinet	
Power stage (59 38 142 K2217) Power stage Light (59 38 720 K2217)	1. Regulator Adjustment 2. Phantom Shim 3. Grad Delay 4. Grad Sens 5. Image Orientation
Fans (30 63 476)	Function check

2.2 Procedures GPA K2209

2.2.1 DAC D1 Test Procedures

Fig. 4: Front of DAC D1



Descriptions:

- X3:
18 bit parallel gradient data and 4MHz clock signal from MMC
- S1/S2:
The DAC output can be switched on/off for every axis; S2/B₀ is not used
- Potentiometers for offset adjustment
- DAC output = nominal value (0-> +/-10V) 10 V corresponds to 300 A

2.2.1.1 a. Check Jumpers and Switch Positions:

- Jumper X16: 2-3 (sign of gradient data, 2-Complement)
- DIP switch S1: both in lower position
- DIP switch S2: both in lower position
- Rotary switch S3: position 7
- Rotary switch S4: position 2 (this configuration results in a clock signal FSYNC = 50 kHz)

2.2.1.2 b. Offset measurement/adjustment:



Prerequisite is the Regulator Adjustment (Tune-up).

- Gradient cabinet has been on for at least 10 minutes.
- Set switches S1/2 to up position ("DAC output zero")
- The corresponding modulator must be on: S1/2 on D4 in down position.
- Connect a DVM to the desired actual value output X3 on D2. Use the QLA-BNC service cable.
- Check the offset: Offset < +/- 0.05 mV
- Use the corresponding R153, 177, 200 on DAC-board D1 to adjust the offset for the X, Y and Z axes if necessary.
 - » If the offset was changed, perform the Tune-up steps Phantom Shim and Gradient Sensitivity

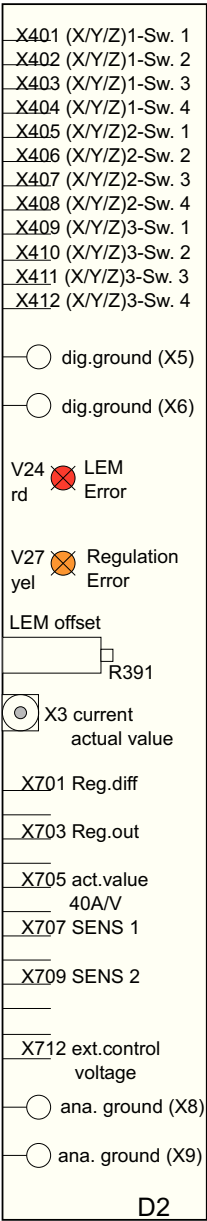
2.2.1.3 c. Gradient Data Test loop GCTX/D1:

Use the Test "Gradient/ DAC/Functional" of the Test Tools platform to confirm the correct gradient data transmission. between MMC/GCTX and GSSU/D1; (a prerequisite for this test is the correct function of the MMC components, refer to MMC-Tests).

2.2.2 Regulator D2 Test Procedure

Fig. 5: Front of Regulator D2

Descriptions:



R391 SENS-Offset adjustment

X3 Actual current value, QLA (1V=40A)

X701 Regulator difference signal

X703 Regulator output signal

X705 Actual current value (1V=40A)

X707 SENS1 current value

X709 SENS2 current value

2.2.2.1 a. Default Regulator Setting by DIP Switches:

DIP-switch default setting:

Switch	S1 (P)	S2 (I)	S1 adjusts the proportional portion of each regulator axis. S2 sets the integral regulator portion of each axis.
1	ON	ON	
2	off	off	
3	off	off	
4	off	off	
5	off	off	
6	off	off	
7	off	off	
8	off	off	

2.2.2.2 b. SENS Offset measurement/adjustment:

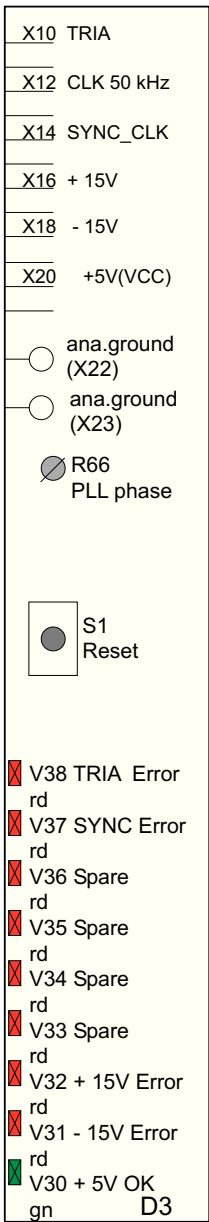
- Gradient cabinet has been on for at least 10 minutes and the regulators are adjusted (see Regulator Adjust/Check below)
- Connect the DVM to X3 ("actual value") on the regulator board. Use the QLA-BNC service cable.
- Set the switch, corresponding to the axis, on the modulator board D4 in upward position ("Modulator off").
- Check the offset: Offset < +/- 0.05 mV; use R191 to adjust the offset if necessary.
 - » If the offset was changed, perform the Tune-up steps Regulator Adjustment, Phantom Shim, Grad Delay and Grad Sens

2.2.2.3 c. Regulator Adjust/Check:

The regulators are set via the Service Software function Tune-up/Regulator Adjust. The QA/Regulator Check function quickly checks the regulator data set.

2.2.3 Synchronization D3 Test Procedures

Fig. 6: Front of Syn-
chronization D3



- Descriptions:**
- X10 Triangle voltage
 - X12 Clock signal CLK, 50kHz
 - X14 SYNC_CLK, 50 kHz (derived from 4MHz system clock)
 - X16 +15V (supplied by E4A, slot 3)
 - X18 -15 V (supplied by E4A, slot 3)
 - X20 +5V (VCC, supplied by E4A, slot 1)
 - R66: Adjustment of the phase locked loop (free running frequency 52 kHz)
 - S1: Hardware supervision reset (Resetting of the error latches)

Error LED's

- 2.2.3.1 a. Check Jumper Settings:
- Jumper X3: 3A - 3B, 6A - 6B, 9A - 9B

2.2.3.2 b. Measurements of Clock Signals and Supply Voltages:

- Triangle voltage:

Amplitude: +/- 9V

Adjustment frequency: 52 kHz

(Adjust with R66; measure with a DMM frequency counter; during adjustment remove the gradient data cable from MMC to D1/X3)

- Supply voltages:

In case of supply voltage error, you can measure the voltages supplied by the E4A directly at the power supply unit terminals, at D3 or D7 (refer also to D7 test procedures).

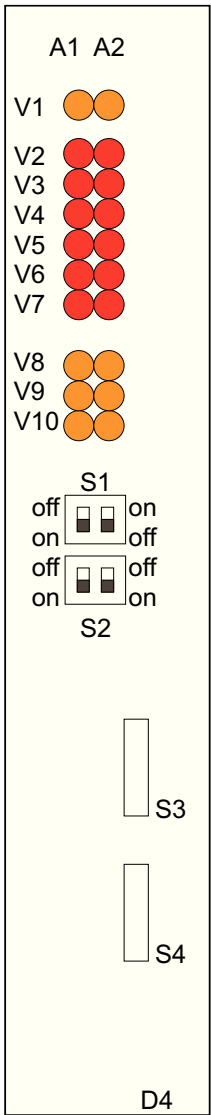
Tolerance ranges (error status if exceeded):

+/- 15V (+/- 13,65V...+/- 16.35V)

+5V (4,55V...5,45V)

2.2.4 Modulator D4 Test Procedures

Fig. 7: Front of Modulator D4



Descriptions:

- LED's (left / right):
- V1 Service mode / Modulator disabled
 - V2 Overtemp. / Drive error (X1 or Y1 or Z1)
 - V3 Spare / Overvoltage X1
 - V4 Overvoltage Y1 / Overvoltage Z1
 - V5 Overcurrent X1 / Duty cycle error X1
 - V6 Overcurrent Y1 / Duty cycle error Y1
 - V7 Overcurrent Z1 / Duty cycle error Z1
 - V8 Current X+ / Current X-
 - V9 Current Y+ / Current Y-
 - V10 Current Z+ / Current Z-
 - S1 left: Modulator X on/off
 - S1 right: Service mode on/off
 - S2 left: Modulator Z on/off
 - S2 right: Modulator Y on/off

2.2.4.1 a. Check of DIP Switch Configurations:

Check the following DIP Switch Configurations:

- DIP switch S1: both in lower position
- DIP switch S2: both in lower position
- DIP switches S3 and S4: Check setting according to the following table:

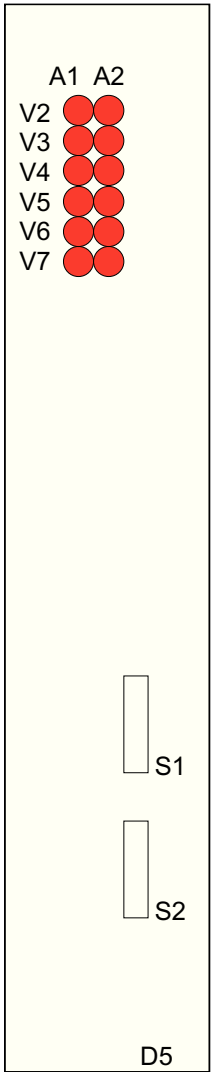
	DIP switch S3	Function	DIP switch S4	Function
1	ON	deadtime1	ON	modsemo x
2	off	deadtime2	ON	modsemo y

	DIP switch S3	Function	DIP switch S4	Function
3	off	mode1	ON	modsemo z
4	off	mode3	ON	sunuse x
5	off	fqa	ON	sunuse y
6	off	mode2	ON	sunuse z
7	ON	fqb	ON	swloss1
8	off	mode4	off	swloss2

(Refer to (→ Interactive Functional Description / MR-000.850.05) for further explanations).

2.2.5 Turbo Modulator D5 Test Procedures

Fig. 8: Front of Modulator D5



- Descriptions:**
- V2 Overtemp. / Drive error (X2 or Y2 or Z2)
 - V3 Spare / Overvoltage X2
 - V4 Overvoltage Y2 / Overvoltage Z2
 - V5 Spare/ Spare
 - V6 Spare/ Spare
 - V7 Spare/ Spare

2.2.5.1 a. Check of DIP Switch Configurations:

DIP switch default setting:

	DIP Switch S1	Function	DIP Switch S2	Function
1	ON	deadtime1	ON	modsemo x
2	off	deadtime2	ON	modsemo y
3	ON	mode1	ON	modsemo z
4	off	mode3	ON	sunuse x
5	ON	not used	ON	sunuse y
6	ON	mode2	ON	sunuse z

	DIP Switch S1	Function	DIP Switch S2	Function
7	ON	not used	ON	swloss1
8	ON	not used	off	swloss2

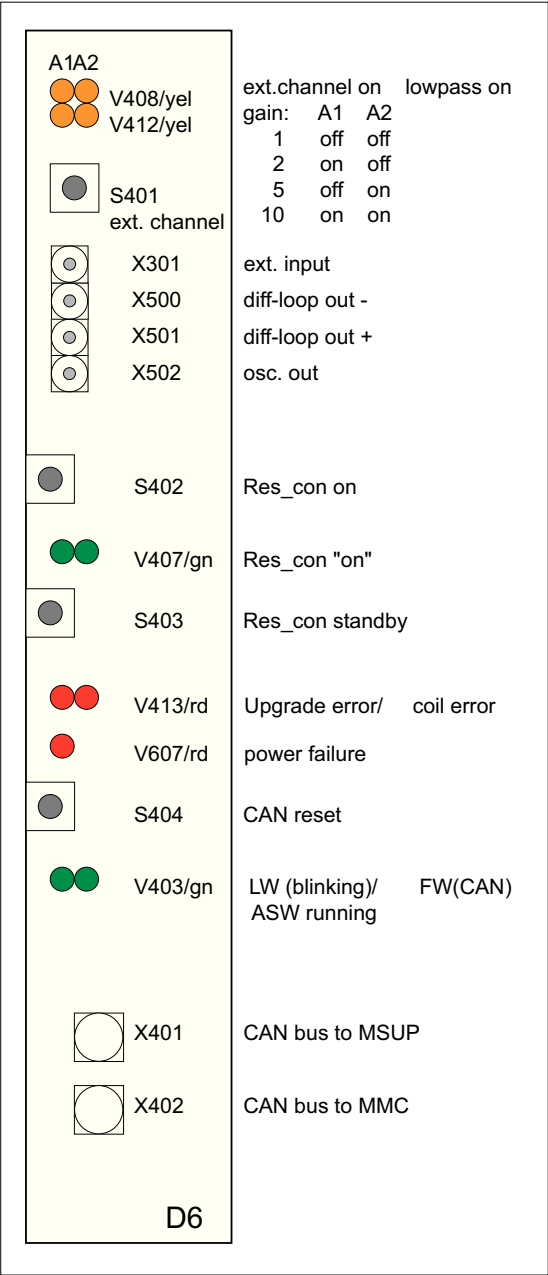
(Refer to the (→ Interactive Functional Description / MR-000.850.05) for further explanations).

2.2.6 Service/CAN D6 Test Procedures

Check Jumper Settings:

- Jumper X601: 1- 2

Fig. 9: Front of Service/CAN D6



Descriptions:

LED's show conditions of external channel settings.

S1: Sets external channel

X301 External signal input

X502 Oscilloscope output

S402: Manual RC "ON"

V407: RC on/off

S403: Manual RC "stand-by"

V413 left: Upgrade component is missing or defective

V413 right: Gradient coil error

S404: Warm start of GPA-processor

V403 left, blinks during normal conditions

2.2.6.1 a. Check communication to RC:

The RC receives the switch-on signal via the fiber optic cable of assembly D7 (U51). Test by pressing S403 and then S402. Verify that the RC is switched to stand-by (LED's "EN" and "RDY" off) and on again (switch-on signal is present at D7 and is transmitted to RC).

2.2.6.2 b. Manual loop Setting:

Fig. 10: LED display on D6 showing the possible gain and low pass filter combinations of the external channel

Setting of external channel on D6/D16:							
V 408 A1= ext.chan. A2= LP V 412 A1= gain A2= gain							
Taster S 401 pressed:							
1*	2*	3*	4*	5*	6*	7*	8*
ext.chan. No LP gain1	LP gain1	no LP gain2	LP gain2	no LP gain5	LP gain 5	no LP gain10	LP gain10
Connector external channel: X 301							

When using the external channel, you can select several combinations of different gain and low-pass filter settings by pressing button S401 n-times according to (→ Fig. 10 Page 107).



If all LED's are off, the automatic loop is activated

2.2.6.3 c. CAN-Bus communication/GPA status:

The present status (Status/warning/errors) of the GPA is displayed completely with the help of the Service Software function **"Gradient/Communication"** in the platform Test Tools.

2.2.7 FOC D7/Power Supply Test Procedures

Assembly D7 is the interface of the GSSU to the power stages, as well as the current measurement and supervision assemblies. Electric signals are converted into FOC signals and vice versa.

In case of errors, this is an easy method for ruling out "monitoring errors":

The status of the error transmission path is tested by pulling the respective fiber optic cable.

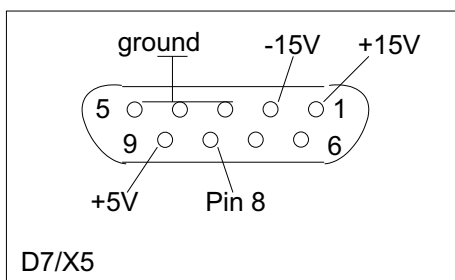
"Light off" means that there is an error and/or one of the corresponding sensors is defective.

(→ Fig. 13 Page 109) shows the configuration of the connections for the fiber optic cables.

Upgrade interface connector:

If there is no upgrade component installed, the pin 8 must be connected to ground.

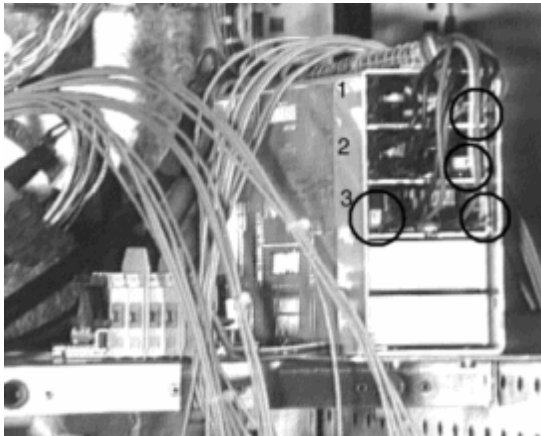
Fig. 11: Upgrade interface connector at D7



2.2.8 Small Signal Power Supply E4A

2.2.8.1 Adjustment of Supply Voltages:

Fig. 12: Adjustment of GSSU power supply voltages



Slot 1: +5V Vcc

Measure the voltage at the terminals of E4A or on D7/X5, Pin 9-5

Slot 2: +15V, supply voltage for D10, Driver supply

Measure the voltage at the terminals of E4A

Slot 3: +/- 15V

Measure the voltage at the terminals or on D7/X5, Pin 1-5 and 2-5.

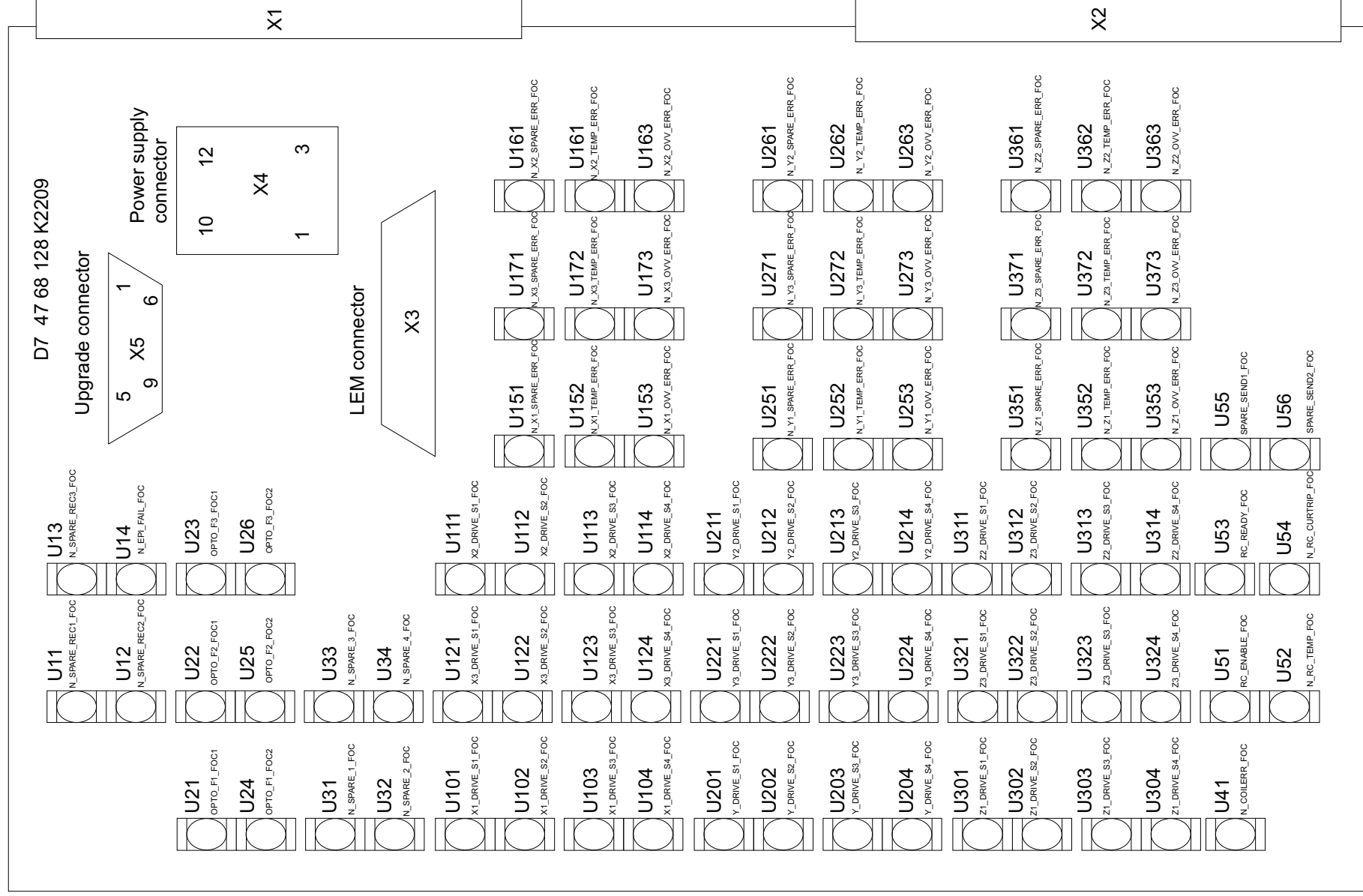
Adjust the voltages with the marked potentiometers (→ Fig. 12 Page 108)

2.2.8.2 Driver supply D10

Check Jumper setting:

- Jumper X4: external clock 1-2 (default); internal clock 2-3.

Fig. 13: D7 K2209 Connector positions

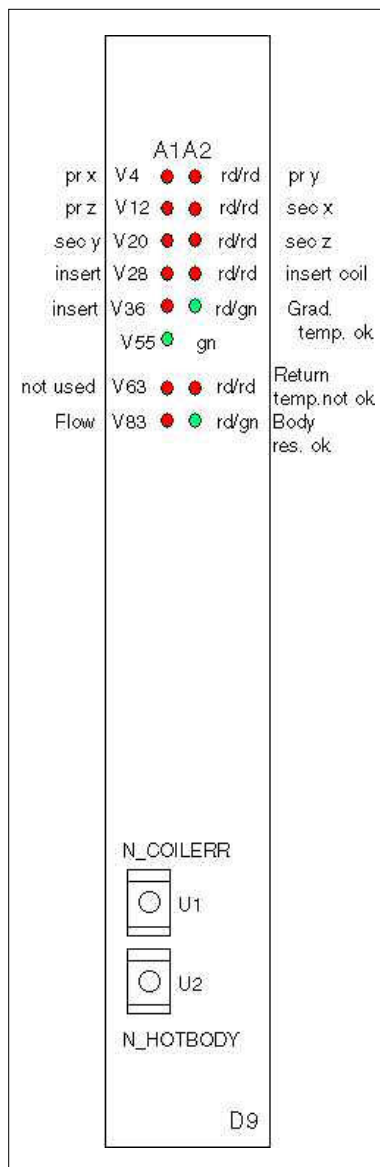


2.2.8.3 Pin Positions of Connectors SENS/X1, D7/X3

PIN-No	Signal	Signal function
1	Spare	n.c.
2	C-CON1_X	Actual value X+
3	GND	Ground
4	C-CON2_X	Actual value X-
5	P15_RET	disconnecting protection
6	C-CON1_Y	Actual value Y+
7	GND	Ground
8	C-CON2_Y	Actual value Y-
9	P_15V	Ground
10	C-CON1_Z	Actual value Z+
11	GND	Ground
12	C-CON2_Z	Actual value Z-
13	GND	Ground
14	P15_X	positive supply voltage (X+ SENS)
15	N15_X	negative supply voltage (X+ SENS)
16	P15_X	positive supply voltage (X- SENS)
17	N15_X	negative supply voltage (X- SENS)
18	P15_Y	positive supply voltage (Y+ SENS)
19	N15_Y	negative supply voltage (Y+ SENS)
20	P15_Y	positive supply voltage (Y- SENS)
21	N15_Y	negative supply voltage (Y- SENS)
22	P15_Z	positive supply voltage (Z+ SENS)
23	N15_Z	negative supply voltage (Z+ SENS)
24	P15_Z	positive supply voltage (Z- SENS)
25	N15_Z	negative supply voltage (Z- SENS)

2.2.9 Supervision D9/Grad.coil Test Procedure

Fig. 14: Front of Supervision D9



Descriptions:

LED's:

pr x,y,z:

NTC-sensor status at primary side of the gradient coil

sec x,y,z:

NTC-sensor status at secondary side of the gradient coil

V55gn: 22V supply voltage ok

V63 rd right:

Return temperature status of cooling water for the body resonator (Temperature too high)

V83 rd left:

Water flow supervision for the body resonator (Flow not ok).

V83 gn right: Body coil ok

N_COILERR:

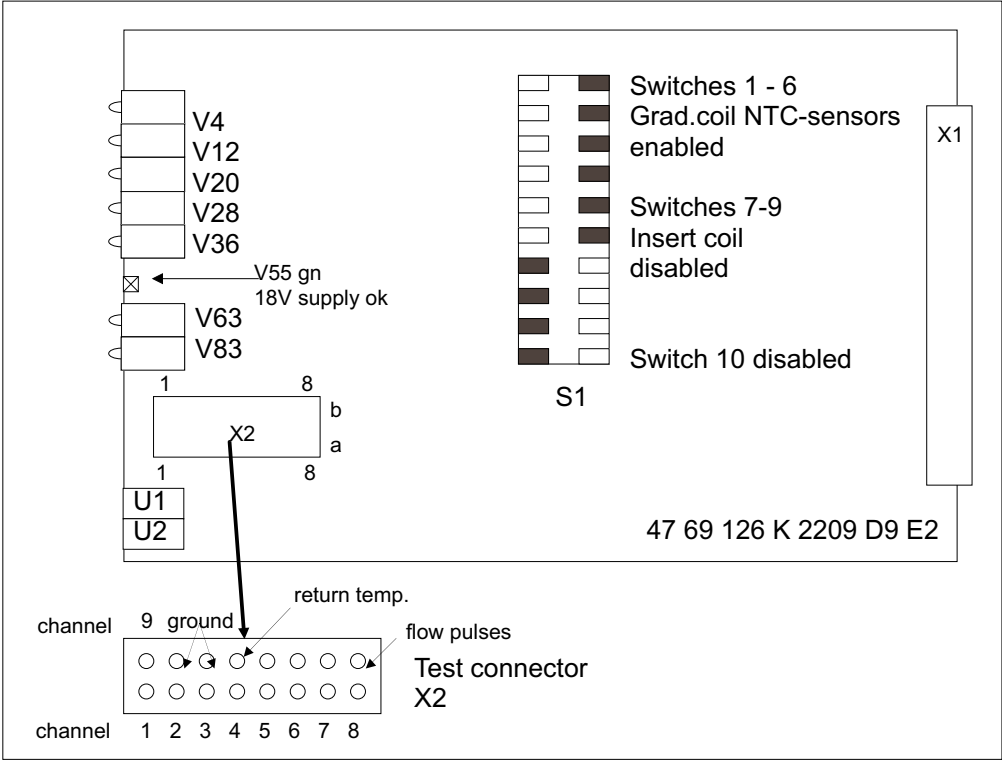
FOC connection to D7/U41 (Light off or glowing weakly: => Error)

N_HOTBODY

FOC connection to BTB1/U5 (Light off => Error/ Body Coil not ok)

The coil supervision board D9 is located in the left part of the patient table control electronics rack.

Fig. 15: Position of DIP switches at D9



Normal condition:

Check Jumpers and Switch Positions:

- Switches 1-6 = ON
- Switches 7-10 = off

(Switch 1 => pr x, switch 2 => pr y, a.s.o.)

An error condition (green LED off) exists if the gradient coil temperature at the location of the NTC-sensor exceeds 83 +/-7 C or the cable to the coil is not connected or the NTC is shorted or broken.

In case of overtemperature and/or shorted NTC the corresponding red LED lights up.

At most two defective NTC-sensors may be deactivated by switching the corresponding DIP-switch S1/D9 to the off position. In this case, the corresponding error LED may be permanently on, but no error message is generated.

In case of problems, measure the resistance of the NTC-sensors at the gradient coil connector X2 or at connector X20/D9:

DIP- Switch at D9	1	2	3	4	5	6
Connector X20/ D9	25-13	24-12	23-11	22-10	21-9	20-8
Grad.coil connector X2	4-5	6-7	8-9	10-11	12-13	14-15
AS39R(S)	Xprim	Yprim	Zprim	Xsec	Ysec	Zsec
AC39T Quantum	Xprim	Yprim	Yprim1	Zprim	Xsec	Ysec
AS Sonata	Xprim	Yprim	Yprim2	Zprim	Xsec	Ysec

DIP- Switch at D9	1	2	3	4	5	6
Connector X20/ D9	25-13	24-12	23-11	22-10	21-9	20-8
Grad.coil connector X2	4-5	6-7	8-9	10-11	12-13	14-15
AC44 (Allegra)	Xprim	Yprim	Zprim	Xsec	Ysec	Zsec
AS Trio	Xprim	Yprim	Yprim2	Zprim	Xsec	Ysec
At most two defective NTC sensors may be deactivated by switching the corresponding DIP- switch D9 to the off position						

Typical resistance of NTC's:

Room temperature: $R > 15\text{k}\Omega$ (approx. $40\text{ k}\Omega$ at $20\text{ }^{\circ}\text{C}$)

Temperature limit: $R < 4\text{ k}\Omega$

In case of a **body coil error**, i.e. cooling water flow too low (refer to **RF-repair instructions** for adjustment) and return cooling water to hot, the red LED's V63 (right) and/or V83 (left) light up.

Measurement of the cooling water temperature sensor resistance in the **body resonator** (return cooling water flow) can be done at connector X21/D9 between Pins 9 and 10.

Resistance values of the temperature sensor:

$R = 2000\text{ }\Omega$ at $20\text{ }^{\circ}\text{C}$

$R > 2140\text{ }\Omega$ -> overtemperature

$R < 2000\text{ }\Omega$ -> OK

2.2.10 RC Test Procedure

According to the table below check the following FOC's on the display board in order to verify a corresponding error message:

- "RDY" off => RC Output voltage lower than 60% of U_{nom}
- "TEMP" off => RC power stage overheated
- "ITRIP" off => At least one of RC output stages in current trip; if this state occurs when the modulators are switched off, a power stage of the GPA is defective: a lighted LED IX1...IZ2 indicates the defective power stage.
- "V.." off -> Corresponding output stage voltage $> 425\text{V}$; if all six FOC's are off, the RC control electronics are defective.

Tab. 1 RC Mark 1

LED	EN	RDY	Water Temp	Air Temp	Phase	V DIP	V PRI	IX1	IX2	IY1	IY2	IZ1	IZ2
LED's continuing on right side										ILIM	P15V	N15V	HF15V
FOC	EN	RDY	TEMP	ITRIP	VX1	VX2	VY1	VY2	VZ1	VZ2			

Tab. 2 RC Mark 2

LED	IX1	IX2	IY1	IY2	IZ1	IZ2	Phase			
	EN	RDY	15V	ILIM	VPRI	VDIP				
	Water Temp	Air Temp	ind Temp	tran temp						
FOC	EN	RDY	TEMP	ITRIP	VX1	VX2	VY1	VY2	VZ1	VZ2

2.2.10.1 Meaning of LED's

LED	Color	Description (LED = ON)
EN	green	RC enabled from GSSU: ON during normal operation, OFF in stand-by mode of RC.
RDY	green	Output voltage of RC is higher than 60% of the nominal output voltage.
WATER TEMP	red	Cooling water temperature too high.
AIR TEMP	red	Air temperature in the GPA cabinet too high.
PHASE	red	Line voltage phase error.
V DIP	red	Line voltage exceeds 528 V.
V PRI	red	Line supply voltage for RC control is lower than 185 V.
IX1... IZ2 (6 ¹)	red	The current trip threshold of the corresponding channel is exceeded. The corresponding power stage of the GPA is probably defective.
ILIM	yel	Primary power stage of RC is in current limit
P 15	green	Supply voltages (15 VDC) for RC are ok. ON during normal and stand-by operation of RC.
N 15	green	
HF 15	green	
ind temp ¹	red	Inductor temperature too high
tran temp ¹	red	Transformer temperature too high
15V ¹	green	Supply voltages (15 VDC) for RC are ok.

1. only for RC Mark 2

2.2.10.2 Measurement /Adjustment of Output Voltage (RC/old type):



Switch off the power supply unit of the RC by fuse first F9 (3*50A) and then F14 in the CCA before working with removed front cover of the RC or making any connection of test cables to the test points of the output voltages

Voltages up to 420 VDC may be present at these test points.



WARNING

There is the danger of destroying the control board of the RC due to attaching high voltages.

- » **When you connect test lines or clips to the test points, make sure that they do not touch adjacent circuit lines.**

With the pot R1, located on the left side of the display board you can adjust the maximum output voltage of the RC: $U_{nom} = 405$ to 410 VDC with modulators switched off. The voltage must be adjusted and measured after installing of a replacement RC (refer to the repair procedures). Through a hole in the front cover you can get access to R1. But in case of six installed power stages you have to remove the front cover in order to measure the output voltage of the RC.

The output voltage of the RC can be measured as described below:

2.2.10.3 1. Three Power Stages are installed:

- Remove the protection cover plate right of power stages.
- Connect FOC's, line voltage and water pipes to the RC.
- Connect the DVM to UB+/UB- in the GPA cabinet (lower fuse holder NH2).
- Switch on the Service mode (S1 right side at D4 in upward position).
- Switch on the RC (first F14 and then F9).

The resonant circuit of the RC can be switched on/off manually by the push-buttons S402 ("on") and S403 ("off") at D6.

- Switch off the modulators (switches S1 left and S2 in upward position).
- Turn pot R1 clockwise to increase the voltage or counter clockwise to decrease the output voltage.

With modulators off, the voltage should be between 405 and 410 VDC.

- After adjustment reassemble the protection cover plate right of power stages.

2.2.10.4 2. Six Power Stages installed/ backside of GPA not removable:

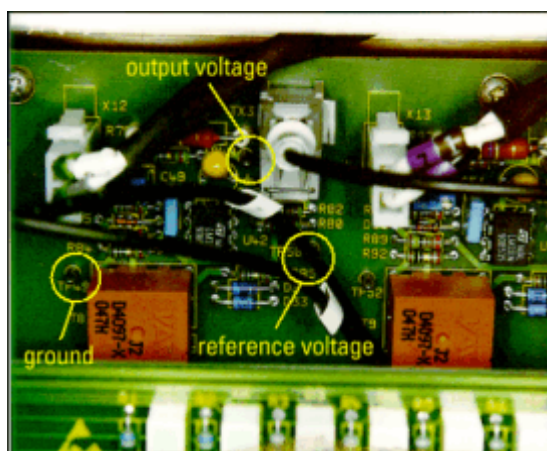
- Remove the power cable, the FOC's and front cover of the RC to get access to the test points which are located above the display board. (refer to the table and figure below).
- Reconnect FOC's, line voltage and water pipes again.
- Measure and adjust the output voltage as described above. (Measure the voltage of one of the six output testpoints only).

Location of test points: (refer to the the figure below)

Test point	X1	X2	Y1	Y2	Z1	Z2
Ground	43	46	49	52	55	58
U _{out}	42	45	48	51	54	57
U _{ref}	44	47	50	53	56	59

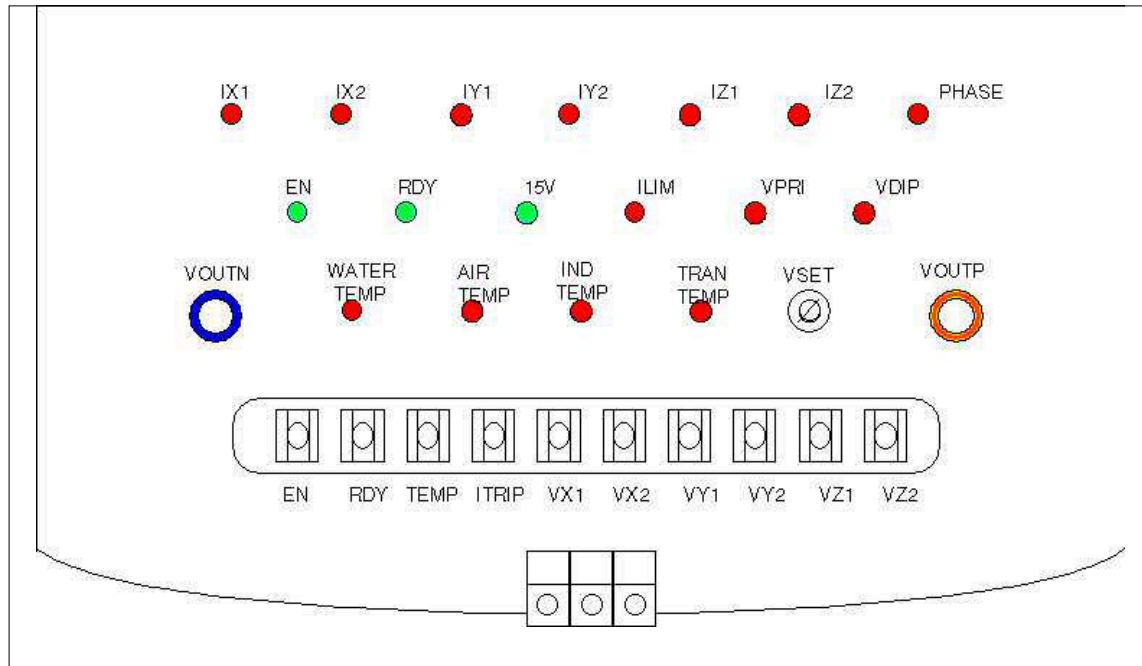
- Optionally you can measure the reference voltage for the output overvoltage supervision: $U_{ref} = 8.6...8.7$ VDC. The resonant circuit of the RC will be switched off, if the output voltage of one output channel exceeds 425 VDC.

Fig. 16: Location of test points on the RC



- If out of tolerance, adjust the voltage with pot VSET at RC to 4,07 VDC.

Fig. 17: Voltage measurement terminals and adjustment pot of RC MK2



Don't forget to deactivate the "Service mode" for normal operation of the MR system.

2.2.11 Final Stage (Papillon type 5939447)

2.2.11.1 Measurement of the resistance against the housing (Leakage Current)

When choke temperature switches had been activated due to a high leakage current of the final stage the faulty amplifier could be identified by the following measurements (only on Papillon final stages, material number: 59 39 447, not on the old style amplifier).

- Remove the final stages (refer to Replacement of Parts)
- Create a short circuit from Ub+ to Ub- terminal of the final stage (with the lead and the 2 clamps) and perform the following measurements (using the standard Fluke DVM):

Measurement points	Nominal values	Measured values
housing > out 1/3	130 kOhm - 180 kOhm	_____
housing > out 2/4	130 kOhm - 180 kOhm	_____
housing > Ub	> 300 kOhm	_____
out 1/3 > out 2/4	> 200 kOhm	_____

- Remove the short circuit from Ub+ to Ub- terminal of the final stage and perform the following measurements:

Measurement points	Nominal values	Measured values
Ub+ > out 1/3	300 kOhm - 900 kOhm	_____
Ub+ > out 2/4	300 kOhm - 900 kOhm	_____
Ub- > out 1/3	300 kOhm - 900 kOhm	_____
Ub- > out 2/4	300 kOhm - 900 kOhm	_____

- » in case the result is out of range (refer to the nominal values listed in the table above), the corresponding amplifier should be replaced as soon as possible. The choke of the affected axis must not be replaced.
- Reinsert and reconnect the final stages.
- Double-check the correct position of the fiber optic connectors.



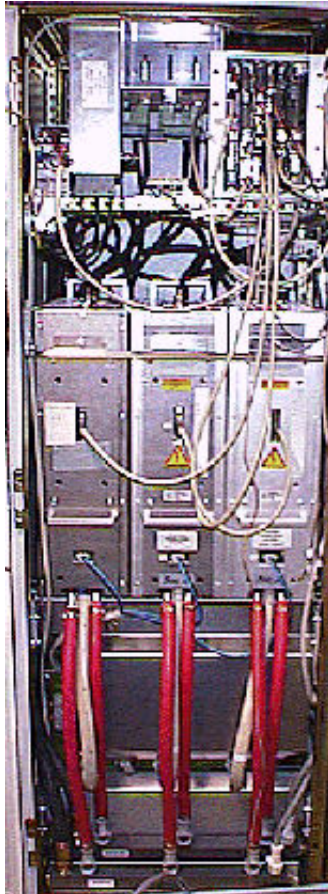
Take care not to damage the fiber optic cables of the final stages while lifting them into the cabinet

Check the fiber optics for loose connectors before reconnecting (loose connectors damage the final stage)

2.3 Procedures GPA K2217

2.3.1 Parts Location

Fig. 18: Front side of GPA_059 cabinet



Components

(from left to right)

Temperature sensor (above E4A)

Power Supply E4A & Fan

Line filter

SENS (in the back)

GSSU & Fan

Cabling, FOC's

Power stages X,Y,Z-axis

NF filters (in the back)

5 fans

Transformer T1 (in the back)

Main contactor K2

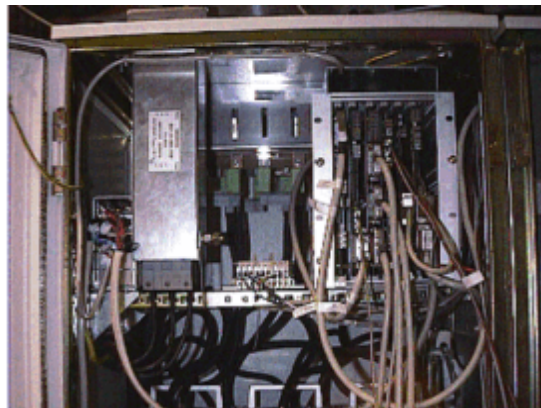
D31, Soft start unit

D32, Rectifier board

Cooling water distributor

2.3.1.1 FRU's in the upper part of the GPA Cabinet

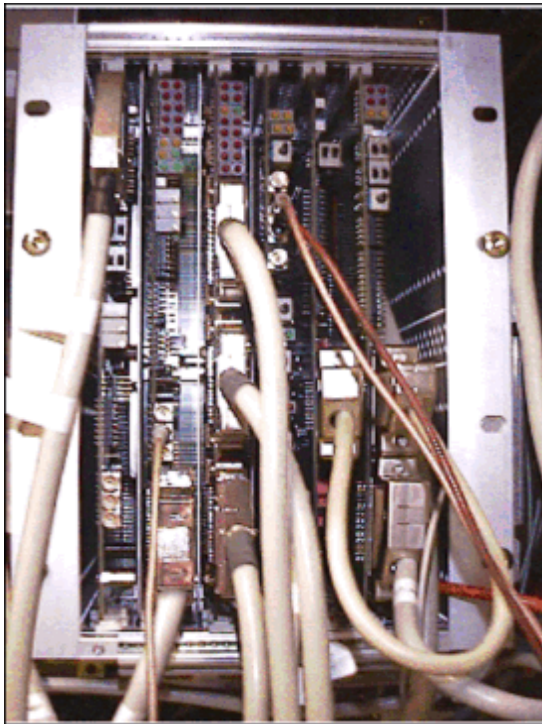
Fig. 19:



Components from left to right

- Power Supply E4A
- Line input filter
- Connector terminal X100
- GSSU
- In the background: Current sensor assembly

Fig. 20: GSSU of GPA K2217

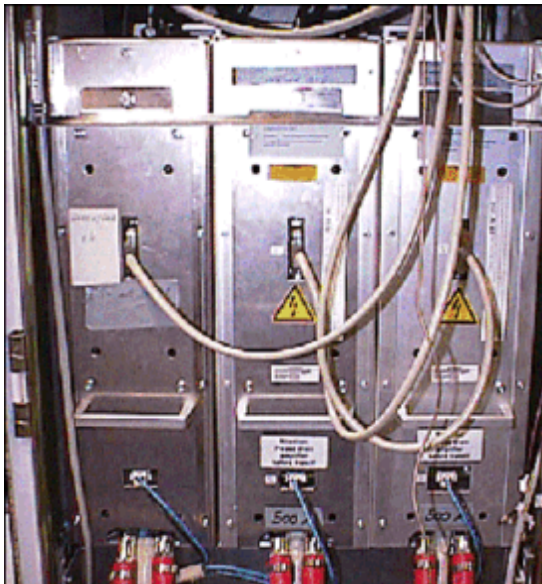


Position of PCB's:

D11(1) - D12(3/4) - D14(2) - D16 - D17
(D13 not used with NUM 4)

2.3.1.2 FRU's in the middle part of the GPA Cabinet

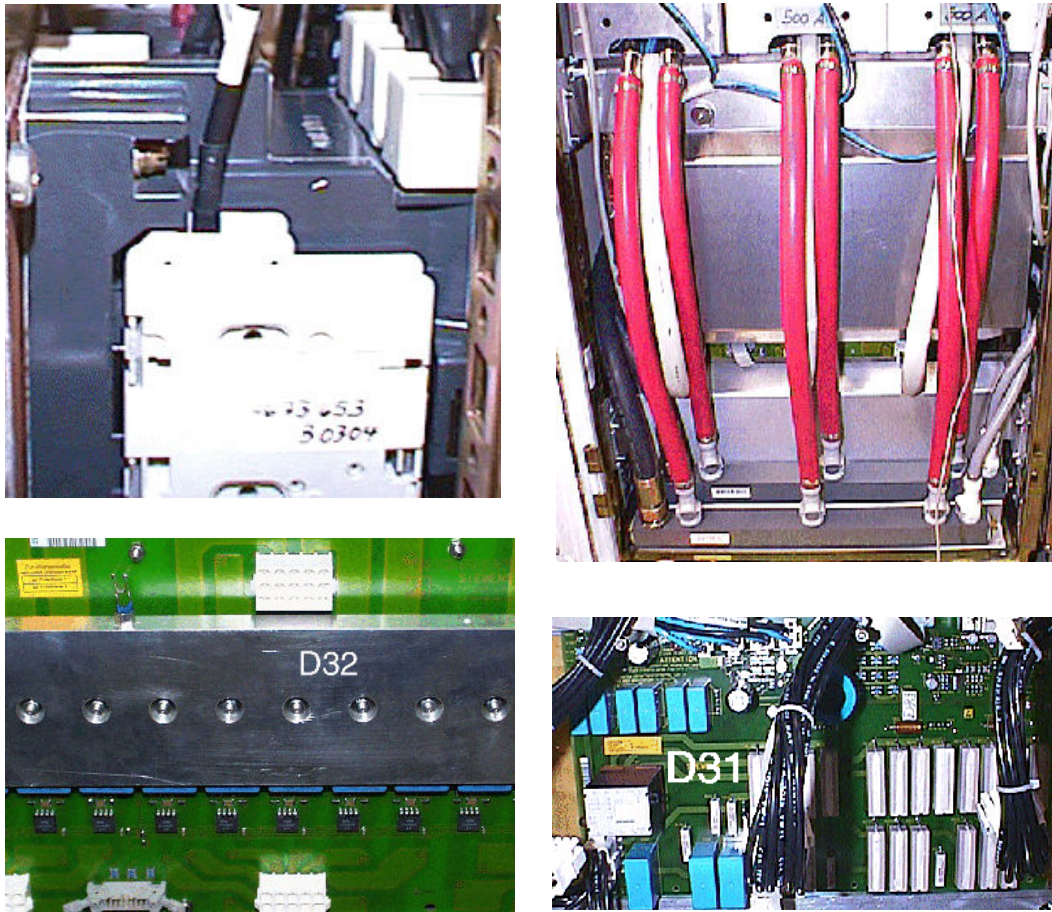
Fig. 21: Middle part of GPA cabinet



X,Y,Z-Power stage with cabling and water pipes

2.3.1.3 FRU's in the lower part of the GPA Cabinet

Fig. 22: Lower part of GPA cabinet (with D32 partially shown left and D31 right and main contactor K2)



Fans below the power stages

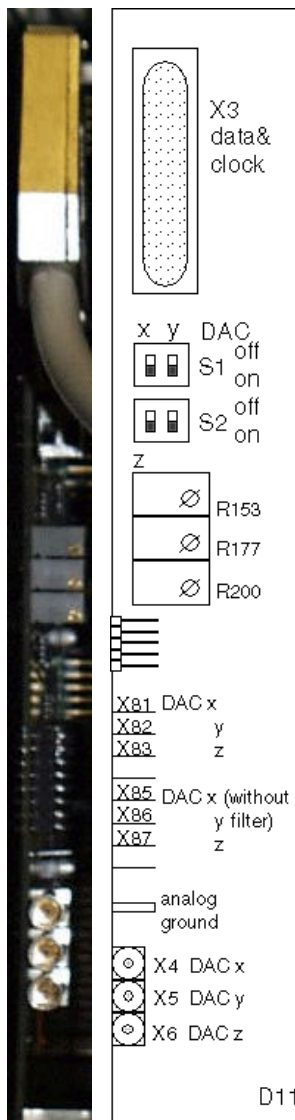
Behind cover plates:

- Rectifier board D32 in front
- Soft start board D31 behind D32
- Main contactor K2
- 25A fuses (15fuses, blow slow)

Water distributor with water pipes

2.3.2 DAC D112/111/11 Test Procedures

Fig. 23: Front of DAC board D11



Descriptions:

X3:

18 bit parallel gradient data and 4MHz clock signal from MMC

S1/S2:

The DAC output can be switched on/off for every axis; S2/2 is not used.

Potentiometers for offset adjustment

DAC output = nominal value (0-> +/-10V) 10 V corresponds to the maximum output current.

2.3.2.1 a. Check Jumpers and Switch Positions:

Jumper X16: 2-3 (sign of gradient data, 2-Complement)

DIP switch S1: both in lower position

DIP switch S2: both in lower position

Rotary switch S3: Position 3

Rotary switch S4: Position 6 (this configuration results in a clock signal FSYNC = 20 kHz)

2.3.2.2 Offset measurement/adjustment:



Prerequisite is the Regulator Adjustment (Tune-up).

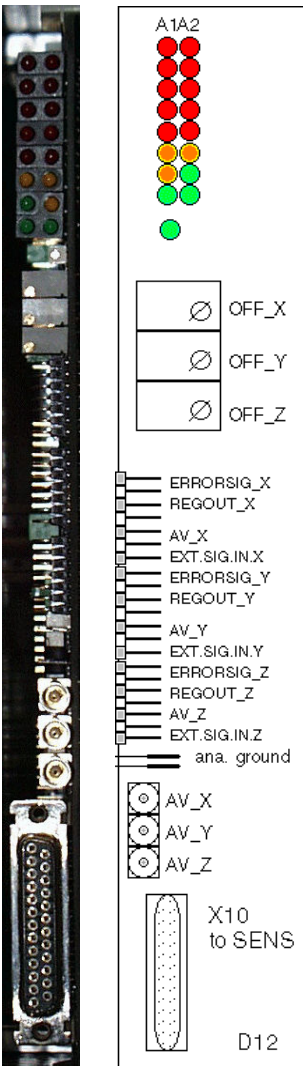
- Gradient cabinet has been on for at least 10 minutes.
- Set switches S1/2 to up position ("DAC output zero")
- The corresponding modulator must be on: S1/2 on D17 in down position.
- Connect a DVM to the desired actual value output X11,12,13 on D12.
Use the QLA-BNC service cable.
- Check the offset: Offset < +/- 0.05 mV
- Use the corresponding R153, 177, 200 on DAC-board D11 to adjust the offset for the X, Y and Z axes if necessary.

2.3.2.3 c. Gradient Data Test loop GCTX/D11:

Use the Test "Gradient/ DAC/Functional" of the Test Tools platform to confirm the correct gradient data transmission. between MMC/GCTX and GSSU/D11; (a prerequisite for this test is the correct function of the MMC components, refer to MMC-Tests).

2.3.3 Regulator D12/123/124 Test Procedure

Fig. 24: Front of Regulator D12



Descriptions:

- LED's (A1 left / A2 right):
V130: IMAX_X / IMAX_Y
V131: IMAX_Z / CAV_ERR_X
V132: CAV_ERR_Y / CAV_ERR_Z
V133: DUCY_X / DUCY_Y
V134: DUCY_Z / OHMLOSS
V135: LOOP_ERR_X / LOOP_ERR_Y
V136: LOOP_ERR_Z / REG_ON_X
V137: REG_ON_Y / REG_ON_Z
V159: current flow

Offset pots

- X: R 155
Y: R 189
Z: R 223

Test points (refer to (→ Interactive Functional Description / MR-000.850.05)

- X11,12,13: Actual current value X,Y,Z
QLA (1V=50A)
X10 SENS connection

2.3.3.1 a. Default Jumper Setting by DIP Switches (Regulator Parameters)

DIP-switch	Pos1	Pos2	Pos3	Pos4	Pos5	Pos6	Pos7	Pos8
S1,S2,S4 S5,S7,S8	ON	off	off	off	off	off	off	off
S3,S6,S9	ON	ON	ON	off	off	off	off	off

2.3.3.2 b. Jumper Settings:

Jumper	Setting	Function
X16, X18, X20	1-2	Nominal value X,Y,Z from D11
	2-3	Nominal value externally

X17, X19, X21	1-2 2-3	Current regulation normally Control from X67,77,87
X22/X23	1-2/1-2	Processor controls DAC gain
	2-3/2-3	DAC maximum range $\leq$ 505A
	open/2-3	DAC maximum range $\leq$ 435A
	2-3/open	DAC maximum range $\leq$ 375A
	without jumper	DAC maximum range $\leq$ 300A
X24	1-2 2-3 do not use this position !	Switching depending from current direction Switching not depending from current direction
X25	18 - 19, 21 - 22, 24 - 25	

2.3.3.3 c. SENS Offset measurement/adjustment:

- Gradient cabinet has been on for at least 10 minutes and the regulators are adjusted (see Regulator Adjust/Check below)
- Connect the DVM to X11,12,13 ("actual value") on the regulator board. Use the QLA-BNC service cable.
- Set the switch, corresponding to the axis, on the I/O board D17 in upward position ("Modulator off").
- Check the offset: Offset $< \pm 0.05$ mV; use offset pots to adjust the offset if necessary.

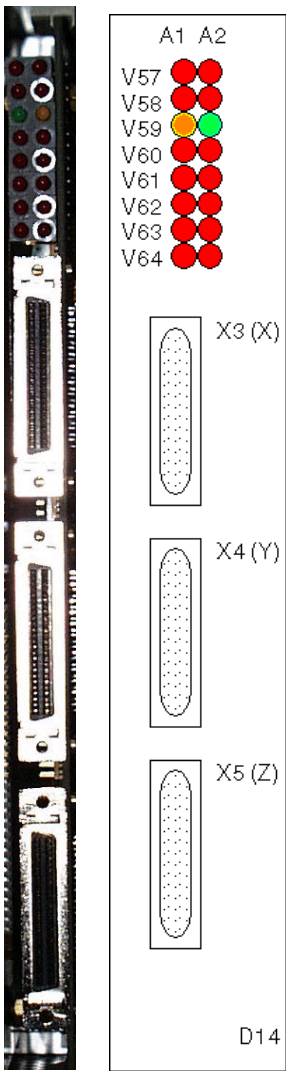
2.3.3.4 d. Regulator Adjust/Check:

In NUMARIS 4 the regulators are set via the Service Software function **TuneUp/Regulator Adjust**.

The **QA/Regulator Check** function quickly checks the regulator data set.

2.3.4 Modulator D14/142 Test Procedures

Fig. 25: Front of Modulator D14



Descriptions:

Error LED's

X3,4,5: Connections to power stages X,Y,Z
Clock signals, control of drivers, power stage transistors and temperature supervision (50-pin connectors)
Note:
If one of these cable connections is defective, LED V62/A2 is on during state 1.

2.3.4.1 a. Meaning of LED's

LED	LED ON
V57/A1	N_CLOCK_ERR (error in 82 MHz clock)
V57/A2	M_MOD_ERR (modulator disabled)
V58/A1	P_15V_ERR (+15V error)
V58/A2	N_15V_ERR (-15V error)
V59/A1	SYNC_ERR_CH (GPA not synchronous to system clock)
V59/A2	5V_ok
V60A1/2	Continuously on: Supervision chip not configured, change D14 (D142)

LED	LED ON		
V60/A1/A2	State 1:A1 ON / A2 OFF	State 2:A1 OFF / A2 ON	State 3:A1 ON / A2 ON
V61/A1	N_INT_INTModulator disabled	N_Spare	MOD SPARE 1
V61/A2	OFF_X, N_MODON_XX-modulator disabled	OFF_Y,N_MODON_YY-modulator disabled	OFF_Z,N_MODON_ZZ-modulator disabled
V62/A1	EMMI Danger of stimulation	Output disabled FOC control disabled	MOD_SPARE 3
V62/A2	Pull protection (cables D14/2 to power stages)	MOD_DEFModulator D14 defective	MOD_SPARE 2
V63/A1	Switching frequency X to high	Frequency Y	Frequency Z
V63/A2	TEMP X Overtemperature PS-X	TEMP YOvertemperature PS-Y	TEMP ZOvertemperature PS-Z
V64/A1	OVV_XOvervoltage PS-Y	OVV_YOver voltage PS-Y	OVV_ZOver voltage PS-Z
V64/A2	Driver voltage Xtoo low	Driver voltage Ytoo low	Driver voltage Ztoo low

After state 3, the LED's V60, A1/2 are off, all 8 LED's below are on.

2.3.4.2 b. Check of Jumper Settings:

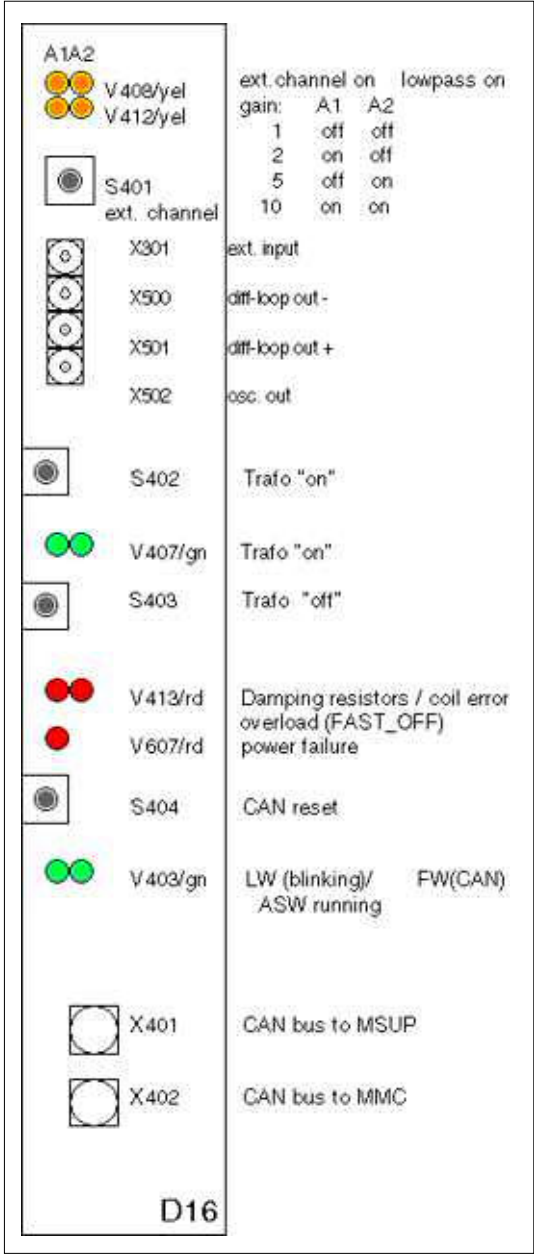
Slew rate settings:

Jumper	Setting	Function
X9/X10	1-2/1-2:	Processor controls slew rate
	2-3/2-3	Maximum slew rate possible
	2-3/open	n/a
	open/2-3	Half slew rate (SR2)
	without jumper SR4 (includes SR2)	Quarter slew rate (SR4)
X11	6 - 7, 18 - 19	
X12	1 - 2	
X13	1 - 2	

(Refer to (→ Interactive Functional Description / MR-000.850.05) for further explanations).

2.3.5 CAN-Service D16 Test Procedures

Fig. 26: Front of Service/Can board D16



Descriptions:

LED's show conditions of external channel settings.

S1: Sets external channel

X301 External signal input

X502 Oscilloscope output

S402: Manual "ON"

V407: Trafo on/off

S403: Manual "OFF"

V413 left: Damping resistors overload

V413 right: Gradient coil error

S404: Warm start of GPA-processor

V403 left, blinking during normal conditions

Jumper X3: 1-2 (regulator D-portion enabled)

2.3.5.1 a. Check communication to main Power Supply Unit:

The main power supply unit receives the switch-on signal via cable from assembly D17 (X1). Test by pressing S403 and then S402. Verify that the main power supply unit is switched to stand-by (LED's "EN" and "RDY" off) and on again after approx. 15 seconds (switch-on signal is present at D17 (X1) and is transmitted to the main power supply unit via cable).

2.3.5.2 b. Check of Jumper Settings:

Jumper X3: 1 - 2

Jumper X601: 1- 2

2.3.5.3 b. Manual loop Configuration:

Fig. 27: LED Display on D6 showing the possible gain and low pass filter combinations of the external channel

Setting of external channel on D6/D16:							
V 408 A1= ext.chan. A2= LP V 412 A1= gain A2= gain							
Taster S 401 pressed:							
1*	2*	3*	4*	5*	6*	7*	8*
ext.chan. No LP gain1	LP gain1	no LP gain2	LP gain2	no LP gain5	LP gain 5	no LP gain10	LP gain10
Connector external channel: X 301							

When using the external channel, you can select several combinations of different gain and low-pass filter settings by pressing button S401 n-times according to (→ Fig. 27 Page 129). Pressing nine times deselects the external channel (all LED's off).

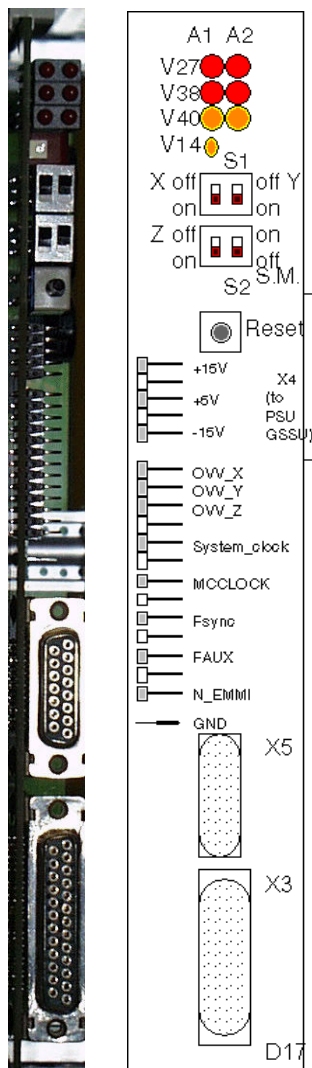
Note: If all LED's are off, the automatic loop is activated.

2.3.5.4 c. CAN-Bus communication/GPA status:

The present status (Status/warning/errors) of the GPA is displayed completely with the help of the Service Software function "**Gradient/Communication**" in the platform Test Tools.

2.3.6 I/O D17/Power supply Test Procedures

Fig. 28: Front of I/O board D17



Descriptions:

LED's (A1 left / A2 right):

V27: Rectifier -Temp. / Transformer-Temp.

V38: Coilerr / Filterfan

V40: EMMI / DB_WDG_ERR (stim.mon.)

V14: Yellow LED: Service mode

Switches for modulator on/off and service mode on/off

Error Reset

Test Pins:

+/-15V, +5V

Overvoltage X,Y,Z

System clock 4 MHz

Modulator clock 20 kHz

Synchron.clock 20 kHz

Auxiliary clock 50 kHz (not used)

X5 Connector to Software stimulation monitor
CCA/D21/X4

X3 Connector to switch-on assembly D31

The I/O board D17 is the interface of the GSSU to the switch-on assembly D31, as well as the SENS (synchronization via three FOC's) and supervision assemblies such as D9 (coil supervision) and the stimulation monitor. Electric signals are converted into FOC signals and vice versa.

Additionally the D17 delivers the supply voltages (from E4A Power Supply) to the GSSU components. X4 on the board is the connector to the E4A.

The four switches at the front of board allow an individual switch off for every modulators and the selection of the service mode.

A push button allows the reset of an error situation.

In case of errors, this is an easy method for ruling out "monitoring errors":

The status of the error transmission path is tested by pulling the respective fiber optic cable.

"Light off" means that there is an error and/or one of the corresponding sensors is defective.

2.3.6.1 Check of Jumpers D17/X9

Jumper	Link	Signal function
X9	11-12	Trafo temperature (no supervision)
	19-20	Filterfan (no supervision)
	21-22	

2.3.6.2 Check of DIP Switches

Switch S1: both down

Switch S2: both down

2.3.6.3 Connector D17/X3 (to Soft-Start unit D31)

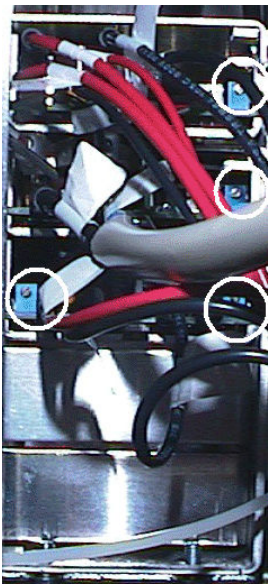
Pin	Meaning	Pin	Meaning
1	K1/K2 switch-on (p)	14	message K2 closed
2	K1/K2 switch-on (n)	15	+15V
3	Trafo overtemperature (p)	16	+15V
4	Trafo overtemperature (n)	17	+15V
5	Rectifier overtemp (p)	18	0V (GND)
6	Rectifier overtemp (n)	19	0V (GND)
		20	shield (to GND)
		21	-15V
		22	-15V
		23	-15V
		24	Extr. power input (p)
		25	Extr. power input (n)

2.3.6.4 D17 FOC's

Connector	Meaning
U1,2,3	Synchronization clock to current sensors
U7	Temperature supervision gradient coil

2.3.7 Small Signal Power Supply E4A Procedures

Fig. 29: Adjustment of supply voltages

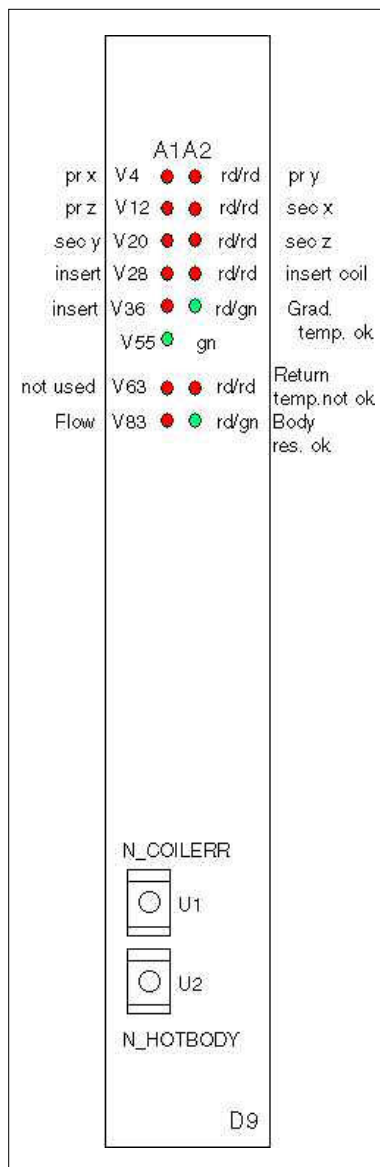


- Slot 1: +5,075 +/- 0,025 VDC
- Measure the voltage on D17 (→ Fig. 28 Page 130)
- Slot 2: +16,5VDC, supply voltage for drivers at D21 on the power stages and fan on top of GSSU.
- Measure the voltage at the terminals of E4A (→ Fig. 29 Page 132).
- Slot 3: +/- 15,05 +/- 0,025VDC
- Measure the voltage on D17 (→ Fig. 28 Page 130).
- Adjust the voltages with switched off modulators using the marked potentiometers (→ Fig. 29 Page 132).

Slot	Terminal at E4A
Slot 1	A = 5V, B = GND, C = 5V (Sense), D = GND(Sense)
Slot 2	E = +Blower, F = GND Blower, G = +Driver supply, H = GND Driver supply
Slot 3	I = +15V, J = GND, K = GND, L = -15V

2.3.8 Supervision D9/Grad.coil Test Procedure

Fig. 30: Front of Supervision D9



Descriptions:

LED's:

pr x,y,z:

NTC-sensor status at primary side of the gradient coil

sec x,y,z:

NTC-sensor status at secondary side of the gradient coil

V55gn: 22V supply voltage ok

V63 rd right:

Return temperature status of cooling water for the body resonator (Temperature too high)

V83 rd left:

Water flow supervision for the body resonator (Flow not ok).

V83 gn right: Body coil ok

N_COILERR:

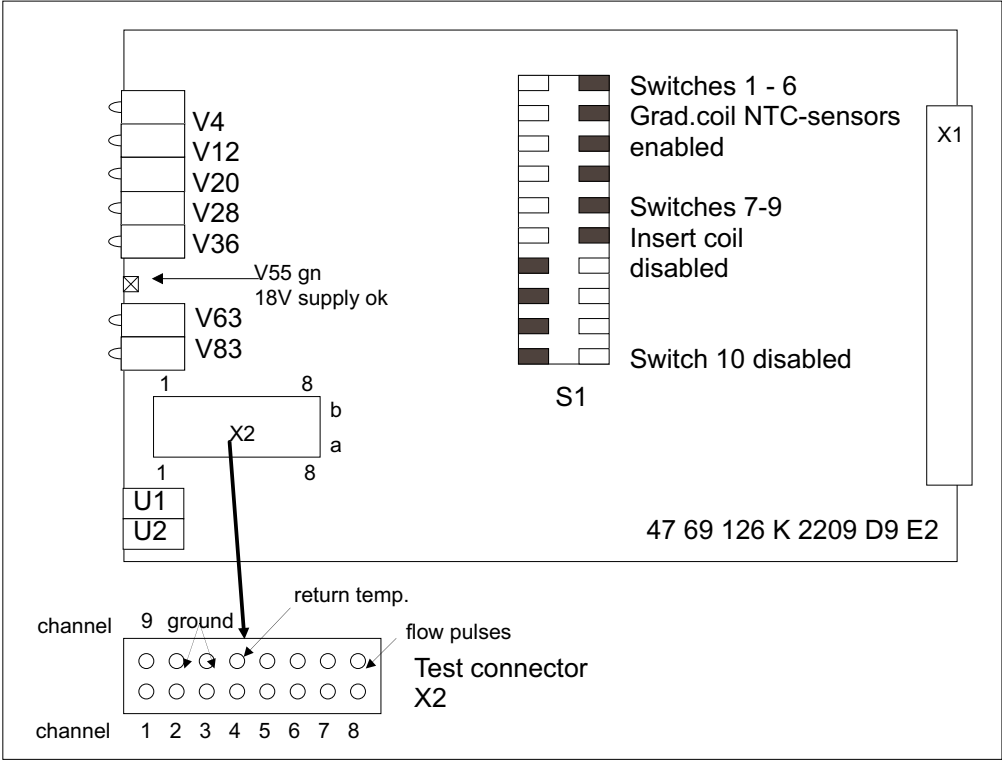
FOC connection to D7/U41 (Light off or glowing weakly: => Error)

N_HOTBODY

FOC connection to BTB1/U5 (Light off => Error/ Body Coil not ok)

The coil supervision board D9 is located in the left part of the patient table control electronics rack.

Fig. 31: Position of DIP switches at D9



Normal condition:

Check Jumpers and Switch Positions:

- Switches 1-6 = ON
- Switches 7-10 = off

(Switch 1 => pr x, switch 2 => pr y, a.s.o.)

An error condition (green LED off) exists if the gradient coil temperature at the location of the NTC-sensor exceeds 83 +/-7 C or the cable to the coil is not connected or the NTC is shorted or broken.

In case of overtemperature and/or shorted NTC the corresponding red LED lights up.

At most two defective NTC-sensors may be deactivated by switching the corresponding DIP-switch S1/D9 to the off position. In this case, the corresponding error LED may be permanently on, but no error message is generated.

In case of problems, measure the resistance of the NTC-sensors at the gradient coil connector X2 or at connector X20/D9:

DIP- Switch at D9	1	2	3	4	5	6
Connector X20/ D9	25-13	24-12	23-11	22-10	21-9	20-8
Grad.coil connector X2	4-5	6-7	8-9	10-11	12-13	14-15
AS39R(S)	Xprim	Yprim	Zprim	Xsec	Ysec	Zsec
AC39T Quantum	Xprim	Yprim	Yprim1	Zprim	Xsec	Ysec
AS Sonata	Xprim	Yprim	Yprim2	Zprim	Xsec	Ysec

DIP- Switch at D9	1	2	3	4	5	6
Connector X20/ D9	25-13	24-12	23-11	22-10	21-9	20-8
Grad.coil connector X2	4-5	6-7	8-9	10-11	12-13	14-15
AC44 (Allegra)	Xprim	Yprim	Zprim	Xsec	Ysec	Zsec
AS Trio	Xprim	Yprim	Yprim2	Zprim	Xsec	Ysec
At most two defective NTC sensors may be deactivated by switching the corresponding DIP- switch D9 to the off position						

Typical resistance of NTC's:

Room temperature: $R > 15\text{k}\Omega$ (approx. $40\text{ k}\Omega$ at $20\text{ }^{\circ}\text{C}$)

Temperature limit: $R < 4\text{ k}\Omega$

In case of a **body coil error**, i.e. cooling water flow too low (refer to **RF-repair instructions** for adjustment) and return cooling water to hot, the red LED's V63 (right) and/or V83 (left) light up.

Measurement of the cooling water temperature sensor resistance in the **body resonator** (return cooling water flow) can be done at connector X21/D9 between Pins 9 and 10.

Resistance values of the temperature sensor:

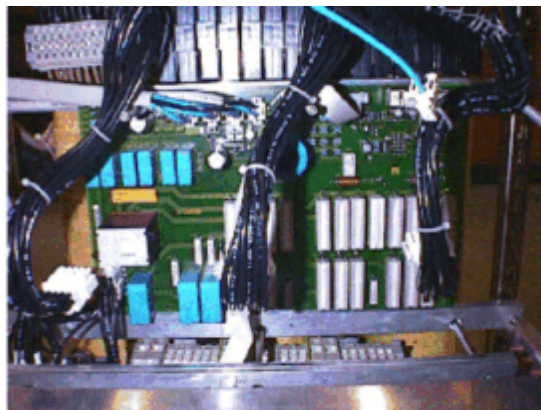
$R = 2000\text{ }\Omega$ at $20\text{ }^{\circ}\text{C}$

$R > 2140\text{ }\Omega$ -> overtemperature

$R < 2000\text{ }\Omega$ -> OK

2.3.9 D31 Soft start Unit Test Procedure

Fig. 32: D31 Soft start unit (D32 rectifier board removed)



Six lighted up LED's at left top of D31 indicate a closed motor protection switch F3/80A in the CCA mainbox, i.e. main voltage is present on D31 (refer to (→ Fig. 36 Page 141)).

**CAUTION**

Never switch on F3 (80A cutter switch in the mainsbox on top of CCA), if the GSSU is switched on.

Otherwise severe damage to the D32 and the power stages may occur.

- » Before switching on F3, switch off the GSSU with circuit breaker F14 in the CCA.

Fig. 33: D31 Soft start unit - Voltage indicating LED's and driver supply connectors X11, 12, 13 (to PS) and X10 (from E4A)



2.3.9.1 Connector X1 to GSSU, D17/X3

Pin	Meaning	Pin	Meaning
1	K1/K2 switch-on (p)	14	Message K2 closed (n)
2	K1/K2 switch-on (n)	15	+15V
3	Trafo overtemperature (p)	16	+ !5V
4	Trafo overtemperature (n)	17	+ 15V
5	Rectifier overtemp (p)	18	0V (GND)
6	Rectifier overtemp (n)	19	0V (GND)
7	Overvoltage Z (p)	20	shield (to GND)
8	Overvoltage Z (n)	21	-15V
9	Overvoltage Y (p)	22	-15V
10	Overvoltage Y (n)	23	-15V
11	Overvoltage X (p)	24	Overload (EPI) (p)
12	Overvoltage X(n)	25	Overload (EPI) (n)
13	Message K2 closed (p)	-	Shield

2.3.9.2 Connectors X11 to X13 and X10

Pin	Meaning	Pin	Meaning
1	+16,5V	2	0V

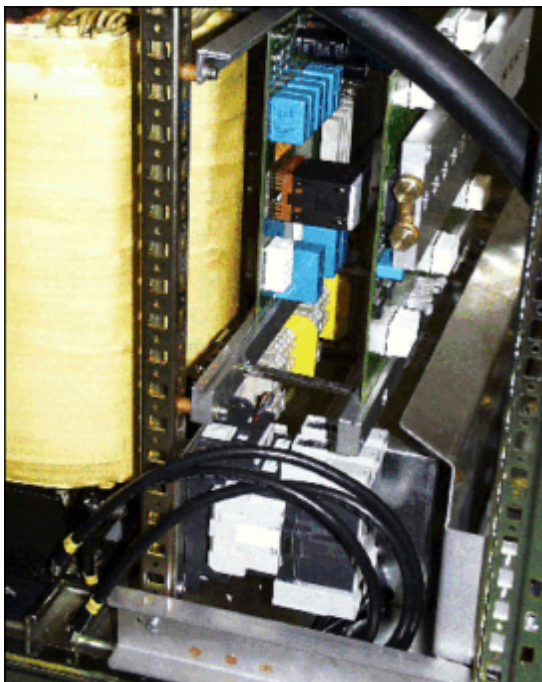
2.3.9.3 Flat Cable Connector X8 to D32/Rectifier Board

Pin	Meaning	Pin	Meaning
1	Connect. supervision	2	
3	+15V	4	+15V
5	0V (GND)	6	0V (GND)
7	PTC on Rectifier(n)	8	PTC on Rectifier (p)
9	Overvoltage X (p)	10	Overvoltage X (n)
11	Overvoltage Y (p)	12	Overvoltage Y (n)
13	Overvoltage Z (p)	14	Overvoltage Z (n)
15		16	Connect. supervis. (GND)

Connector X2 to K2 main Contactor

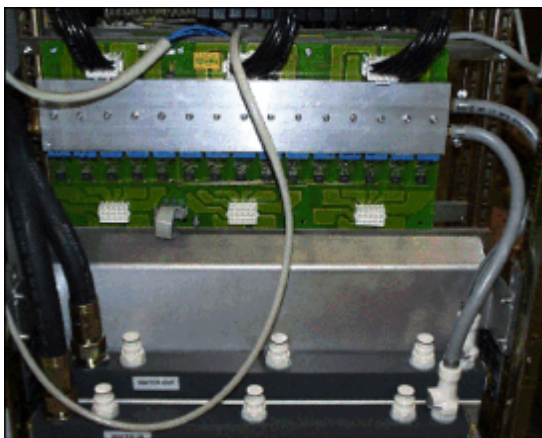
Pin	Meaning	Pin	Meaning	Pin	Meaning
1	K2_A1_old	2	K2_A2	3	K2_A1_new
4	K2 mess. cont.	5	K2 mess.cont.	6	
7		8		9	
10	Transformer L1	11	Transformer L2	12	Transformer L3
13	Line L1	14	Line L2	15	Line L3

Fig. 34: D31 Connector X2 - Main contactor K2



2.3.10 D32 Rectifier Test Procedure

Fig. 35: Rectifier board, ACV power connectors from T1 and cooling water pipes plugged in



CAUTION

Never switch on F3 (80A cutter switch in the mainsbox on top of CCA), if the GSSU is switched on.

Otherwise severe damage to the D32 and the power stages may occur.

- » Before switching on F3, switch off the GSSU with circuit breaker F14 in the CCA.

2.3.10.1 Connector X1 to D31/Softstart unit

Pin	Meaning	Pin	Meaning
1	Connect. supervision	2	
3	+15V	4	+15V
5	0V (GND)	6	0V (GND)
7	PTC on Rectifier(n)	8	PTC on Rectifier (p)
9	Overvoltage X (p)	10	Overvoltage X (n)
11	Overvoltage Y (p)	12	Overvoltage Y (n)
13	Overvoltage Z (p)	14	Overvoltage Z (n)
15		16	Connect. supervis. (GND)

2.3.10.2 Connector X11(X), X21(Y), X31(Z) ACV supply of rectifiers

Pin	Meaning X11 (X)	Meaning X21 (Y)	Meaning X31 (Z)
1	2U1	2W1	2V1
2,3	2U2	2W2	2V2
4	3V1	3U1	3W1
5,6	3V2	3U2	3W2
7	4W1	4V1	4U1
8,9	4W2	4V2	4U2
10	5U1	5W1	5V1
11,12	5U2	5W2	5V2
13	6V1	6U1	6W1
14,15	6V2	6U2	6W2

2.3.10.3 Measurement of Supply Voltages for Power Stages

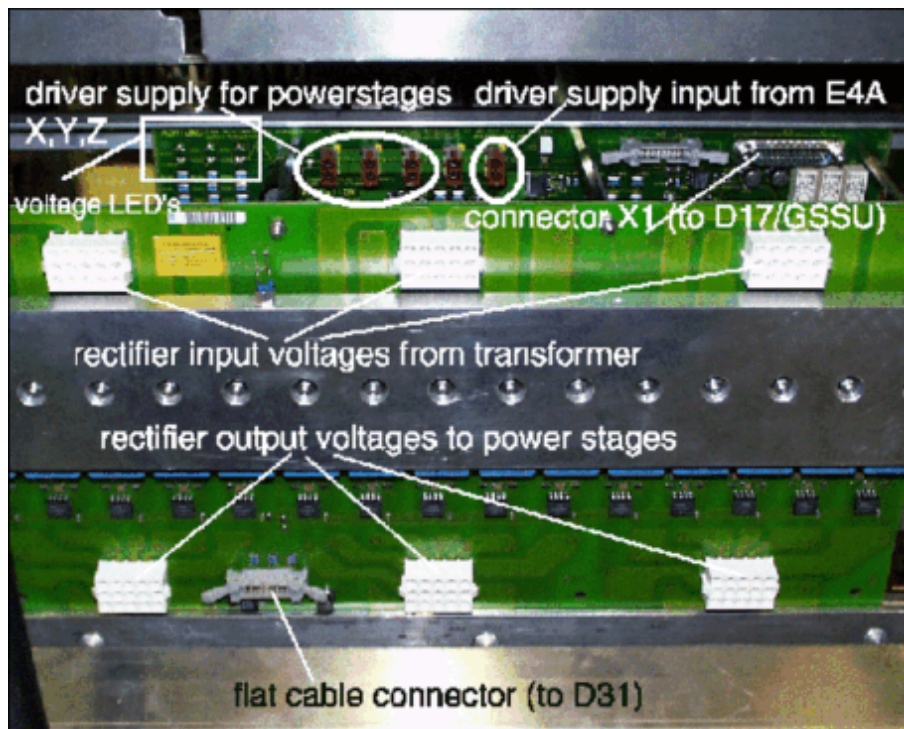
- Switch off K2/T1 with D16/S403.
- Switch off the cutter switch F3/3*80A in the CCA mainsbox.
- Pull one of the output connectors X12, X22, or X 32, (where you want to measure the output voltage).
- Connect a DVM (range 1000 VDC) with appropriate clips to the connector pins at D32 according to the table below.
- Switch off the modulators (D17 S1 and S2 left in upward position).
- Switch on the Service mode (D17 S2 right in upward position).

- Switch on F3/3*80A.
- Switch on K2/T1 with D16/S402.
- Read the output voltage at DVM.
- Switch off K2/T1 with D16/S403.
- Switch off F3/3*80A in the CCA mainsbox.
- Remove the measurement equipment.
- Plug in the removed connector again.
- Switch on F3/3*80A in the CCA mainsbox.
- Switch on K2/T1 with D16/S402.
- Switch on the modulators (D17 S1 and S2 left in downward position).
- Switch off the Service mode (D17 S2 right in downward position).

2.3.10.4 Connector X21(X), X22(Y), X32 (Z) DCV supply of Power Stages

Pin	Meaning	Pin	Meaning
1	DC voltage X,Y,Z_U1+	3	DC voltage X,Y,Z_U1-
4	DC voltage X,Y,Z_U2+	6	DC voltage X,Y,Z_U2-
7	DC voltage X,Y,Z_U3+	9	DC voltage X,Y,Z_U3-
10	DC voltage X,Y,Z_U4+	12	DC voltage X,Y,Z_U4-
13	DC voltage X,Y,Z_U5+	15	DC voltage X,Y,Z_U5-

Fig. 36: Plug positions on D32/D31



2.3.11 GPA Fans Test Procedure

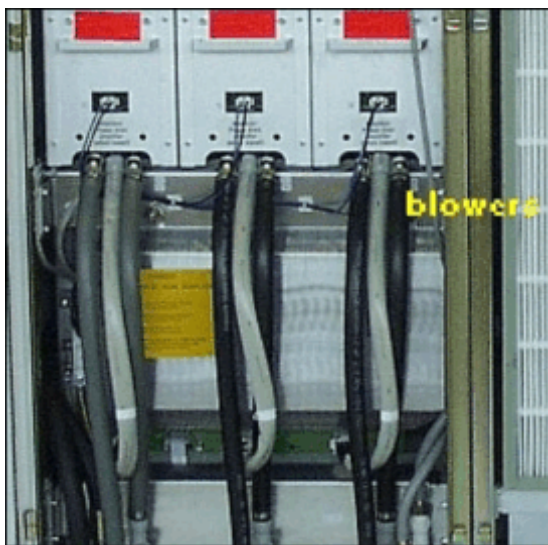
2.3.11.1 Check of Fan Function:

All five fans must work properly



The fan function of the first K2217 cabinets was controlled by temperature switches. With Update Instruction MR004/00/R these cabinets had been modified to "gradient power amplifier blower-permanently on".

Fig. 37: Gradient Amplifier Blower



2.4 TestTools - Gradient Loop Tests

2.4.1 Test usage and interpretation of results

The tests are available in the Service Software platform **TestTools**.

2.4.1.1 MMC - DAC Gradient Data Test (DAC/Functional)

(Loop MMC/GCTX - GSSU/ D1/D11 - MMC/GCTX)

Test of 18 bit parallel data transmission and control signals via cable.

In case of errors, perform the MMC-tests, check the cable connection between GCTX and GPA. In addition check the D1 K2209/D11 K2217.

2.4.1.2 DAC - Nominal Value (DAC/Functional)

(Loop MPCU/GCTX - D1/D11/DAC - D6/D16 - RFSU/ADC)

Nominal value for gradient current from DAC for x,y,z axes; Stability and linearity measurement with several parameter variations, such as amplitude and timing.

If some of the test fail, check D1 K2209/D11 K2217 (DAC), D6 K2209/D16 K2217, cables to the RFSU, receiver board. Perform the MMC tests.



If the test fails due to a regulation (loop) error, switch on the "service mode" (right switch S1 upward on D4 or right switch S2 on D17). Now you can test the correct function of the DAC's.

2.4.1.3 Actual current Value (X,Y,Z-axis/Functional)

(Loop MPCU/GCTX - D1/D11/DAC - D2/D12 - D14 - ((D3) - D4/5 - D7) - PS - SENS- D2/D12 - D6/D16 - RFSU/ADC)

Actual value for gradient current from D2/D12, actual value amplifier for x,y,z axes; Stability and linearity measurement.

If some of the test fail, check D2 K2209/D12 K2217 (regulator board) with "Regulator Check", D6 K2209/D16 K2217 (loopmux), cables to the RFSU, receiver board. Perform the MMC tests.

Check the following GPA K2209 components:

D4/5 K2209 (modulator board), D3 K2209 (synchronization board), D7 K2209 (FOC board) and the FOC's to the power stages. Moreover check for error messages concerning the D10 K2209 (driver supply board), power stages, the current sensors (SENS) and the RC; refer to the corresponding (→ Troubleshooting Charts GPA K2209 / Page 13), (→ Troubleshooting Charts KAS K2217 / Page 48).

For further information refer to the Online Help of the gradient tests.

2.5 Normal/Standby Operation

2.5.1 Status of Components K2209

Gradient Small Signal Unit Components	
D6 Service/CAN-board Normal Mode:	2 LED's "RC on" (V407,green) on; left LED (V403,green) is blinking (loadware is ok, application software (ASW) is running)
Stand-by:	2 LED's (V407,green) off -> RC in stand-by; left LED (V403,green) is blinking (loadware is ok, ASW is running)
D1 DAC	Switches S1/2 in downward position
D2 Regulator	no error LED on
D3 Synchron.	LED (V30, green) on -> VCC +5V ok
D4/5 Modulator	no red error LED on; LED's (V8-10, yellow) indicate gradient (+/-) pulsing during a measurement
	Switches D4 S1/S2 "Modulator on" in downward position; Switch S1 "Service mode off" in downward position
Stand-by:	D4, LED V5/A2 "Modulator disabled" (V1/A2,yellow) is on
D10 Driver supply	Green LED is on -> Output voltage ok
D9 Supervision (located in Pat. tab. control)	LED "Grad. temp.ok" (V36/A2, green) on; LED "Body res. ok" (V83/A2, green) on.
GPA Small Signal Power Supply: Fan is running	
CCA cabinet: Circuit breakers F14,15 and F9 are switched on Fan's beside the RC and at the top (inside) of the cabinet are running	
Power stage: FOC "T" is on (red light).	

Gradient Small Signal Unit Components

Resonant Converter (normal mode):

Green LED's "EN", "RDY", "P15V", "N15V", "HF15V" are on

Resonant Converter in stand-by:

Green LED's "P15V", "N15V", "HF15V" are on, "EN" and "RDY" are off

2.5.2 Status of Components K2217

Gradient Small Signal Unit Components

D16 CAN-Service-board Normal Mode:	2 LED's "Trafo on" (V407,green) on; left LED (V403,green) is blinking (loadware is ok, application software (ASW) is running)
Stand-by:	2 LED's (V407,green) off -> "Trafo off"; left LED (V403,green) is blinking (loadware is ok, ASW is running)
D11 DAC	Switches S1/2 in downward position
D12 Regulator Normal Mode:	LED's V136/137 green on
Stand-by:	LED's V136/137 green off
D14 Modulator Normal Mode:	green LED V59 on -> VCC +5V ok
Stand-by:	D14, LED "Modulator disabled" (V61/A2, red) is on in state 1, 2 and 3.
D17 I/O	Switches S1/S2 "Modulator on" in downward position; Switch S1 "Service mode off" in downward position
D9 Supervision (located in Pat. tab. control)	LED "Grad. temp.ok" (V36/A2, green) on; LED "Body res. ok" (V83/A2, green) on.
GPA Small Signal Power Supply: Fan is running	

Gradient Small Signal Unit Components

CCA cabinet: Circuit breakers F14 are switched on;
Cutter switch F3/3*80 A in the CCA mainsbox (top of CCA) is switched on.

Power stages:

If the six green LED's behind the six holes in the front of the PS are on, the capacitors are charged.

2.6 GPA Power-on Process

After "Power On" or "Reset") all CAN units run through a self-test and start the firmware. The firmware checks to whether the loadware has been loaded. It also checks the loadware code. The loadware is started if the consistency test is successful.

When the gradient processor is in the loadware mode, a logon command is sent every 500 milliseconds to the MPCU.

The gradient processor is the link between the gradient power amplifier and the rest of the system. As a CAN unit, it is connected with the CAN bus to the system.

In loadware mode, the gradient processor is ready for operation with the exception of sending commands to the MPCU.

The gradient processor is fully operational after the MPCU has validated the version of the gradient processor via the login request and the Software application has been started via a CAN command.

In other words, after the CAN unit has been booted and the loadware started, the application software is not fully available. However, after the login request by the MPCU, the application software is fully operational.

The application software of the gradient processor allows for the reliable operation of the overall component, namely the gradient power amplifier. The most important tasks are:

- Setting the PI regulators for the three gradient axis
- Setting the different test loops for the Service Software (gradient stability/linearity tests)
- Service mode, switch on/off
- Setting the Power on/Standby mode

2.7 GPA Status (Example K2209)

Using the test Gradient/GPA status/Comm/Status in the platform Test Tools, you will get the following listing as an example:

```
Starting Test GetGpaStatus ...  
Status of GPA is ok.  
  
...  
GPA_STATUS: Resonantconverter current trip =no  
GPA_STATUS: Resonantconverter standby      =no  
GPA_STATUS: Resonantconverter defect       =no  
GPA_STATUS: Drivercurrentsupply           =ok  
  
...
```

and so on.

In case of errors, the corresponding status is indicated with "!!!". In this case, refer to the "Details" of the Event Log, respectively to the corresponding (→ Troubleshooting Charts GPA K2209 / Page 13), (→ Troubleshooting Charts KAS K2217 / Page 48).

2.8 Description of GPA Service mode

Service mode can be activated either by the Service Software or by setting manual (right) switch S1 on board D4 to the up position.

If service mode is active, the following errors will be handled as warnings, e.g. the modulators and regulators will not be disabled:

- K2209: RC not ready ($U_{\text{act}} < 60\%$ of U_{nom})
- K2209: Synchronization error (D3, LED V37 red/ D14 LED V3 red)
- K2209/K2217: Regulation error x,y,z (D2, LED V27 yellow/ D12, LED's V6/V7 yellow)
- K2209/K2217: No reaction of modulator

Version 02:

- Link error cleared.

Version 03:

- Minor editorial changes.

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Print No.: MR-000.840.16.03.02 | Replaces: MR-000.840.16.02.02
Doc. Gen. Date: 07.15 | Language: English
Part No.: n.a.
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